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Trigger-Based Declarative Conformance Checking

Master Thesis

presented by
Duygu Mustafa
434509

First Examiner: Prof. Dr. Wil M. P. van der Aalst
Second Examiner: Prof. Dr. Andreas Schuppert

Advisor: Christopher Schwanen and Harry Beyel

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Abstract

Conformance checking relates observed event data to a normative process specification in order to detect and diagnose deviations. Many real-world processes are governed not by a rigid procedure but by declarative rules that constrain the order of activities together with the data values, resources, and timing conditions under which they apply. Checking such rules on real event logs raises two recurring problems. First, procedural conformance checking techniques evaluate entire traces and return aggregate fitness scores, obscuring which individual rule was violated and when. Second, a single case identifier often bundles several independent enactments of the same process, so trace-level analysis conflates episodes that should be assessed separately.

This thesis proposes a declarative, multi-perspective conformance checking framework that addresses both problems. Normative rules are expressed as Multi-Perspective Declare (MP-Declare) constraints spanning control-flow, data, and temporal dimensions. Rather than evaluating whole traces, the framework introduces a trigger-based episode extraction procedure that segments full event logs into individual process episodes, localizing constraint evaluation to the context in which each rule applies. Standard MP-Declare semantics are extended with a target-selection mode for repeatedly recorded measurements in which constraint fulfillment is determined by the most recent measurement within the temporal window and not by any satisfying occurrence. Together, these contributions yield deviation reports that are localized to individual episodes and specific constraints rather than to entire traces.

The framework is instantiated and evaluated on mechanical ventilation weaning in intensive care units (ICUs), using event logs derived from Medical Information Mart for Intensive Care IV (MIMIC-IV). Evaluation on synthetic logs confirms that the checker largely reproduces the expected semantics and shows that episode-scoped evaluation better preserves conformance verdicts and outcome associations than case-level analysis. In the clinical study, episode-level evaluation distinguishes documented deviations from gaps in clinical documentation and preserves differences between successive weaning attempts that aggregate trace-level analysis obscures. These findings demonstrate that episode-scoped, multi-perspective conformance checking provides more precise and interpretable assessments of guideline adherence in recurrent clinical processes.

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List of Abbreviations

APRV	airway pressure release ventilation
ASP	answer set programming
BPMN	Business Process Model and Notation
CI	confidence interval
CIG	computer-interpretable guideline
CPAP	continuous positive airway pressure
EHR	electronic health record
FWER	family-wise error rate
ICU	intensive care unit
LPM	local process model
LTL	Linear Temporal Logic
LTL_f	Linear Temporal Logic on Finite Traces
MFOTL	Metric First-Order Linear Temporal Logic
MIMIC-IV	Medical Information Mart for Intensive Care IV
MP-Declare	Multi-Perspective Declare
MV	mechanical ventilation
NIV	non-invasive ventilation
PEEP	positive end-expiratory pressure
PSV	pressure support ventilation
RASS	Richmond Agitation–Sedation Scale
SAS	Riker Sedation–Agitation Scale
SBT	spontaneous breathing trial
SIMV	synchronized intermittent mandatory ventilation

1 Introduction

Many organizational and knowledge-intensive processes are governed by normative guidelines: clinical protocols, service-management policies, or regulatory procedures that prescribe which activities should be performed, in what order, and under which conditions. Whether the actual behavior of practitioners conforms to these guidelines is rarely self-evident, since a persistent gap separates prescribed procedure from recorded practice [42, 49]. As the day-to-day execution of such processes is increasingly captured in information systems as event data, *process mining* offers a means to close this gap analytically. Within process mining, *conformance checking* compares recorded behavior against a normative model and reports where the two diverge, providing an automated and scalable alternative to manual audits for adherence monitoring [5, 59].

The value of conformance checking depends critically on how the normative model is expressed. *Procedural* models enumerate the permitted execution paths and, when confronted with the flexibility inherent in knowledge work, either over-constrain legitimate variation or collapse the result into a single aggregate fitness score that reveals little about *which* obligation was breached [44]. *Declarative* models such as Declare instead adopt an *open-world assumption*, specifying only the behaviors that are required or forbidden and permitting everything not explicitly ruled out. This approach accommodates the variability of flexible processes far more naturally [16, 80]. Many real-world guidelines are moreover *multi-perspective*. Their rules interleave control-flow (event ordering), data conditions (thresholds over recorded measurements), and temporal constraints (deadlines and waiting periods). Multi-Perspective Declare (MP-Declare) extends the declarative formalism to capture all three dimensions simultaneously [16].

A further difficulty is one of *scope*. Conformance checking conventionally treats an entire case trace, from a case's first recorded event to its last, as the unit of evaluation [16, 59]. Object-centric process mining relaxes this assumption by relating events to multiple interacting objects rather than to a single case [3], but it presupposes that these objects are explicitly recorded in the event data. Yet even a single case frequently traverses several conceptually distinct phases, each opening its own locally scoped compliance obligations that arise, must be met, and expire within that phase. An IT incident may be reopened repeatedly, each reopening starting a fresh handling cycle; an insurance claim may undergo successive reassessments; an intensive-care patient may undergo several weaning attempts during one admission.

The weaning example makes the pitfall concrete. A guideline obligation such as "every spontaneous breathing trial must be preceded by a passing readiness assessment" is scoped, in intent, to a single weaning attempt. Consider an admission with two attempts (Figure 1.1): in the first, the patient is trialed despite a *failing* readiness assessment and the trial fails; in the second, a passing readiness assessment precedes a successful trial.

Evaluated over the whole trace, the premature trial of the first attempt violates the obligation, and since a violation, once incurred, cannot be repaired by any later behavior, the single trace-level verdict condemns the entire case as non-conformant, a verdict that is coarse in both directions. It does not localize the deviation to the first attempt, and it withholds credit from the second attempt, which followed the guideline exactly. A case with one flawed attempt among many becomes indistinguishable from one in which every attempt was mishandled [44, 72]. Scoping the evaluation to each attempt instead yields attempt-level verdicts, the first violated and the second satisfied, that pinpoint where care departed from the guideline and how often. The corrective is to make the recurrent *episode*, rather than the whole case, the unit of evaluation.

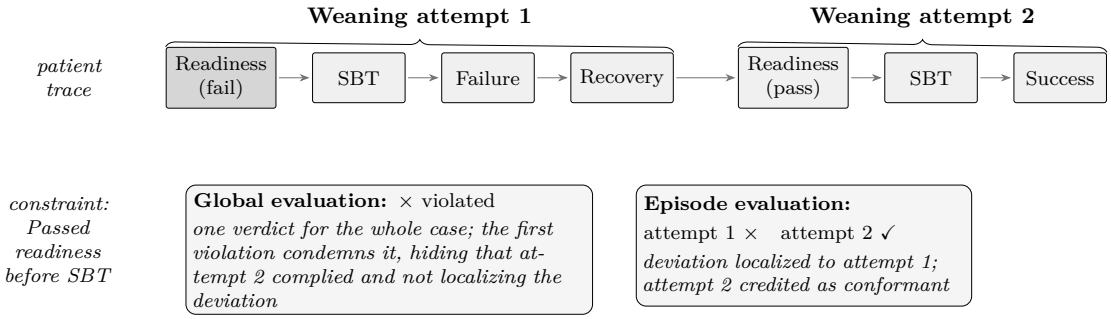


Figure 1.1: A patient trace with two weaning attempts, evaluated globally and per episode under the obligation "every spontaneous breathing trial (SBT) must be preceded by a passing readiness assessment".

This thesis develops a declarative, multi-perspective conformance-checking framework built around exactly this episode-scoped view. The method is organized as a four-step pipeline: a trigger-based procedure that segments each case trace into episodes, a bounded evaluation context constructed per constraint, an MP-Declare evaluation step, and an outcome association analysis. Two further observations about guideline-governed, measurement-intensive processes shape the latter steps. First, many of the quantities such guidelines regulate are not discrete actions but repeatedly recorded *measurements*. Consider a rule such as "after the intervention, blood pH must be above threshold". It is about where the quantity stands at the end of the observation period, yet standard MP-Declare matching is existential over target events, so a single transiently compliant reading fulfills the rule even when a later reading contradicts it. The framework therefore extends MP-Declare with a target-selection mode that consults the last measurement in the window. Second, a guideline rule is itself a hypothesis about good conduct, an assertion that adhering leads to better episodes. The outcome association stage tests this hypothesis against the data, estimating per rule whether adherence is statistically associated with the favorable episode outcome, and thereby checks not only the log against the rules but the rules against reality. The method is thereby domain-agnostic. All domain knowledge, the activity mapping, the constraint set, the episode triggers, and the outcome definition enters through externalized configuration. Transferring the

framework to a new process therefore requires authoring configuration, not modifying the method.

We instantiate and evaluate the framework on a demanding real-world instance of the recurrent-episode problem: the weaning of intensive-care patients from mechanical ventilation. Weaning is guideline-governed [10, 14, 89], multi-perspective (readiness thresholds, trial ordering, and timing deadlines) [10], and naturally episodic, as a single admission may contain several spontaneous breathing trials [14], each a distinct compliance episode. It thus exercises the method while grounding the evaluation in data of genuine clinical consequence. The full clinical background and the guideline formalization are deferred to the case study in Chapter 6. The present chapter treats weaning only as the running example that motivates the general design.

1.1 Problem Statement

Normative guidelines for flexible processes encode a rich combination of ordering constraints, data conditions, temporal deadlines, and state-dependent applicability conditions. Event data recorded by information systems captures the actual behavior of such processes at scale, but is inherently complex. Observations are irregularly recorded, the same activity recurs under conceptually different circumstances depending on case context, and not every deviation from a guideline reflects erroneous execution, as some are warranted adaptations to case-specific circumstances. Existing conformance-checking approaches assess only structural alignment between observed and prescribed sequences, and therefore neither capture the full multi-perspective nature of such guidelines, spanning control-flow, data, and temporal dimensions, nor distinguish critical violations from minor or warranted deviations, rendering their output insufficient for actionable compliance monitoring. Moreover, evaluation at the level of the full case conflates the multiple episodes a case may contain into a single trace, obscuring deviations that are meaningful only at the level of an individual episode. Figure 1.1 illustrates this last point on the weaning example: a case containing one non-conformant and one conformant attempt receives a single violated verdict, which neither localizes the deviation nor credits the attempt that followed the guideline.

1.2 Research Questions

RQ1. How can the normative rules of a guideline-governed process be formally modeled as MP-Declare constraints that incorporate control-flow, data, and temporal dimensions while preserving the open-world assumption?

RQ2. How can locally scoped compliance episodes be systematically identified and delimited from full case traces through trigger-based segmentation, and to what extent does constraint evaluation at the episode level improve the precision and interpretability of conformance diagnostics compared to global, trace-level evaluation?

RQ3. To what extent does the proposed framework achieve adequate coverage and precision when instantiated on a real-world event log, evaluated here on intensive care unit (ICU) weaning data from Medical Information Mart for Intensive Care IV (MIMIC-IV) [47], and what clinically relevant guideline deviations does it identify?

1.3 Research Goals

1. Develop a systematic method for translating the normative rules of a guideline-governed process into a formal set of MP-Declare constraints addressing the control-flow, data, and temporal perspectives.
2. Develop a trigger-based segmentation procedure that divides full case traces into locally scoped compliance episodes, and evaluate whether localizing MP-Declare constraint checking to these episodes reduces irrelevant violations and improves the actionability of deviation reports relative to global evaluation.
3. Instantiate and evaluate the framework on ICU weaning data from MIMIC-IV to generate structured diagnostics of guideline adherence, demonstrating the method on a real-world, multi-perspective, episodic process.

1.4 Contributions

This thesis makes the following contributions.

- **A trigger-based episode extraction procedure** that segments full case traces into locally scoped compliance episodes, each opened by a recurrent domain-meaningful event and bounded by a configurable observation window, an end-event condition, or the onset of the next episode. The constraint model is left intact, so the open-world semantics of Declare is preserved at the episode level (Section 4.1).
- **A bounded evaluation context** that resolves the boundary friction created by cutting episodes out of continuously recorded processes, through constraint-scoped pre- and post-context windows that admit the measurements and the late-recorded evidence an episode boundary would otherwise cut off, and derived history attributes that supply the summarized history a condition refers to (Section 4.2).
- **An extension of MP-Declare with target selection**, a mode for repeatedly recorded measurements in which an obligation can be discharged by the last measurement in the window instead of by any satisfying occurrence. The extension is conservative, reducing to standard MP-Declare semantics when it is not configured (Section 4.3).
- **An outcome association stage** that treats each constraint as a hypothesis about good conduct and estimates, per constraint, whether adherence is associated with

per-constraint

the favorable episode outcome. This yields a verdict on rule quality and not on rule adherence alone, and makes faithfully transcribed but invalid quality indicators visible (Section 4.4).

- **A verdict catalogue for MP-Declare checkers**, a reusable suite of 95 hand-written micro-traces, each paired with one constraint and its expected verdict. The catalogue covers every implemented template in satisfied, violated, and, where the semantics permit, vacuous cases, and varies the dimensions that real logs exercise only sparsely: activation and correlation conditions, forward and backward time windows, and target-selection modes (Section 5.1.3).
- **An evaluation on synthetic logs with planted ground truth** establishing that the checker returns the verdicts its semantics prescribe across all 24 implemented templates, and isolating the effect of the scoping choice by holding events, constraints, and statistical machinery fixed (Section 5.2.1).
- **A formalization of a clinical weaning guideline as** 21 **MP-Declare constraints, and its evaluation on MIMIC-IV**, yielding adherence diagnostics over 66 848 episodes drawn from 35 692 ICU stays, including a decomposition of violations by cause that separates documentation gaps from clinical deviation (Chapter 6).
- **An open, configuration-driven implementation** of the full pipeline, in which the trigger specification, constraint definitions, context windows, and outcome specification are versioned configuration artifacts rather than code, so that transferring the framework to a new process requires no modification of the method. The implementation is accompanied by an interactive front end for authoring these configurations, running the pipeline, and inspecting episode verdicts and constraint–outcome associations.[1]

1.5 Thesis Structure

The remainder of this thesis is structured as follows. Chapter 2 surveys related work and positions the contribution. Chapter 3 introduces the preliminaries on event data, Declare and MP-Declare, conformance checking, and statistical tests. Chapter 4 presents the four-step framework, from trigger-based episode extraction to outcome association analysis. Chapter 5 evaluates the framework on synthetic logs with planted ground truth and on a public event log. Chapter 6 applies it to ICU weaning in MIMIC-IV. Chapter 7 discusses the findings and trade-offs, and Chapter 8 concludes the thesis.

[1] `https://github.com/USER/REPO`

2 Related Work

This chapter positions the thesis within related work on conformance checking (Section 2.1), declarative process modeling and its conformance techniques (Section 2.2), process mining and guideline adherence in healthcare (Section 2.3), and localized process analysis (Section 2.4).

2.1 Conformance Checking

Conformance checking, the comparison of observed process behavior in event logs against a normative model, is a core discipline of process mining [18]. Early techniques replay each trace on a Petri net and count missing and remaining tokens [81], which yields a fitness score but only coarse diagnostics. Alignments address this by computing an optimal pairing of log moves and model moves under a cost function, so that each deviation is located as a concrete skipped or inserted activity [7], and De Leoni and van der Aalst [57] extend them to check control-flow, data, time, and resources simultaneously. Behavioral profiles offer a lighter alternative that compares ordering and exclusiveness relations between pairs of activities rather than complete runs [92]. A parallel line of work checks logs against individual rules instead of a complete model. Van der Aalst et al. [4] verify Linear Temporal Logic (LTL) properties directly on event logs, and Ramezani et al. [77] formalize compliance rules derived from regulations as Petri-net patterns and align logs against them, reporting for each rule where a violation occurred.

Despite these advances, procedural approaches share a fundamental limitation in high-variability settings: the model must enumerate all permissible execution paths explicitly [6, 30], which leads to so-called spaghetti-like models for processes with many alternative routes, as commonly observed for healthcare processes [66, 78]. In domains such as intensive care, where patient trajectories diverge substantially and clinician discretion is central, a procedural model either oversimplifies the normative behavior or becomes too complex to be interpretable or maintainable [71].

2.2 Declarative Process Modeling and Conformance

The foundations of the declarative approach to process management were established by Pesic and van der Aalst [75], who proposed the ConDec language (later renamed Declare) as a paradigm shift away from imperative, flow-based workflow systems. Rather than prescribing how a process must be executed, Declare specifies a compact set of behavioral constraints, grounded in Linear Temporal Logic on Finite Traces (LTL_f) [36], within which all executions that do not violate any constraint are permitted. This

open-world assumption is a defining property. Any behavior not explicitly forbidden is implicitly allowed. Di Ciccio and Montali [22] provide a comprehensive reference for the full lifecycle of declarative process mining, spanning reasoning (consistency checking and dead activity detection), discovery (inferring constraints from event logs while excluding vacuous satisfactions), and monitoring (classifying in-progress traces as permanently or temporarily satisfied or violated, using RV-LTL [12] semantics).

The expressiveness of control-flow-only Declare is, however, acknowledged as insufficient for many real-world settings [86]. Burattin et al. [16] extend the language to a multi-perspective formalism, MP-Declare, grounded in Metric First-Order Linear Temporal Logic (MFOTL), permitting activation conditions on data payloads, correlation conditions between activating and target events, and metric time constraints expressed as intervals. A corresponding multi-perspective discovery methodology is presented by Schönig et al. [83], who demonstrate that control flow, data, time, and resource dimensions can be simultaneously mined from event logs.

Conformance checking techniques have followed this shift to the declarative setting. De Leoni et al. [58] proposed a dedicated framework for aligning event logs and Declare models, enabling not only conformance diagnostics but also event-log preprocessing (e.g., cleaning, repair) prior to further analysis. Bergami et al. [13] extend alignment-based conformance checking to data-aware Declare by reducing the problem to a data-agnostic alignment task solved via cost-optimal automated planning, thereby providing both deviation quantification and optimal trace repair strategies. The multi-perspective formalism of Burattin et al. [16] complements these alignment-based approaches with an automata-driven framework for MP-Declare conformance that is linear in the number of traces, constraints, and events per trace, with worst-case quadratic complexity. The framework explicitly distinguishes fulfillments, violations, and vacuous satisfactions per constraint per trace. Casas-Ramos et al. [19] recently demonstrated that alignment-based conformance checking can be extended to rich data-aware Declare specifications with general data types through a novel combination of A* search and SMT solving, achieving efficiency without sacrificing expressiveness.

Alternative computational substrates have been proposed based on answer set programming (ASP), both for data-aware Declare in general [21] and for non-automata encodings of LTL-based declarative specifications [51]. However, these approaches still define the semantics of LTL_f via finite-state automata, which implies that formulas must first be transformed into automata before ASP reasoning can take place, thereby retaining the automata construction as a computational bottleneck [23]. As such, automata operations already introduce a substantial computational burden, adding an ASP layer on top would increase complexity without providing additional diagnostic value for per-constraint temporal analysis in the present setting. For these reasons, and because the automata-based MP-Declare framework already supports metric temporal constraints and fine-grained activation-state diagnostics, this work adopts the automata approach of Burattin et al. [16] and treats ASP-based techniques as complementary, but out of scope.

What these approaches report is *whether* a constraint holds, not *why* it failed. Alignment-based methods return a cost and an optimal repair [13, 19, 59], and the

automata-based MP-Declare framework returns a fulfillment, a violation, or a vacuous satisfaction per constraint per trace [16]. For a data condition evaluated over routinely collected clinical data, however, a condition that comes out false admits two incompatible readings: the process departed from the rule, or the quantity the rule refers to was never recorded. A binary verdict cannot separate them, and neither family of methods attempts to. Work on log imperfection approaches the problem from a different direction, resolving partial orders and bounding conformance over uncertain event data [1, 63, 73], which quantifies the uncertainty of a verdict without attributing a particular undischarged obligation to a cause. The framework developed here records, for every activation it cannot discharge, which situation applied, so that a documentation gap and a departure from the guideline are reported as distinct findings rather than as one non-conformance rate. Chapter 6 reports the resulting decomposition.

To the best of our knowledge, existing applications of MP-Declare have not yet explored trigger-based, episode-level conformance in high-variability clinical settings. This thesis adopts the MP-Declare formalism of Burattin et al. [16] and extends it with a locality-aware evaluation pipeline.

2.3 Healthcare

This section reviews the methodological literature on formalizing clinical guidelines and assessing adherence to them, since it is this body of work that the thesis positions itself against. The clinical background of ventilator weaning, together with the epidemiological evidence motivating the case study, is deferred to Sections 6.1 and 6.2.

2.3.1 Modeling Healthcare Processes

Before process mining techniques can be applied to measure protocol adherence, clinical guidelines must be transformed into formal, machine-interpretable models. Peleg [74] provides a comprehensive methodological review of computer-interpretable guideline (CIG) languages. An early empirical study by Mulyar et al. [67] evaluated four prominent CIG languages against 43 control-flow patterns from workflow management, finding that all four languages offer direct support for basic control-flow patterns but fail to support multiple-instance patterns, and that their expressive power broadly aligns with traditional workflow languages. This finding motivates the adoption of Declare in the clinical domain, as it natively supports patterns that procedural CIG languages struggle to express, especially in contexts involving highly variable or concurrent care activities.

Healthcare processes span a spectrum from highly structured administrative workflows to unstructured, knowledge-intensive treatment and guideline processes [17]. In structured settings, such as laboratory order handling, behavior is relatively predictable and can be captured by procedural models that closely guide and reflect execution [17]. In contrast, diagnostic and therapeutic processes must accommodate substantial variability, clinician discretion, and patient-specific adaptations, which makes purely procedural modeling brittle and often unreadable when variability is encoded explicitly [50, 80].

Moreover, modeling time-sensitive medical processes is intrinsically challenging because ICU event data often suffer from timing errors, unsynchronized clocks, and device- or human-induced recording imprecision, leading to substantial temporal uncertainty in the observed traces [38]. Declarative approaches, such as CIGDec and Declare, have therefore been advocated as more suitable for these unstructured medical processes, since they specify admissible and forbidden behaviors through constraints instead of enumerating all permissible paths [68, 80]. Recent work further shows that variability itself can be modeled explicitly using healthcare-specific patterns, providing clinicians with an accessible abstraction layer that can later be translated into formal process models [50].

While the suitability of declarative modeling for clinical guidelines is thus well-established, the question of *how* guidelines are translated into formal constraints remains an open methodological choice. Recent work has explored automated translation of natural language into formal temporal-logic specifications. NL2LTL [34] demonstrates the feasibility of mapping textual patterns to Declare-style LTL templates, but is limited to simple, patternized commands and does not support the complex data conditions or metric temporal constraints required by clinical guidelines. Fionda et al. [31] show more rigorously that current LLMs tend to over-constrain declarative models and lose semantic equivalence as specification complexity grows. ConformalNL2LTL [91] adds probabilistic correctness guarantees to NL-to-LTL translation, but remains evaluated in constrained robotic domains far removed from the multi-perspective structure of ICU weaning protocols. Given these limitations (insufficient handling of data and temporal richness, semantic fragility at scale, and unacceptable translation uncertainty in a safety-critical setting) this thesis encodes weaning guidelines manually into MP-Declare through expert-driven formalization rather than adopting automated NL-to-LTL pipelines.

2.3.2 Predictive Modeling of Weaning Outcomes

A largely separate strand of work sidesteps guideline formalization altogether and instead trains supervised machine learning models to predict weaning and extubation outcomes directly from routinely collected ICU data. Fritsch et al. [33] develop a machine learning model to predict the duration of unassisted spontaneous breathing during prolonged weaning from ventilator and clinical parameters, reporting only mediocre accuracy and concluding that setting realistic spontaneous-breathing targets remains difficult even with dedicated predictive modeling. Xu et al. [93] similarly construct a risk-prediction model for "difficult weaning" from 36 clinical variables collected across 487 ICU patients, representative of a broader family of single-center classifiers trained to flag patients at risk of extubation failure. Bakos et al. [11] synthesize this literature in a systematic review and meta-analysis, concluding that despite promising discriminatory performance, limited external validation, heterogeneity, and inconsistent reporting preclude routine clinical adoption. These approaches share a common framing: they optimize for outcome discrimination using an arbitrary combination of predictors, typically through opaque ensemble or deep learning models, without reference to any normative guideline. Consequently, they can flag that an outcome is unlikely, but not whether the documented

care pathway leading up to it adhered to, or departed from, the recommended procedure. This thesis instead targets exactly that latter question, at the resolution of individual guideline obligations, complementing rather than competing with outcome-prediction approaches.

2.3.3 Adherence to Clinical Guidelines

The application of process mining conformance checking to measure adherence to clinical guidelines has matured considerably over the past decade, yet persistent limitations remain across both methodology and clinical utility.

Rovani et al. [80] apply Declare-based conformance checking to urology data and introduce a model repair procedure in which a training log is used to adjust the normative model when deviations are clinically justified, demonstrating that declarative models accommodate the natural variability of care without penalising legitimate deviations from a rigid pathway. However, their framework operates at the level of full patient traces and does not incorporate data conditions or temporal constraints, leaving multi-perspective violations undetected. Grüger et al. [40] investigate the HL7 Arden Syntax for declarative guideline conformance checking and show that standard alignments typically yield little medical value, proposing instead manually constructed alignments that are clinically interpretable, yet their approach requires substantial manual effort and does not scale to large electronic health record (EHR) datasets. Lichtner et al. [62] address the scalability dimension by designing a system for automated near-real-time monitoring of adherence to evidence-based guideline recommendations within a hospital information system, though their rule-based approach lacks the expressiveness to capture ordering and temporal dependencies between clinical events. Van der Lubbe et al. [64] propose a concordance score model derived from EHR data to quantify guideline adherence as a continuous metric, offering a complementary perspective to deviation diagnostics, but their scoring scheme abstracts away the process structure entirely and cannot identify which specific guideline steps were violated or in what sequence.

Taken together, these studies reveal a consistent pattern. Each addresses one dimension of the adherence assessment problem while leaving others unresolved. Oliart et al. [72] confirm this through a scoping review of 20 years of work, showing that most approaches rely on rule-based alignment or simple threshold checks, that process mining remains underused despite its ability to capture complex temporal patterns, and that patient context or clinical rationale behind deviations is rarely modeled. Hoang et al. [44] extend this analysis with a more recent scoping review focused on EHR-based adherence assessment, finding that even process mining-based approaches largely lack contextual sensitivity and seldom incorporate patient-specific factors or clinician justifications, leaving the clinical meaning of deviations unresolved across the field.

Munoz-Gama et al. [71] offer a systematic characterization of why these limitations persist, synthesizing a four-year effort involving 55 researchers into a systematic characterization of the open challenges of process mining in healthcare. They identify high process variability, patient-centred focus, data quality issues, and the need for interpretability as defining features that require targeted methodological adaptations, precisely the

challenges that the case studies above individually encounter. This collective position reinforces the view that many guideline-governed processes are inherently flexible, multi-perspective, and difficult to assess with traditional, trace-level conformance techniques. In particular, prior work on declarative and multi-perspective conformance checking has shown that evaluating entire case traces and reporting aggregate fitness scores obscures which specific obligations were breached and in which clinical context they occurred. Building on these insights, the framework proposed in this thesis adopts an episode-scoped, multi-perspective declarative conformance checking approach that integrates data conditions and temporal windows with trigger-based sub-trace segmentation, aiming to produce more localized and clinically interpretable diagnostics of guideline adherence.

2.4 Localized Process Mining and Decomposed Conformance Checking

A fundamental limitation of global process mining approaches (whether imperative or declarative) is their requirement to model the entire process holistically. In the ICU weaning context, a clinician's primary interest may lie not in the full ICU episode but in specific care sub-processes triggered by particular clinical states (e.g., the sequence of events following a failed spontaneous breathing trial (SBT)) [10]. This motivates more targeted, locally scoped process mining approaches.

Bose and van der Aalst [15] introduce a burst-based notion of a case in their analysis of a Dutch academic hospital log, where long patient traces are segmented into temporally dense "bursts" separated by idle periods. Each burst is treated as a homogeneous sub-case suitable for discovery and alignment. This explicitly acknowledges that clinically meaningful behavior often unfolds in short, intense episodes embedded within much longer care trajectories.

Building on the idea that only fragments of a log may exhibit regular structure, Tax et al. [87] propose local process models (LPMs), small Petri nets that capture behavioral patterns of interest over a subset of activities without aiming to cover the full case from start to end. Leemans and van der Aalst [55] pursue a complementary direction with frequent episode discovery, mining partially ordered episodes that represent recurring local patterns in event logs and showing how such episodes can be used for conformance checking against partial orders rather than full models. Together, these works converge on the view that local patterns (bursts, LPMs, episodes) are often more interpretable and diagnostically useful than an overgeneral global model, especially in high-variability domains such as healthcare. Yzquierdo-Herrera et al. [94] explicitly frame this as sub-process discovery for process diagnostics, proposing a method that segments aligned traces into "building blocks" representing sub-processes and organizing these into a hierarchy that reflects dependencies and correlations between sub-parts of the process. Their results show that anomalies can be framed in the context of specific sub-processes, addressing a limitation of purely global diagnostics. Conceptually, this is closely aligned with the ICU weaning setting, where clinically relevant deviations (e.g., premature extubation) occur within a specific sub-process rather than across the entire ICU stay.

Across these contributions there is broad agreement that local behavioral units (bursts, sub-processes, episodes) are essential to make sense of complex, heterogeneous logs. However, all of them are grounded in imperative representations (Petri nets or partial orders). None of these localized discovery approaches are formulated in terms of declarative constraint templates, nor do they exploit the inherently local activation semantics of Declare constraints.

On the conformance side, a related but distinct line of work decomposes models and logs to perform conformance checking at the level of sub-processes. Munoz-Gama et al. [70] present Single-Entry Single-Exit decomposed conformance checking, where a global model is partitioned into SESE fragments and conformance is computed on these fragments before being composed into a global fitness value. The primary motivation is scalability and improved diagnostics. Violations can be localized to specific fragments, which effectively act as structural sub-processes. The same authors extend this direction [69] with hierarchical and "in the large" conformance checking, where models are structured into hierarchies or partitions and conformance is assessed within and across these parts. Lee et al. [54] later recompose fragment-wise alignments into a globally correct or bounded-approximate alignment, tackling earlier concerns that decomposition might distort fitness measures. All these works share the notion that conformance checking can and should be applied at the level of sub-models to gain scalability and more focused diagnostics. The direction of decomposition sets these approaches apart from the episode-based evaluation developed in this thesis. Decomposed and hierarchical conformance checking partitions the *model* top-down into structural fragments, whereas episode-based evaluation leaves the constraint model intact and segments only the *trace*.

What is notably missing in this body of work is a declarative, rule-based notion of locality. In declarative process mining, locality is inherent in the semantics of constraints (each constraint is only evaluated when its activation condition holds), yet existing studies treat conformance globally at the level of a full Declare specification [16, 59].

3 Preliminaries

This chapter introduces the concepts and notation on which the remainder of the thesis builds. Section 3.1 fixes the basic mathematical notation, and Section 3.2 defines event data and event logs. Section 3.3 presents the declarative modeling language Declare, its semantics in linear temporal logic on finite traces, and its multi-perspective extension MP-Declare. Section 3.4 describes how conformance to such constraints is checked, and Section 3.5 introduces the statistical tests used to relate conformance verdicts to process outcomes.

3.1 Mathematical Foundations

Definition 3.1 (Set)**.** A *set* X is a collection of unique objects. We denote $X = \{x_1, x_2, x_3, \dots x_n\}$ as a finite set with $|X| = n$ elements. The empty set is represented by $\emptyset$.

Definition 3.2 (Subset)**.** A set Y is a *subset* of a set X if every element of Y is also an element of X. We write $Y \subseteq X$.

Definition 3.3 (Union)**.** For two sets X and Y, we denote the union as $X \cup Y = \{z \mid z \in X \vee z \in Y\}$.

Definition 3.4 (Cartesian product)**.** For two sets X and Y, the *cartesian product* is $X \times Y = \{(x, y) \mid x \in X \wedge y \in Y\}$, i.e., the set of all ordered pairs whose first component is an element of X and whose second component is an element of Y.

Definition 3.5 (Function)**.** A *function* f between a set X and a set Y, denoted as $f : X \to Y$, maps each element of X to exactly one element of Y.

Definition 3.6 (Partial function)**.** A *partial function* $f : X \nrightarrow Y$ maps each element of a subset of X, called the domain of f, to exactly one element of Y. For elements outside its domain, f is undefined.

Definition 3.7 (Sequence)**.** A *sequence* of length n over a set X is an ordered collection of elements, defined by the function $\sigma : \{1, 2, \dots n\} \to X$, where $\sigma(i)$ denotes the element at position i. The length is denoted as $|\sigma| = n$. We use the following notation for sequences: $\sigma = \langle x_1, \dots, x_n \rangle$. The empty sequence is represented by $\langle\rangle$. For a set X, we denote X^* as the set of all possible sequences over X.

3.2 Event Data

Process mining [2] involves discovering process models, checking conformance against actual event data, and enhancing existing models by analyzing and optimizing event logs captured by information systems. It bridges the gap between data science and process science by providing tools to discover, monitor, and improve business processes.

In a process, the execution of an activity is recorded as an event, which includes the time when it occurred within the context of a specific process instance, known as a case. Beyond the activity and the timestamp, an event may carry additional attributes, such as the resource that performed the activity or a measured value. The collection of such events constitutes an event log, serving as the primary input for process mining techniques. The following definitions formalize universes, events, event logs, and traces.

Definition 3.8 (Universes)**.** We assume the existence of a universe of events $\mathcal{E}$, together with the universe of activities $\mathcal{U}_{act}$, the universe of timestamps $\mathcal{U}_{time}$, the universe of case identifiers $\mathcal{U}_{case}$, the universe of attribute names $\mathcal{U}_{attr}$, the universe of attribute values $\mathcal{U}_{val}$, and the universe of durations $\mathcal{U}_{dur}$.

Definition 3.9 (Time domain)**.** Timestamps are totally ordered, and durations are the non-negative distances between them: $t' - t \in \mathcal{U}_{dur}$ for all $t, t' \in \mathcal{U}_{time}$ with $t \leq t'$, and $t + d,\ t - d \in \mathcal{U}_{time}$ for all $t \in \mathcal{U}_{time}$ and finite $d \in \mathcal{U}_{dur}$. We write 0 for the least and $+\infty$ for the greatest duration, the latter used only as an interval bound. Likewise, we use $+\infty$ as an upper bound on timestamps, with $t < +\infty$ for all $t \in \mathcal{U}_{time}$.

Definition 3.10 (Event)**.** For each attribute name $x \in \mathcal{U}_{attr}$, let $\pi_x : \mathcal{E} \rightharpoonup \mathcal{U}_{val}$ be a partial function assigning to an event the value it records for x, undefined if the event records no such value. Three projections are total on $\mathcal{E}$ and are called *mandatory*:

- $\pi_{act} : \mathcal{E} \to \mathcal{U}_{act}$, the activity executed by the event;
- $\pi_{time} : \mathcal{E} \to \mathcal{U}_{time}$, the timestamp at which the event occurred;

- $\pi_{case} : \mathcal{E} \to \mathcal{U}_{case}$, the case in whose context the event occurred.

The *payload* of an event $e \in \mathcal{E}$ is the partial function $\pi_{pay}(e) : \mathcal{U}_{attr} \rightharpoonup \mathcal{U}_{val}$ with $\pi_{pay}(e)(x) = \pi_x(e)$, whose domain consists of the non-mandatory attribute names on which π_x is defined at e.

Definition 3.11 (Event log, trace)**.** An *event log* is a finite set of events $L \subseteq \mathcal{E}$. The set of cases occurring in L, and the set of events belonging to a case c, are

$$\mathit{cases}(L) = \{\, \pi_{case}(e) \mid e \in L \,\}, \qquad L_c = \{\, e \in L \mid \pi_{case}(e) = c \,\}.$$

For a case $c \in \mathit{cases}(L)$, the *trace* of c is the sequence $\sigma_c = \langle e_1, \ldots, e_n \rangle \in \mathcal{E}^*$ enumerating L_c such that $\pi_{time}(e_i) \leq \pi_{time}(e_j)$ for all $1 \leq i \leq j \leq n$. Events bearing identical timestamps are ordered by a fixed total order on $\mathcal{E}$, in this thesis lexicographically by activity label and then by source item identifier, so that σ_c is uniquely determined. Throughout this thesis, an event is *before* (*after*) another iff its timestamp is strictly smaller (greater); events sharing a timestamp are neither before nor after each other. The tie-breaking order only determines positions, i.e., which event counts as adjacent to another and which is first or last among simultaneous events.

Unless stated otherwise, the term *trace* refers to such a sequence of events throughout the remainder of this thesis. Table 3.1 shows an event log in tabular form, with one column per projection.

Table 3.1: Example event log with four events over two cases. Each column is one attribute projection; the *Department* column is a non-mandatory attribute and thus part of the payload.

Event	π_{case}	π_{act}	π_{time}	π_{dept}
e_1	ord-904	Order Placed	15.08.2024	Sales
e_2	ord-904	Item Packed	16.08.2024	Warehouse
e_3	ord-905	Order Placed	16.08.2024	Sales
e_4	ord-904	Shipped	20.08.2024	Logistics

The log of Table 3.1 is the set $L = \{e_1, e_2, e_3, e_4\}$ with $\mathit{cases}(L) = \{\text{ord-904}, \text{ord-905}\}$. Selecting and ordering by timestamp yields the traces $\sigma_{\text{ord-904}} = \langle e_1, e_2, e_4 \rangle$ and $\sigma_{\text{ord-905}} = \langle e_3 \rangle$. The payload of e_1 is defined on the single attribute *dept*, with $\pi_{pay}(e_1)(\mathit{dept}) = \text{Sales}$.

3.3 Declare

Declare [75] is a declarative process modeling language. Instead of describing all permitted execution paths explicitly, as procedural notations such as Petri nets or Business Process Model and Notation (BPMN) do, a Declare model consists of a set of *constraints* that every process execution must respect. Any behavior that does not violate a constraint is allowed. This *open-world* semantics makes Declare well suited for flexible, high-variability processes. Constraints are instantiated from a repertoire of *templates*, parameterized patterns with a formal semantics in LTL_f. We follow the presentation of Di Ciccio and Montali [22].

Throughout this section, we abstract from timestamps and data and view a trace as the sequence of its activity labels. For a trace $\sigma = \langle e_1, \ldots, e_n \rangle \in \mathcal{E}^*$ in the sense of Definition 3.11, its *control-flow projection* is $\sigma^{act} = \langle \pi_{act}(e_1), \ldots, \pi_{act}(e_n) \rangle \in \mathcal{U}_{act}^*$. We fix a finite alphabet of activities $\Sigma \subseteq \mathcal{U}_{act}$ and, slightly overloading notation, write $\sigma \in \Sigma^*$ with $\sigma(i)$ denoting the activity occurring at instant i.

3.3.1 Linear Temporal Logic on Finite Traces

LTL_f [22, 36] has the same syntax as classical LTL but is interpreted over finite traces, matching the finite nature of recorded process executions. Following [22], we include past-tense modalities.

Definition 3.12 (Syntax of LTL_f)**.** Well-formed LTL_f formulae over an alphabet Σ are built from the activities in Σ, the unary temporal operators $\bigcirc$ (next) and $\ominus$ (yesterday), and the binary temporal operators $\mathcal{U}$ (until) and $\mathcal{S}$ (since), according to the following grammar, where $a \in \Sigma$:

$$\varphi ::= a \mid (\neg\varphi) \mid (\varphi_1 \wedge \varphi_2) \mid (\bigcirc\varphi) \mid (\varphi_1 \,\mathcal{U}\, \varphi_2) \mid (\ominus\varphi) \mid (\varphi_1 \,\mathcal{S}\, \varphi_2)$$

Definition 3.13 (Semantics of LTL_f)**.** Let $\sigma \in \Sigma^*$ be a trace of length n and let $1 \leq i \leq n$ be an instant. An LTL_f formula φ is satisfied at instant i of σ, written $\sigma, i \models \varphi$, inductively defined as follows:

- $\sigma, i \models a$ iff $\sigma(i) = a$;
- $\sigma, i \models \neg\varphi$ iff $\sigma, i \not\models \varphi$;
- $\sigma, i \models \varphi_1 \wedge \varphi_2$ iff $\sigma, i \models \varphi_1$ and $\sigma, i \models \varphi_2$;
- $\sigma, i \models \bigcirc\varphi$ iff $i < n$ and $\sigma, i+1 \models \varphi$;
- $\sigma, i \models \ominus\varphi$ iff $i > 1$ and $\sigma, i-1 \models \varphi$;
- $\sigma, i \models \varphi_1 \,\mathcal{U}\, \varphi_2$ iff $\sigma, j \models \varphi_2$ for some j with $i \leq j \leq n$, and $\sigma, k \models \varphi_1$ for all k with $i \leq k < j$;

- $\sigma, i \models \varphi_1 \mathcal{S} \varphi_2$ iff $\sigma, j \models \varphi_2$ for some j with $1 \leq j \leq i$, and $\sigma, k \models \varphi_1$ for all k with $j < k \leq i$.

A trace σ satisfies φ, written $\sigma \models \varphi$, iff $\sigma, 1 \models \varphi$.

Note that $\bigcirc$ is a *strong* next. Specifically, $\bigcirc\varphi$ requires that a next instant exists. Consequently, in LTL_f, $\neg \bigcirc \varphi$ is not equivalent to $\bigcirc\neg\varphi$, as the former also holds at the last instant of a trace. From the basic operators, the usual Boolean abbreviations (*true*, *false*, $\vee$, $\rightarrow$) are derived, together with the following temporal abbreviations:

- $\mathit{end} \equiv \neg \bigcirc \mathit{true}$ and $\mathit{start} \equiv \neg \ominus \mathit{true}$, holding at the last and first instant of a trace, respectively;
- $\Diamond\varphi \equiv \mathit{true}\, \mathcal{U}\, \varphi$ (*eventually*): φ holds at the current or a future instant;
- $\Box\varphi \equiv \neg\Diamond\neg\varphi$ (*globally*): φ holds from the current instant until the end;
- $\blacklozenge\varphi \equiv \mathit{true}\, \mathcal{S}\, \varphi$ (*once*): φ holds at the current or a past instant;
- $\boxminus\varphi \equiv \neg\blacklozenge\neg\varphi$ (*historically*): φ holds from the start up to the current instant.

Example 3.14. Let $\sigma = \langle a, b, b, c, d \rangle$ and consider $\varphi_1 = \Diamond d$, $\varphi_2 = \Box(b \rightarrow \Diamond d)$, and $\varphi_3 = \bigcirc b$. Then $\sigma \models \varphi_1$, since d occurs at instant 5. Also, $\sigma \models \varphi_2$, since every occurrence of b is eventually followed by d. Finally, $\sigma \models \varphi_3$, since b occurs at instant 2.

3.3.2 Declare Templates and Specifications

A Declare *template* is a predicate $k(x_1, \ldots, x_m)$ over activity variables $x_1, \ldots, x_m$ whose semantics is given as an LTL_f formula $\varphi_{k(x_1,\ldots,x_m)}$ over those variables. The number m of variables is called the *arity* of the template. Templates of arity one are called *existence* templates, templates of arity two *relation* templates. We further call a relation template *unordered* if its formula only requires that the related activity occurs at some instant of the trace, before or after the current one ($\blacklozenge y \vee \Diamond y$), as in RespondedExistence, CoExistence, and their negations. All other relation templates are *ordered*, since their formula restricts the related activity either to the future ($\Diamond$, $\bigcirc$) or to the past ($\blacklozenge$, $\ominus$), as in Response and Precedence. Among the ordered templates, the *alternate* and *chain* variants additionally constrain the events between the two occurrences. Throughout this thesis, the two activities of a relation constraint are assumed to be distinct. A *constraint* is obtained by instantiating the variables of a template with concrete activities. Table 3.2 lists the templates used in this thesis together with their LTL_f semantics.

Definition 3.15 (Declare specification, constraint)**.** A *Declare specification* is a triple $DS = (\mathit{Rep}, \Sigma, K)$, where Rep is a finite, non-empty set of templates, Σ is

Table 3.2: Semantics of the Declare templates implemented in this thesis [22], together with their activation and target formulae (cf. Section 3.3.3).

Template	LTL_f semantics	Activation	Target	Explanation
Existence templates				
AtLeastOne(x)	$\Box(\mathit{start} \rightarrow \Diamond x)$	*start*	$\Diamond x$	x occurs at least once.
Absence(x)	$\Box(x \rightarrow \bot)$	x	$\bot$	x never occurs.
AtMostOne(x)	$\Box(x \rightarrow \neg \bigcirc \Diamond x)$	x	$\neg \bigcirc \Diamond x$	x occurs at most once.
Init(x)	$\Box(\mathit{start} \rightarrow x)$	*start*	x	The trace starts with x.
End(x)	$\Box(\mathit{end} \rightarrow x)$	*end*	x	The trace ends with x.
Relation templates				
RespondedExistence(x, y)	$\Box(x \rightarrow \blacklozenge y \vee \Diamond y)$	x	$\blacklozenge y \vee \Diamond y$	If x occurs, y occurs somewhere in the trace.
Response(x, y)	$\Box(x \rightarrow \Diamond y)$	x	$\Diamond y$	Every x is eventually followed by y.
AlternateResponse(x, y)	$\Box(x \rightarrow \bigcirc(\neg x \,\mathcal{U}\, y))$	x	$\bigcirc(\neg x \,\mathcal{U}\, y)$	Every x is followed by y, with no further x in between.
ChainResponse(x, y)	$\Box(x \rightarrow \bigcirc y)$	x	$\bigcirc y$	Every x is immediately followed by y.
Precedence(x, y)	$\Box(y \rightarrow \blacklozenge x)$	y	$\blacklozenge x$	y occurs only if x occurred before.
AlternatePrecedence(x, y)	$\Box(y \rightarrow \ominus(\neg y \,\mathcal{S}\, x))$	y	$\ominus(\neg y \,\mathcal{S}\, x)$	Every y is preceded by x, with no further y in between.
ChainPrecedence(x, y)	$\Box(y \rightarrow \ominus x)$	y	$\ominus x$	y occurs only if x immediately precedes it.
Coupling templates				
CoExistence(x, y)	$\Box(x \rightarrow \blacklozenge y \vee \Diamond y)$ $\wedge\ \Box(y \rightarrow \blacklozenge x \vee \Diamond x)$	x y	$\blacklozenge y \vee \Diamond y$ $\blacklozenge x \vee \Diamond x$	x and y either both occur or neither does.
Succession(x, y)	$\Box(x \rightarrow \Diamond y)$ $\wedge\ \Box(y \rightarrow \blacklozenge x)$	x y	$\Diamond y$ $\blacklozenge x$	Every x is eventually followed by y, and every y is preceded by x.
AlternateSuccession(x, y)	$\Box(x \rightarrow \bigcirc(\neg x \,\mathcal{U}\, y))$ $\wedge\ \Box(y \rightarrow \ominus(\neg y \,\mathcal{S}\, x))$	x y	$\bigcirc(\neg x \,\mathcal{U}\, y)$ $\ominus(\neg y \,\mathcal{S}\, x)$	x and y occur in strict alternation.
ChainSuccession(x, y)	$\Box(x \rightarrow \bigcirc y)$ $\wedge\ \Box(y \rightarrow \ominus x)$	x y	$\bigcirc y$ $\ominus x$	x and y occur only as adjacent pairs $\langle x, y \rangle$.
Negative relation templates				
NotRespondedExistence(x, y)	$\Box(x \rightarrow \Box\neg y \wedge \boxminus\neg y)$	x	$\Box\neg y \wedge \boxminus\neg y$	If x occurs, y never occurs in the trace.
NotResponse(x, y)	$\Box(x \rightarrow \Box\neg y)$	x	$\Box\neg y$	After any x, y never occurs.
NotChainResponse(x, y)	$\Box(x \rightarrow \neg \bigcirc y)$	x	$\neg \bigcirc y$	No x is immediately followed by y.
NotPrecedence(x, y)	$\Box(y \rightarrow \boxminus\neg x)$	y	$\boxminus\neg x$	No y is preceded by x.
NotChainPrecedence(x, y)	$\Box(y \rightarrow \neg \ominus x)$	y	$\neg \ominus x$	No y is immediately preceded by x.
NotCoExistence(x, y)	$\Box(x \rightarrow \neg(\blacklozenge y \vee \Diamond y))$	x	$\neg(\blacklozenge y \vee \Diamond y)$	x and y never both occur in the trace.
NotSuccession(x, y)	$\Box(x \rightarrow \Box\neg y)$ $\wedge\ \Box(y \rightarrow \boxminus\neg x)$	x y	$\Box\neg y$ $\boxminus\neg x$	No x is followed by y anywhere later in the trace.
NotChainSuccession(x, y)	$\Box(x \rightarrow \neg \bigcirc y)$ $\wedge\ \Box(y \rightarrow \neg \ominus x)$	x y	$\neg \bigcirc y$ $\neg \ominus x$	x and y never occur as an adjacent pair $\langle x, y \rangle$.

a finite, non-empty set of activities, and K is a finite set of *constraints*, i.e., pairs $(k(x_1, \ldots, x_m), \kappa)$ of a template $k(x_1, \ldots, x_m) \in \mathit{Rep}$ and a mapping κ assigning each variable x_i to an activity $\kappa(x_i) = a_i \in \Sigma$. We compactly denote such a constraint as $k(a_1, \ldots, a_m)$.

Definition 3.16 (Constraint formula, specification formula)**.** The *formula* of a constraint $k(a_1, \ldots, a_m)$, written $\varphi_{k(a_1,\ldots,a_m)}$, is the LTL_f formula obtained from $\varphi_{k(x_1,\ldots,x_m)}$ by replacing each x_i with a_i. A trace $\sigma \in \Sigma^*$ *satisfies* the constraint if $\sigma \models \varphi_{k(a_1,\ldots,a_m)}$. Otherwise, it *violates* the constraint. The formula of a specification $DS = (\mathit{Rep}, \Sigma, K)$ is $\varphi_{DS} = \bigwedge_{\rho \in K} \varphi_\rho$. A trace σ is *accepted* by DS if $\sigma \models \varphi_{DS}$, and *rejected* otherwise.

Example 3.17. Consider the constraint $\textsc{Response}(a, b)$ with $\varphi_{\textsc{Response}(a,b)} = \Box(a \to \Diamond b)$. Every occurrence of a must eventually be followed by an occurrence of b. The traces $\langle b \rangle$ and $\langle a, c, a, b \rangle$ satisfy the constraint, whereas $\langle a \rangle$ and $\langle a, b, a, c \rangle$ violate it, since their last occurrence of a is not followed by any b.

3.3.3 Activation and Target

The LTL_f semantics of a constraint determines *whether* a trace satisfies it, but it does not distinguish *how*. In Example 3.17, both $\langle b \rangle$ and $\langle a, c, a, b \rangle$ satisfy $\textsc{Response}(a, b)$, yet the first trace never interacts with the constraint at all. It contains no occurrence of a that would demand a subsequent b. This phenomenon is known as *vacuous satisfaction* [22]. To capture the pragmatic interaction between traces and constraints, each constraint is complemented with an *activation*, a triggering condition, and a *target*, the condition whose satisfaction the activation demands. If a constraint is never activated in a trace, the satisfaction of its target is not enforced. Following [22], activation and target are given as two additional LTL_f formulae that must be coherent with the constraint formula in the following sense.

Definition 3.18 (Activation and target)**.** Let k be a constraint over activities Σ with formula φ_k. An *activation* α_k and a *target* τ_k of k are LTL_f formulae such that, for every trace $\sigma \in \Sigma^*$:

$$\sigma \models \varphi_k \quad \text{iff} \quad \sigma \models \Box(\alpha_k \to \tau_k)$$

Table 3.2 lists activations and targets for the considered templates. For instance, for $\textsc{Response}(a, b)$ we have $\alpha = a$ and $\tau = \Diamond b$. With this assignment, every occurrence of a activates the constraint and requires the target $\Diamond b$ to hold at that instant. For $\textsc{Precedence}(a, b)$, it is the occurrence of b that activates the constraint, requiring that a has occurred before ($\tau = \blacklozenge a$). The distinction between activation and target is central to

conformance checking (Section 3.4), where the instants at which a constraint is activated are classified by whether the target holds there.

3.3.4 MP-Declare

Standard Declare constrains only the control-flow perspective, i.e., the ordering of activities. Real-life processes, and clinical processes in particular, additionally involve *data* (attribute values attached to events) and *time* (metric distances between events, beyond mere ordering). MP-Declare [16] extends Declare templates with conditions over these perspectives, formally grounded in MFOTL. We adopt the notation of Burattin et al. [16]. Recall from Definition 3.10 that every event e carries a payload $\pi_{pay}(e)$ assigning values to a finite set of attributes. The conditions of MP-Declare are predicates over these payloads and over the events' timestamps.

An MP-Declare constraint is built on a Declare template, whose activation and target now take the payloads of events into account, and refines it with three conditions: an *activation condition*, a relation that must be true when the activation occurs on the payload of the activation event only; a *correlation condition*, a relation that must hold when the target occurs and can relate the payloads of the two events; and a *time condition*, restricting the metric temporal distance between the two.

Definition 3.19 (MP-Declare constraint)**.** An *MP-Declare constraint* is a tuple $c = (k(a_1, \ldots, a_m), \varphi_a, \varphi_c, \varphi_\tau)$, where:

- $k(a_1, \ldots, a_m)$ is a Declare constraint with activation α and target τ;
- φ_a is the *activation condition*, a predicate over the payload of an event matching the activation α;
- φ_c is the *correlation condition*, a predicate evaluated upon occurrence of an event matching the target τ, over the payload of that event and the payload of the activating event;
- $\varphi_\tau = [t_0, t_1]$ with $t_0, t_1 \in \mathcal{U}_{dur}$, $t_0 < t_1$, is the *time condition*, an interval bounding the admissible temporal distance $\Delta(e_\alpha, e_\tau)$ between the activating event e_α and the target event e_τ. If no time condition is specified, $\varphi_\tau = [0, +\infty)$.

Since durations are non-negative (Definition 3.9), the temporal distance $\Delta(e_\alpha, e_\tau)$ is measured in the direction in which the template looks for its target:

$$\Delta(e_\alpha, e_\tau) = \begin{cases} \pi_{time}(e_\tau) - \pi_{time}(e_\alpha) & \text{if } \tau \text{ refers to the future and } \pi_{time}(e_\tau) > \pi_{time}(e_\alpha), \\ \pi_{time}(e_\alpha) - \pi_{time}(e_\tau) & \text{if } \tau \text{ refers to the past and } \pi_{time}(e_\tau) < \pi_{time}(e_\alpha), \\ \text{undefined} & \text{otherwise.} \end{cases}$$

Here, τ refers to the future if it uses $\Diamond$ or $\bigcirc$ (e.g., Response), to the past if it uses $\blacklozenge$

or $\ominus$ (e.g., Precedence), and to both if it uses both (e.g., RespondedExistence), in which case whichever line applies is used. A target event sharing the activation's timestamp is neither before nor after it (Definition 3.11), so Δ is undefined for such a pair. A target event satisfies φ_τ iff $\Delta(e_\alpha, e_\tau)$ is defined and $\Delta(e_\alpha, e_\tau) \in \varphi_\tau$. For instance, Precedence(a, b) with $\varphi_\tau = [0, 24\,\mathrm{h}]$ requires an occurrence of a within the 24 hours preceding each occurrence of b.

The semantics of an MP-Declare constraint refines that of its underlying template. Specifically, the constraint is activated only by events that match α *and* whose payload satisfies φ_a. Each such activation requires the existence of a target event matching τ whose payload, jointly with the activating payload, satisfies φ_c, and whose temporal distance to the activation lies within the interval φ_τ [16]. For example, the data-aware response

$$\textsc{Response}(a, b) \text{ with } \varphi_a : \pi_x(e_\alpha) > 5,\ \varphi_c : \pi_y(e_\tau) < 10,\ \varphi_\tau = [0, 12\,\mathrm{h}]$$

where e_α denotes the activating event, an occurrence of a, and e_τ a candidate target event, an occurrence of b. The constraint requires that whenever a occurs with attribute x exceeding 5, an occurrence of b with attribute y below 10 must follow within twelve hours. Notably, occurrences of a with $x \leq 5$ do not activate the constraint. An *MP-Declare specification* is a Declare specification whose constraints are MP-Declare constraints.

3.4 Conformance Checking

Conformance checking [2, 81] compares observed process behavior, recorded in an event log, against a normative process model, in order to detect and diagnose deviations. In the procedural setting, conformance is typically established by replaying traces on the model or by computing alignments that map observed onto modeled behavior [57, 81]. In the declarative setting considered in this thesis, the normative model is an (MP-)Declare specification. Conformance is assessed per constraint. A trace conforms to a specification iff it satisfies every constraint in it.

A binary satisfied/violated verdict per trace is, however, of limited diagnostic value. Following Burattin et al. [16], we exploit the activation semantics of Section 3.3.3 to classify, for each constraint, every activation within a trace individually.

Definition 3.20 (Activation instance, fulfillment, violation)**.** Let σ be a trace and c an MP-Declare constraint with activation α, target τ over the target activity b, and conditions φ_a, φ_c, φ_τ (Definition 3.19).

- An *activation instance* of c is an instant i of σ such that $\sigma, i \models \alpha$ and the payload of $\sigma(i)$ satisfies φ_a.

- For an activation instance i, a *qualifying target* is an instant j such that

$$\sigma(j) = b, \qquad \varphi_c(\pi_{pay}(\sigma(i)), \pi_{pay}(\sigma(j))), \qquad \Delta(\sigma(i), \sigma(j)) \in \varphi_\tau.$$

  We write τ^i for the formula τ in which b is read as "b is a qualifying target for i".

- An activation instance i is a *fulfillment* if $\sigma, i \models \tau^i$, and a *violation* otherwise.

- The trace σ *satisfies* c iff it contains no violation of c, and satisfies c *vacuously* iff it contains no activation instance of c.

Reading b as a qualifying target binds the correlation and time conditions to the same event. For a positive template such as Response, with $\tau = \Diamond b$, a fulfillment requires one later event that carries b *and* satisfies both conditions. For a negative template such as NotResponse, with $\tau = \Box\neg b$, it requires that no later event carrying b satisfies both. For templates without a target activity, such as Absence, $\tau^i = \tau$. A plain Declare constraint is the special case $\varphi_a = \varphi_c = \mathit{true}$ and $\varphi_\tau = [0, +\infty)$, for which τ^i differs from τ only in disregarding target events that share the activation's timestamp.

This per-activation classification yields fine-grained diagnostics. Rather than reporting that a trace violates a specification, conformance checking reports, for each constraint, *which* activations were fulfilled and which were violated, and distinguishes traces that genuinely comply from those that merely never trigger a constraint.

Example 3.21. Consider $c = \text{Response}(a, b)$ and the trace $\sigma = \langle a, c, b, a, c\rangle$. The constraint is activated at instants 1 and 4. Instant 1 is a fulfillment, since b occurs at instant 3. Instant 4 is a violation, since no b follows. Hence σ violates c, with one fulfillment and one violation. The trace $\langle c, b, c\rangle$ satisfies c vacuously, as it contains no occurrence of a.

3.5 Statistical Tests

The statistical outcome analysis in Section 6.4.4 relies on hypothesis testing to assess whether constraint violations are associated with adverse episode-level outcomes. This section introduces the statistical concepts used throughout the analysis, following the standard treatment of categorical data analysis [9].

Hypothesis testing and p-values. Informally, a statistical hypothesis test asks whether an observed pattern in the data, for instance, that patients whose treatment deviates from the guideline die more often, could plausibly have arisen by chance alone. The *null hypothesis* H_0 states that no true association exists (e.g., violation status and outcome are independent), while the *alternative hypothesis* H_1 states that an association exists.

Definition 3.22 (p-value)**.** Let T be a test statistic with observed value $\hat{t}$. The *p-value* is the probability, computed under the assumption that H_0 holds, of observing a test statistic at least as extreme as the one actually observed:

$$p = \Pr\left(T \geq \hat{t} \mid H_0\right).$$

The p-value ranges over $[0, 1]$. A small p-value (conventionally $p < 0.05$, the *significance level* α) indicates that the observed data would be unlikely under the null hypothesis, and H_0 is rejected in favor of H_1. A large p-value does not prove the null hypothesis. Instead, it merely indicates insufficient evidence against it. Importantly, the p-value quantifies statistical evidence, not the size or clinical relevance of an effect.

Fisher's exact test [32]. When both the exposure (constraint violated or not) and the outcome (e.g., in-hospital mortality) are binary, their independence can be tested exactly with Fisher's test. Informally, it asks: given the row and column totals, how likely is a split at least as unbalanced as the observed one?

Definition 3.23 (Fisher's exact test)**.** Let n episodes be classified by two binary variables, violation status (violated / not violated) and outcome (present / absent), giving the 2×2 contingency table

	Outcome present	Outcome absent	
Violated	a	b	$a+b$
Not violated	c	d	$c+d$
	$a+c$	$b+d$	n

where $a + b$ and $c + d$ are the row totals and $a + c$ and $b + d$ the column totals (the *margins*). The null hypothesis H_0 states that violation status and outcome are independent. Conditional on the margins, the table is fully determined by the cell count a, and under H_0 the random variable X giving this count follows the hypergeometric distribution

$$\Pr(X = k \mid H_0) = \frac{\binom{a+b}{k}\binom{c+d}{a+c-k}}{\binom{n}{a+c}}.$$

The two-sided p-value is the total probability of all tables with the same margins that are at most as probable as the observed one:

$$p = \sum_{k\,:\,\Pr(X=k\mid H_0) \leq \Pr(X=a\mid H_0)} \Pr(X = k \mid H_0).$$

Unlike the χ^2 test, Fisher's test is *exact*. Specifically, it does not rely on a large-sample approximation and therefore remains valid for small cell counts, which frequently occur

when individual constraints are rarely violated.

Multiple testing and the Bonferroni correction [26]. When m hypotheses are tested simultaneously—one per constraint—the probability of at least one false rejection (the *family-wise error rate*, FWER) grows with m. For example, at $\alpha = 0.05$, testing 20 independent true null hypotheses yields a false positive with probability $1 - 0.95^{20} \approx 0.64$.

Definition 3.24 (Bonferroni correction)**.** Given m hypothesis tests and a target family-wise error rate α, the Bonferroni correction rejects hypothesis i only if its p-value satisfies

$$p_i \leq \frac{\alpha}{m}.$$

Equivalently, adjusted p-values $\tilde{p}_i = \min(m \cdot p_i, 1)$ are compared against α.

The correction guarantees FWER $\leq \alpha$ by the union bound, regardless of dependence between tests. It is conservative, in that with many tests, it reduces statistical power and may fail to detect true associations. Associations reported as significant after Bonferroni correction therefore carry strong evidence, whereas non-significance may reflect either absence of an effect or lack of power.

Effect sizes: odds ratio and risk ratio. Statistical significance alone does not convey how strongly a violation is associated with an outcome. Two effect size measures for 2×2 tables are used. Informally, the *risk ratio* compares how much more likely the outcome is among violating episodes, while the *odds ratio* compares the corresponding odds.

Definition 3.25 (Risk ratio)**.** Using the contingency table above, the risks of the outcome in the violated and non-violated groups are $R_1 = a/(a + b)$ and $R_0 = c/(c + d)$, respectively. The *risk ratio* (or relative risk) is

$$RR = \frac{R_1}{R_0} = \frac{a/(a+b)}{c/(c+d)} = \frac{a(c+d)}{c(a+b)}.$$

Definition 3.26 (Odds ratio)**.** The odds of the outcome in the violated and non-violated groups are a/b and c/d, respectively. The *odds ratio* is

$$OR = \frac{a/b}{c/d} = \frac{a \cdot d}{b \cdot c}.$$

Both measures range over $(0, \infty)$ and are interpreted relative to the neutral value 1. A value of 1 indicates no association. Values greater than 1 indicate that the outcome is more frequent (RR) or has higher odds (OR) among violating episodes. Values below 1 indicate a protective association. For example, $RR = 2$ means the outcome is twice as

frequent among violating episodes. The odds ratio always lies farther from 1 than the risk ratio and approximates it only when the outcome is rare (*rare disease assumption*). By contrast, the risk ratio is generally the more directly interpretable of the two, while the odds ratio is well-defined in a broader range of study designs and is the effect measure reported by Fisher's exact test implementations. Effect sizes are reported together with confidence intervals. Throughout this thesis, the interval for a risk ratio is the large-sample interval obtained on the log scale and back-transformed, as introduced by Katz et al. [48], which is the conventional construction for the cohort-style tables considered here. An interval excluding 1 corresponds to a statistically significant association.

4 Trigger-Based Declarative Conformance Checking

The framework developed in this chapter scopes conformance checking to *episodes*. From a full case trace, it extracts sub-traces segmented by occurrences of a domain-meaningful trigger event, and evaluates each of them against a multi-perspective declarative model. The episode, rather than the trace, thus becomes the fundamental unit of conformance assessment, as motivated in Chapter 1.

Throughout this chapter, let $L \subseteq \mathcal{E}$ be an event log as defined in Definition 3.11, over a finite activity alphabet $\Sigma \subseteq \mathcal{U}_{act}$. For a case $c \in cases(L)$, we write $\sigma_c = \langle e_1, \ldots, e_n \rangle$ for its trace and abbreviate $t_j = \pi_{time}(e_j)$ for the timestamp of its j-th event. Constraints are MP-Declare constraints in the form $c^* = (k(b_1, \ldots, b_m), \varphi_a, \varphi_c, \varphi_\tau)$ as defined in Definition 3.19, evaluated under the activation semantics of Definition 3.20.

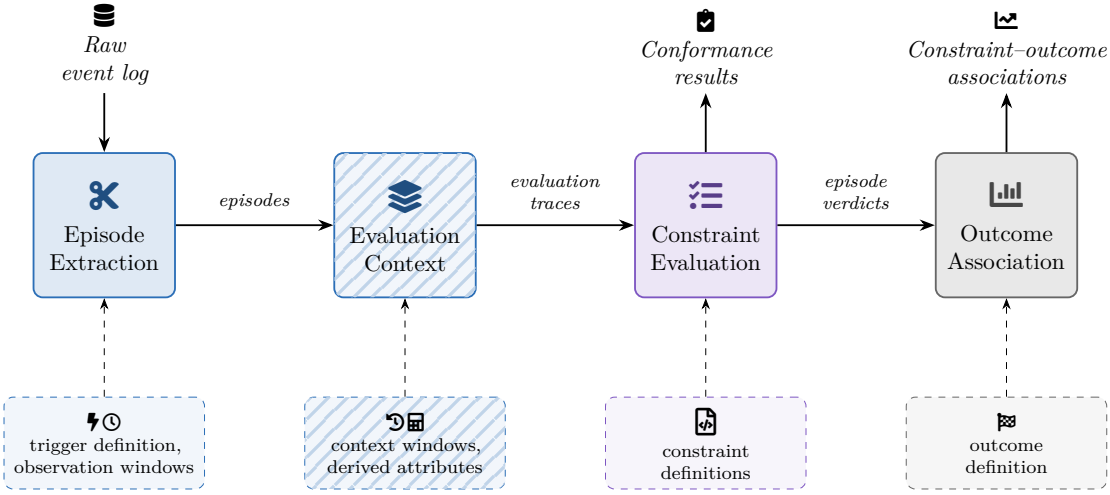


Figure 4.1: Overview of the four-step conformance checking pipeline. Solid arrows denote data flow, labelled with the artifact each step produces and the definition that introduces it; dashed boxes denote the configuration artifacts through which domain knowledge enters.

The framework expects its input event log in the form described in Chapter 3: cases as timestamp-ordered traces of events, each carrying an activity label and a payload of attribute–value pairs. Every attribute x referenced by a constraint condition must satisfy that $\pi_x(e)$ is defined on the events that expose its value, and every normatively relevant occurrence must be present as an explicit labeled event. Producing such a log

from heterogeneous source systems is a domain-specific preprocessing step outside the method. For the ICU weaning case, it is described in Section 6.2.5.

On this input, the framework operates as a pipeline of four steps, shown in Figure 4.1. *Trigger-based episode extraction* segments each case trace into episodes, each opened by an occurrence of the trigger event and bounded by a configurable observation window or an explicit end-event condition. *Evaluation context construction* extends each episode, per constraint, into the evaluation trace that constraint may see. *Constraint evaluation* checks every episode against a set of MP-Declare constraints, classifying each activation instance as a fulfillment or a violation under the semantics of Chapter 3, extended in this thesis with a target-selection mode. *Outcome association analysis* relates the per-episode verdicts to a specified episode outcome, estimating per constraint whether adherence is statistically associated with the favorable outcome. Domain knowledge enters only through the configuration artifacts of Figure 4.1; transferring the framework to a new domain requires authoring these, not modifying the method.

Section 4.1 formalizes episode extraction. Section 4.2 defines the evaluation context. Section 4.3 presents constraint evaluation and its target-selection extension. Section 4.4 defines outcome association analysis, building on the episode attributes and inter-episode rules of Section 4.4.1.

Running Example

A running example accompanies the definitions of this chapter: a grill at a garden party, on which batches of food are grilled one after another. Table 4.1 lists an event log L over the activity alphabet $\Sigma = \{G, B, T, S\}$ consisting of a single case c, the grill, and Figure 4.2 shows its trace σ_c on a time axis. The activity G puts a batch on the grill and records the number of pieces in the attribute *pcs*. B brushes the grate clean, T reads the grill thermometer and records the temperature in degrees Celsius in the attribute *temp*, and S serves the batch. A batch that is not served before the next one goes on the grill has been binned. Timestamps are given in minutes.

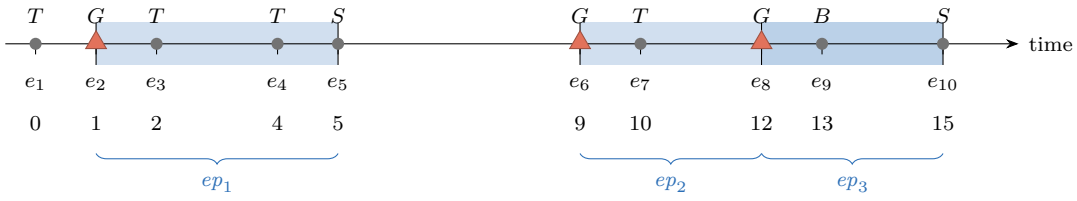


Figure 4.2: The trace σ_c of the running example log of Table 4.1. Coral triangles are trigger events, grey circles all other events; the numbers below the axis are the event timestamps. Blue regions are the three episodes the trace induces. Each includes its trigger and ends just before the event that closes it: the end events e_5 and e_{10} for ep_1 and ep_3, the next trigger e_8 for ep_2.

Table 4.1: The running example log L. All events belong to a single case c, a grill; activity labels abbreviate *grill batch* (G), *brush grate* (B), *thermometer reading* (T), and *serve batch* (S). The attribute *temp* is recorded by events with activity label T, the attribute *pcs* by events with activity label G, and each is undefined otherwise. Timestamps are in minutes.

Event	π_{act}	π_{time}	π_{temp}	π_{pcs}
e_1	T	0	290	—
e_2	G	1	—	6
e_3	T	2	310	—
e_4	T	4	260	—
e_5	S	5	—	—
e_6	G	9	—	7
e_7	T	10	250	—
e_8	G	12	—	2
e_9	B	13	—	—
e_{10}	S	15	—	—

4.1 Trigger-Based Episode Extraction

This stage establishes the scope of the conformance checking. Each trace from the event log gets divided into episodes based on occurrences of repeating domain-meaningful events. The declarative specification is then evaluated separately within each resulting episode exactly as it would over an entire trace. Consequently, the open-world semantics of Declare remain intact at the episode level. The events that initiate these episodes are referred to as *trigger events*. The trigger specification is a domain-specific input to the framework, as described in Chapter 4.

Definition 4.1 (Trigger event)**.** Let $L \subseteq \mathcal{E}$ be an event log over the activity alphabet $\Sigma \subseteq \mathcal{U}_{act}$. A *trigger specification* is a pair $(a_{trig}, \varphi_{trig})$ consisting of an activity label $a_{trig} \in \Sigma$ and a Boolean predicate φ_{trig} over payloads, the latter defaulting to *true*. An event $e \in L$ is a *trigger event* iff $\pi_{act}(e) = a_{trig}$ and $\varphi_{trig}(\pi_{pay}(e))$ holds.

Example 4.2. Under the trigger specification (G, true) of the running example, every batch put on the grill opens an episode, so the trigger events of Table 4.1 are e_2, e_6, and e_8. As φ_{trig} is trivially satisfied, every event carrying the label G qualifies.

The optional payload predicate further restricts which occurrences of a_{trig} qualify as trigger events, based on information contained in the event payload. For example, a trigger specification may require a measurement event to record a value exceeding a

configured magnitude. Smaller measurements with the same activity label would then not initiate an episode.

A single case may contain multiple trigger events, with each occurrence initiating a separate episode. Given a trigger specification, an episode is the portion of a case trace that falls within the observation scope initiated by one trigger event.

Definition 4.3 (Episode)**.** Let $\sigma_c = \langle e_1, \ldots, e_n \rangle$ be the trace of a case $c \in \mathit{cases}(L)$ (Definition 3.11), let $(a_{trig}, \varphi_{trig})$ be a trigger specification (Definition 4.1), and let $e_i \in \sigma_c$ be a trigger event with timestamp t_i. The *episode* induced by e_i is the sub-sequence

$$\mathit{ep}(e_i) = \langle e_j \in \sigma_c \mid t_i \leq t_j < t_{end} \rangle,$$

which preserves the order of σ_c, and whose *episode end* t_{end} is determined by the episode boundary configuration as

$$t_{end} = \min(t_{obs},\ t_\varphi,\ t_{next}), \qquad \begin{aligned} t_{obs} &= t_i + W_{obs}, \\ t_\varphi &= \min\{\, t_j \mid e_j \in \sigma_c,\ t_j > t_i,\ \varphi_{end}(e_j) \,\}, \\ t_{next} &= \min\{\, t_j \mid e_j \in \sigma_c,\ t_j > t_i, \\ &\qquad e_j \text{ is a trigger event} \,\}, \end{aligned}$$

Here, $W_{obs} \in \mathcal{U}_{dur}$ with $W_{obs} > 0$ is the configured observation-window length, and φ_{end} is the configured end-event condition. If either mechanism is not configured or no matching event exists, its boundary is set to $+\infty$. The boundary t_{next} is determined by the next trigger event in the same case and is not configurable. It ensures that an episode ends when the next trigger occurs, even if neither configurable boundary has been reached. If no subsequent trigger occurs and neither configurable boundary is finite, the episode extends to the end of the trace. All three bounds are exclusive, so that the episode covers the half-open interval $[t_i, t_{end})$.
Trigger events sharing a timestamp induce the same episode and are treated as a single trigger. The distinct episodes induced by the trigger events in σ_c, ordered by the timestamps of their trigger events, form the *episode sequence* $E(\sigma_c) = \langle \mathit{ep}_1, \ldots, \mathit{ep}_r \rangle$ of case c. We write $\hat{e}_k$ for the trigger event inducing ep_k, the first in σ_c if several share its timestamp, so that $\mathit{ep}_k = \mathit{ep}(\hat{e}_k)$.

Example 4.4. Suppose $\varphi_{end}(e) \equiv \pi_{act}(e) = S$, so that serving a batch closes its episode, and W_{obs} is unconfigured, so that $t_{obs} = +\infty$ throughout. The three trigger events identified in Example 4.2 then induce the following episodes.

- e_2 at $t = 1$: $t_\varphi = 5$ from e_5 and $t_{next} = 9$ from e_6. Hence, $t_{end} = 5$ and $\mathit{ep}_1 = \langle e_2, e_3, e_4 \rangle$.
- e_6 at $t = 9$: $t_\varphi = 15$ from e_{10} and $t_{next} = 12$ from e_8. Hence, $t_{end} = 12$ and $\mathit{ep}_2 = \langle e_6, e_7 \rangle$. The episode therefore ends at the next trigger event, before the

end-event condition is satisfied: the second batch is never served, as the third goes on the grill first.

- e_8 at $t = 12$: $t_\varphi = 15$ and $t_{next} = +\infty$. Hence, $t_{end} = 15$ and $ep_3 = \langle e_8, e_9 \rangle$.

The resulting episode sequence is $E(\sigma_c) = \langle ep_1, ep_2, ep_3 \rangle$. The end events e_5 and e_{10} close ep_1 and ep_3 without belonging to them, just as e_8 closes ep_2 and instead opens ep_3.

Figure 4.3 illustrates the extraction of a single episode. The three boundary times are considered simultaneously, and the earliest one determines the episode end. The two configurable boundary mechanisms serve complementary purposes. A fixed-duration window is appropriate when an obligation is defined relative to a fixed period following the trigger, whereas an end-event condition is appropriate when the end of the relevant phase is represented by an observable event. Both mechanisms may be configured simultaneously when an episode should end at whichever boundary occurs first. The next trigger event provides a structural boundary rather than a configurable one. When a new trigger occurs before the current episode reaches either configurable boundary, the current episode ends at that trigger and the new trigger initiates the subsequent episode. This keeps episodes pairwise disjoint, so that no event belongs to two episodes. An event of a later episode can serve as a target of an earlier one only where a post-context window (Section 4.2.1) explicitly admits it. The trigger event is included in the episode, while the end event, the next trigger, and the window bound t_{obs} are excluded. Events occurring before the trigger or at or after t_{end} therefore do not belong to the episode, and the episodes of a case are pairwise disjoint. An end event belongs to no episode unless a later trigger shares its timestamp; where a constraint needs it as a target, the post-context window (Section 4.2.1) makes it visible. Membership is decided by timestamp alone. Events sharing the trigger's timestamp belong to its episode regardless of their position in σ_c, so simultaneously recorded events are never split across episodes. Within an episode, events retain their positional order from σ_c (Definition 3.11). If an end event coincides with the next trigger, it falls into the subsequent episode, where it does not act as an end event, since t_φ only considers events strictly after the trigger.

Episodes of the same case are evaluated independently of one another. The conformance verdict of an episode depends on its events and on the bounded evaluation context described in Section 4.2, but not on the verdicts of other episodes. Each episode is evaluated as a separate obligation. The episodes of a case are pairwise disjoint, so each event belongs to at most one episode; events before the first trigger, or between an episode's end and the next trigger, belong to none. The effect of the context windows on these properties is described in Section 4.2.1.

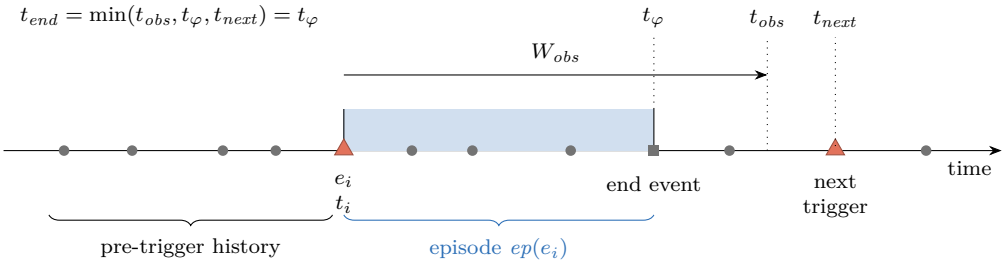


Figure 4.3: Trigger-based extraction of one episode from a case trace. Grey circles are events, coral triangles trigger events, the square an event satisfying the end-event condition φ_{end}. The trigger e_i opens the episode (shaded blue); the episode end t_{end} is the earliest of the three boundary candidates, here t_φ, so the end event closes the episode without belonging to it.

4.2 Constructing the Evaluation Context

Episode extraction determines which events are included within an episode. Some constraint conditions, however, depend on information outside the episode, such as a measurement recorded shortly before the trigger or a quantity derived from earlier events in the case. This information is not itself subject to activaton of the episode's constraints but it provides context for their evaluation.

The framework distinguishes two mechanisms for providing such context. The *evaluation trace* (Section 4.2.1) determines which events are visible to a constraint by extending the episode with bounded, constraint-scoped context windows. The *derived history attributes* (Section 4.2.2) provide values computed from the visible history rather than recorded directly, such as elapsed durations or the occurrence of an earlier event. Both mechanisms are parameterized by configuration as described in Chapter 4.

4.2.1 Constraint-Scoped Evaluation Traces

The episode of Definition 4.3 is the default unit of evaluation. Unless configured otherwise, a constraint is evaluated using exactly the events of the episode. When an episode is extracted from a continuously recorded case trace, however, relevant evidence may lie just outside its boundaries. A data condition may depend on a measurement recorded shortly before the trigger, while a response obligation may be fulfilled by an event occurring shortly after the episode end. In both cases, the relevant information lies outside the episode despite being temporally close to it. The framework addresses this by extending the events visible to a constraint through bounded, constraint-scoped context windows.

Definition 4.5 (Context windows)**.** Let c^* be an MP-Declare relation constraint (Definition 3.19) with distinct activation and target labels $a \neq b$ whose target depends only on occurrences of b, i.e., is of the form $\Diamond b$, $\blacklozenge b$, $\blacklozenge b \vee \Diamond b$, or a negation thereof;

a coupling constraint is treated as its two relation conjuncts. A *context window specification* for c^* is a pair $(W_{pre}, W_{post}) \in \mathcal{U}_{dur} \times \mathcal{U}_{dur}$ of durations (Definition 3.9), both defaulting to 0. The *pre-context window* makes the events up to W_{pre} before the trigger of an episode visible to c^*, as data and as past targets, except occurrences of a, which would activate c^* outside the episode. The filter is label-based, so it also removes occurrences of a that do not satisfy φ_a; as $a \neq b$, these are never targets of c^*. The *post-context window* makes the occurrences of b up to W_{post} after the episode end visible as future targets. For all other constraints, the evaluation trace is the episode itself. The effect of both windows on visibility is defined by Definition 4.6.

Definition 4.6 (Evaluation trace)**.** Let $ep(e_i)$ be the episode induced by a trigger event e_i of the trace σ_c, with trigger timestamp t_i and episode end timestamp t_{end} (Definition 4.3), and let c^* be an MP-Declare constraint (Definition 3.19) with activation activity label $a \in \Sigma$, target activity label $b \in \Sigma$, and context window specification (W_{pre}, W_{post}) (Definition 4.5). The *evaluation trace* of $ep(e_i)$ under c^* is the timestamp-ordered concatenation

$$\sigma_{ev}(c^*, e_i) = \underbrace{\langle e_j \in \sigma_c \mid t_i - W_{pre} \leq t_j < t_i,\ \pi_{act}(e_j) \neq a \rangle}_{\text{pre-context (excludes } a\text{)}} \cdot ep(e_i) \cdot \underbrace{\langle e_j \in \sigma_c \mid W_{post} > 0,\ e_j \notin ep(e_i),\ t_{end} \leq t_j \leq t_{end} + W_{post},\ \pi_{act}(e_j) = b \rangle}_{\text{post-context (targets only)}}.$$

The upper bound of the post-context is inclusive, matching the time conditions (Definition 3.19). Its lower bound is the exclusive episode end, so the post-context begins exactly where the episode stops and admits occurrences of b at t_{end}, including an end event labeled b. For $W_{post} = 0$, the default, no post-context is formed.

Example 4.7. The constraint c_1^* of the running example requires the grate to be brushed after a large batch, since residue from such a batch makes the next one stick and char. Formally, $c_1^* = (\textsc{Response}(G, B),\ \varphi_a,\ \mathit{true},\ [0, 5])$ with activation condition $\varphi_a(e) \equiv \pi_{pcs}(e) \geq 3$: after a batch of at least three pieces, the grate must be brushed within five minutes. Its context windows are $W_{pre} = 6$ and $W_{post} = 2$; the post-context window admits a brushing up to two minutes after the batch has left the grill. Applied to the episodes of Example 4.4, it yields

$$\begin{aligned}
\sigma_{ev}(c_1^*, e_2) &= \langle e_1 \rangle \cdot ep_1 \cdot \langle\rangle = \langle e_1, e_2, e_3, e_4 \rangle, \\
\sigma_{ev}(c_1^*, e_6) &= \langle e_4, e_5 \rangle \cdot ep_2 \cdot \langle e_9 \rangle = \langle e_4, e_5, e_6, e_7, e_9 \rangle, \\
\sigma_{ev}(c_1^*, e_8) &= \langle e_7 \rangle \cdot ep_3 \cdot \langle\rangle = \langle e_7, e_8, e_9 \rangle.
\end{aligned}$$

The pre-context window $[3, 9)$ of the second episode admits e_4 and e_5, neither of

which carries the activation label G. The pre-context window $[6, 12)$ of the third episode contains both e_6 and e_7, but e_6 carries c_1^*'s own activation label G and is therefore excluded, leaving only e_7. The post-context window $[12, 14]$ of the second episode admits e_9 but not e_8, whose label is not B; the post-context window $[5, 7]$ of the first episode and $[15, 17]$ of the third contain no event labeled B. For the constraint c_2^* of Example 4.11, both context windows are 0, so its evaluation traces coincide with the episodes, $\sigma_{ev}(c_2^*, \hat{e}_k) = ep_k$ for $k = 1, 2, 3$. Figure 4.4 illustrates the resulting evaluation trace and the asymmetry of the two windows.

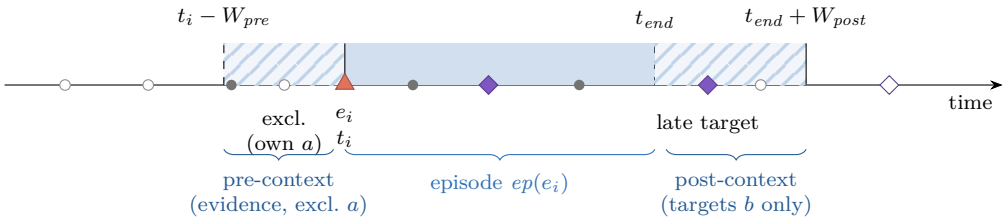


Figure 4.4: The evaluation trace $\sigma_{ev}(c^*, e_i)$ of Definition 4.6. Filled shapes belong to the evaluation trace, open shapes do not; violet diamonds are events bearing the constraint's target label b, grey circles are other events, the coral triangle is the trigger. The episode is shaded blue, the two context windows hatched blue. The pre-context window admits any event as evidence except a further occurrence of the constraint's own activation label a; the post-context window admits only events bearing the target label b.

Since the pre-context contains no occurrence of a and the post-context only occurrences of $b \neq a$, every activation instance of c^* in $\sigma_{ev}(c^*, e_i)$ lies within $ep(e_i)$; context events serve only as targets or data. For a positive template, a target in the post-context can fulfil an activation; for a negative template such as NotResponse, it can violate it. Episodes remain pairwise disjoint, while the evaluation traces of adjacent episodes may overlap: a shared event is a context event in one and an episode event in the other, and it can activate a constraint only in its own episode.

Because every activation lies within its episode, the windows also determine whether the time condition is fully observable. A past-directed constraint with $\varphi_\tau = [t_0, t_1]$ sees its full time condition if $W_{pre} \geq t_1$, and a future-directed one if $W_{post} \geq t_1$; smaller windows truncate the time condition at the episode boundary. Which constraints receive a non-zero window, and how large, is domain knowledge.

4.2.2 Derived History Attributes

Data conditions read attribute values recorded on the events of the evaluation trace. Some required quantities are not recorded directly and must instead be computed from the visible history. Examples include the elapsed time since the case started, whether

an earlier event occurred in a particular way, or the time elapsed since a designated event. The framework provides such quantities as *derived history attributes*, computed on demand from the evaluation trace.

Definition 4.8 (Derived history attributes)**.** A *history specification* is a partial function

$$D : \mathcal{U}_{attr} \rightharpoonup (\mathcal{E}^* \times \mathcal{U}_{time} \rightharpoonup \mathcal{U}_{val}),$$

assigning to each attribute name d in its domain a *derivation function* $D(d)$ that maps a sequence of events and a reference timestamp to a value.
Let $\sigma_{ev} = \sigma_{ev}(c^*, e_i)$ be the evaluation trace of an episode under an MP-Declare constraint c^* (Definition 4.6). For an event $e \in \sigma_{ev}$ with timestamp $t = \pi_{time}(e)$, write $\sigma_{\leq t} = \langle e' \in \sigma_{ev} \mid \pi_{time}(e') \leq t \rangle$ for the history visible at e, i.e., all events of σ_{ev} up to and including the timestamp of e. The *derived history state* at e under D is the partial function $\delta_e : \mathcal{U}_{attr} \rightharpoonup \mathcal{U}_{val}$ with $\mathrm{dom}(\delta_e) \subseteq \mathrm{dom}(D)$, defined by

$$\delta_e(d) = D(d)(\sigma_{\leq t},\, t) \qquad \text{for } d \in \mathrm{dom}(D),$$

where $\delta_e(d)$ is undefined whenever the right-hand side is.

Example 4.9. Let $D(d)(\sigma, t) = t - \pi_{time}(\sigma(1))$, the minutes elapsed since the first event of the visible history. Over $\sigma_{ev}(c_1^*, e_6) = \langle e_4, e_5, e_6, e_7, e_9 \rangle$ of Example 4.7, evaluating at the activating event e_6 gives $\sigma_{\leq 9} = \langle e_4, e_5, e_6 \rangle$ and $\delta_{e_6}(d) = 9 - 4 = 5$; evaluating at the target event e_9 gives $\sigma_{\leq 13} = \sigma_{ev}$ and $\delta_{e_9}(d) = 13 - 4 = 9$.

The reference time is the timestamp of the *evaluated event*, so duration-style attributes change as the episode unfolds. The visible history is the constraint's *evaluation trace*; pre-trigger events are included only when admitted by the constraint's pre-context window (Section 4.2.1). Thus, derived history attributes are computed for each condition evaluation from the history visible to the constraint.

The history specification determines which derived history attributes are available to conditions. The construction of δ_e is fixed, while the derivation functions determine which aspects of history are available. Typical derivations include the elapsed time since the first event of the visible history or since the most recent occurrence of a designated activity; whether an event of a designated label occurred earlier in the visible history, such as an unsuccessfully concluded prior event; the elapsed duration of the phase opened within the current episode; and the current value of a monitored categorical setting.

Derived history attributes are referenced in conditions through the same predicate language as recorded attributes. The evaluation stage therefore does not distinguish between recorded and derived attributes. Attributes that require the full case prefix are handled separately. For example, the ordinal position of an episode within its case is computed at extraction time and carried as an episode attribute for reporting and stratification; it is not visible to activation conditions. Both context mechanisms are

computed over the evaluation trace, so constraint evaluation cost depends on the length of that trace rather than the full case trace.

4.3 Constraint Evaluation and Target Selection

Constraint evaluation follows MP-Declare as defined in Definition 3.19. Each domain rule is represented by an MP-Declare constraint with an activation condition φ_a, a correlation condition φ_c, and a time condition φ_τ. Chapter 6 instantiates this translation for the ICU weaning guideline. Evaluation proceeds per activation: each activation instance within an episode's evaluation trace (Section 4.2.1) is classified as a fulfillment or violation according to Definition 3.20.

Standard MP-Declare semantics evaluate the correlation condition existentially over candidate target events: an activation is fulfilled if at least one candidate within the temporal window satisfies the condition. This is suitable when any qualifying occurrence is sufficient, but can misrepresent rules over repeated measurements. If an early measurement satisfies the condition but a later one does not, existential matching still yields a fulfillment, even when compliance depends on the final observed value.

To accommodate these cases, constraints get extended with a target-selection parameter $\mathit{target_selection} \in \{\mathit{any}, \mathit{earliest}, \mathit{latest}\}$. The modes select any satisfying candidate, the chronologically first candidate, or the chronologically last candidate, respectively.

Definition 4.10 (Target selection)**.** Let c^* be an MP-Declare constraint (Definition 3.19) with template Response or Precedence, target activity label $b \in \Sigma$, correlation condition φ_c, and time condition φ_τ, and let σ_{ev} be its evaluation trace (Definition 4.6). Let i be an activation instance of c^* with activating event e. The *candidate targets* of i are

$$M_i = \langle\, m \in \sigma_{ev} \mid \pi_{act}(m) = b,\ \Delta(e,m) \text{ defined},\ \Delta(e,m) \in \varphi_\tau \,\rangle = \langle m_1, \ldots, m_k \rangle,$$

in their order in σ_{ev}. By the definition of Δ, these lie after e for Response and before e for Precedence. Let $\psi(m)$ hold iff $\pi_{pay}(e)$ and $\pi_{pay}(m)$ jointly satisfy φ_c. Then

$$\begin{aligned} \mathit{any}\colon &\quad i \text{ is a fulfillment iff } \psi(m_j) \text{ holds for some } m_j \in M_i, \\ \mathit{earliest}\colon &\quad i \text{ is a fulfillment iff } \psi(m_1) \text{ holds}, \\ \mathit{latest}\colon &\quad i \text{ is a fulfillment iff } \psi(m_k) \text{ holds}, \end{aligned}$$

and i is a violation otherwise, in particular if $M_i = \langle\rangle$. The mode is configured per constraint and defaults to *any*, which coincides with Definition 3.20.

Example 4.11. The constraint c_2^* of the running example requires the grill to have settled below 300 °C by the last reading taken while a batch is on it. Formally, $c_2^* = (\textsc{Response}(G, T),\ \mathit{true},\ \varphi_c,\ [0, 5])$ with $\varphi_c \equiv \pi_{temp}(e_\tau) < 300$ and both context windows 0. The activation at e_2 has the candidate targets $\langle e_3, e_4 \rangle$ with temperatures 310 and 260: the grill was too hot when the batch went on but was turned down in time. Under *latest*, only e_4 is evaluated and the activation is a fulfillment; under *earliest*, only e_3 is evaluated and it is a violation; under *any*, it is a fulfillment witnessed by e_4. The activation at e_6 is fulfilled by its single candidate e_7 under every mode, and the activation at e_8, without any candidate, is violated under every mode.

The selection modes can also be characterized in the LTL_f vocabulary of Section 3.3.1.

Proposition 4.12 (LTL_f characterization of target selection)**.** Let i be an activation instance with candidate targets M_i and predicate ψ as in Definition 4.10. For an instant j of σ_{ev}, let β hold at j iff $\sigma_{ev}(j) \in M_i$, and let β^+ hold at j iff additionally $\psi(\sigma_{ev}(j))$ holds. Then, under each mode, i is a fulfillment iff $\sigma_{ev}, i \models \phi$ for the corresponding formula ϕ:

Mode	Response	Precedence
any	$\Diamond\, \beta^+$	$\blacklozenge\, \beta^+$
earliest	$(\neg\beta)\, \mathcal{U}\, \beta^+$	$\blacklozenge\, (\beta^+ \wedge \neg \ominus \blacklozenge\, \beta)$
latest	$\Diamond\, (\beta^+ \wedge \neg \bigcirc \Diamond\, \beta)$	$(\neg\beta)\, \mathcal{S}\, \beta^+$

Remark 4.13. The predicates β and β^+ are defined relative to the activation being checked. For example, consider two activations of c_2^* at times 0 and 3, and a thermometer reading at time 6. The reading falls outside the first activation's window $[0, 5]$ but inside the second's, making it a candidate target only for the second activation.
In standard Declare, target eligibility does not depend on the activation, so a formula such as $\Box(a \rightarrow \Diamond\, b)$ can express the constraint for all activations at once. By contrast, $\Box(a \rightarrow \Diamond\, \beta^+)$ is ill-defined here because it does not specify which activation determines β^+. The formulas in Proposition 4.12 therefore characterize one activation at a time. Expressing the constraint for all activations in a single formula requires referring each target back to its activating event, which the underlying MFOTL formalism of MP-Declare supports through first-order variables [16].

The *any* formulas require at least one satisfying candidate. The *earliest* and *latest* formulas additionally require that the selected candidate be the first or last in the relevant temporal order. Under *any*, evaluation can stop at the first satisfying candidate; *earliest*

and *latest* require identifying the selected candidate. Earlier observations remain in the evaluation trace but do not determine the verdict unless selected by the configured mode. For PRECEDENCE, the chronologically last candidate is the observation closest to the activation. Figure 4.5 illustrates a case in which the modes produce different verdicts.

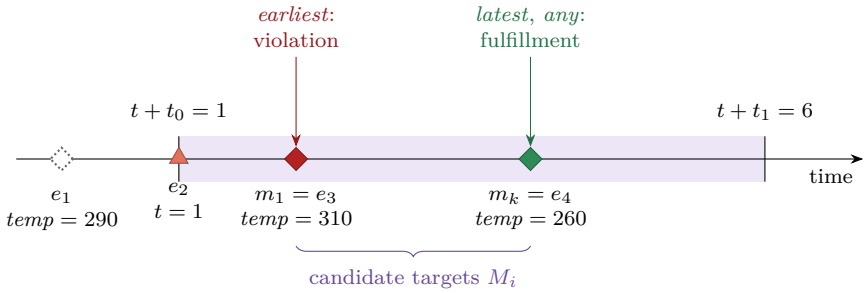


Figure 4.5: Target selection on the candidate sequence of Example 4.11. The coral triangle is the activating event e_2 at time $t = 1$; diamonds are thermometer readings, events carrying the target label T, green if they satisfy the correlation predicate $\psi(m) \equiv \pi_{temp}(m) < 300$ and red otherwise. The violet region is the window $\varphi_\tau = [0, 5]$ relative to t; the dotted diamond is e_1, which carries the target label but precedes the activation and is therefore no candidate for the future-directed template RESPONSE. The activation is fulfilled under *latest* and *any*, both witnessed by $m_k = e_4$, and violated under *earliest*, which consults $m_1 = e_3$ alone.

The appropriate mode depends on the semantics of the rule. For past-directed constraints such as the readiness assessment constraints in Chapter 6, *latest* selects the most recent measurement before the activation. The case study does not use *earliest*. Moreover, target selection is configured per constraint and applies only to RESPONSE and PRECEDENCE templates. It leaves the template semantics, activation and correlation conditions, and temporal intervals unchanged. The default mode *any* preserves the standard existential semantics of MP-Declare [16].

4.4 Outcome Association Analysis

The constraint evaluation stage determines whether each episode complies with its MP-Declare rules. The final stage examines whether compliance is associated with a favorable outcome, thereby assessing the relationship between rule adherence and process outcomes. This stage relies on two constructions. First, each episode is assigned attributes, including a designated outcome attribute. The outcome used for a constraint need not belong to the same episode as the constraint. For example, a rule prescribing a normative measure in an episode is assessed against that episode's outcome, whereas the effect of a corrective measure within an episode with an unfavourable outcome is evaluated using the outcome of the *next* episode. Second, the per-episode conformance verdicts serve as the exposure

variable in a statistical test of their association with the corresponding outcome.

4.4.1 Episode Attributes and Inter-Episode Rules

The preceding sections fix where a constraint is evaluated: within one episode, over its evaluation trace. Beyond its events, each episode carries *episode attributes* established during extraction and aggregation. These are the ordinal position of the episode within its case and a designated outcome. The definition below formalizes this construct and designates one instance, the outcome attribute, for association analysis.

Definition 4.14 (Episode attributes)**.** Let $E(\sigma_c) = \langle ep_1, \ldots, ep_r \rangle$ be the episode sequence extracted from the trace of a case $c \in \mathit{cases}(L)$ in trigger order (Definition 4.3). An *episode attribute* is a partial function

$$\gamma : E(\sigma_c) \rightharpoonup D_\gamma,$$

where D_γ is the attribute's value domain. For an episode $ep \in E(\sigma_c)$, $\gamma(ep)$ is defined if and only if the value of γ is observable or derivable for ep.
The *outcome attribute* is the episode attribute

$$\omega : E(\sigma_c) \rightharpoonup \{\mathit{pos}, \mathit{neg}\}$$

designated for outcome-association analysis. Its derivation from the trace σ_c and the episode boundaries is specified as domain input, analogously to the trigger specification.

Example 4.15. The outcome attribute of the running example assigns *pos* to episodes closed by the end-event condition, whose batch was served, and *neg* to episodes closed by the following trigger, whose batch was binned, so $\omega(ep_1) = \mathit{pos}$, $\omega(ep_2) = \mathit{neg}$, and $\omega(ep_3) = \mathit{pos}$. The constraint c_1^* is activated at e_2 and e_6, whose attribute *pcs* satisfies φ_a, but not at e_8. For the activation at e_2, no event labeled B in $\sigma_{ev}(c_1^*, e_2)$ falls within $[1, 6]$, so $X_1 = \mathit{violated}$; for the activation at e_6, the event e_9 falls within $[9, 14]$ and is admitted by the post-context window, so $X_2 = \mathit{satisfied}$. In ep_3 the constraint is not activated and X_3 is vacuous. Assessed against their own episodes, the two non-vacuous verdicts contribute the pairs $(\mathit{violated}, \mathit{pos})$ and $(\mathit{satisfied}, \mathit{neg})$.

Relating verdicts to episode attributes requires specifying *which episode's* attributes a constraint is assessed against. By default, a constraint is assessed against its own episode: the association analysis pairs $(X_i, \omega(ep_i))$, where $X_i \in \{\mathit{satisfied}, \mathit{violated}, \mathit{vacuous}\}$ is the constraint's verdict in episode ep_i. Vacuous episodes are excluded, since non-activation identifies a different population, not a compliance level. This default is the correct reference for most constraints.

Reactive obligations require a different reference. The attributes affected by compliance are those of the *successor* episode. Such a rule is also typically activated by the very event that determines its own episode's outcome. That outcome is then constant by construction across the episodes in which the rule applies, and assessing the rule against its own episode yields a degenerate comparison. The framework therefore allows a constraint to retain its episode-local evaluation while associating its verdict with an attribute of the successor episode. Such constraints are designated as *inter-episode rules*.

Definition 4.16 (Inter-episode rule)**.** Let $E(\sigma_c) = \langle ep_1, \ldots, ep_r \rangle$ be the episode sequence of a case $c \in \mathit{cases}(L)$, and let ω be its outcome attribute (Definition 4.14). An MP-Declare constraint c^* is an *inter-episode rule* if its verdict in an episode is associated with the outcome of the immediately succeeding episode.
For an episode $ep_i = ep(\hat{e}_i)$, let X_i denote the verdict obtained by evaluating c^* over the evaluation trace $\sigma_{ev}(c^*, \hat{e}_i)$ of Definition 4.6. The associated prospective outcome is

$$Y_i = \omega(ep_{i+1}),$$

defined if and only if

$$i < r \quad \text{and} \quad ep_{i+1} \in \mathrm{dom}(\omega).$$

Thus, an episode ep_i contributes the pair (X_i, Y_i) to the outcome-association analysis for c^* only when Y_i is defined. Episodes without a succeeding episode or with an undefined outcome for their succeeding episode are excluded from that analysis, and their number is reported alongside the estimates.

Example 4.17. Designating c_1^* an inter-episode rule leaves the verdicts of Example 4.15 unchanged and replaces the referenced outcome by that of the successor episode. The pairs become $(X_1, \omega(ep_2)) = (\mathit{violated}, \mathit{neg})$ and $(X_2, \omega(ep_3)) = (\mathit{satisfied}, \mathit{pos})$. This matches the rationale of the rule: the grate left unbrushed after the first batch precedes the binned second batch, and the brushing after the second batch precedes the served third. The third episode has no successor, so Y_3 is undefined; it is vacuous in any case.

Although Definition 4.16 is stated for the outcome attribute, the same construction applies to any episode attribute. An inter-episode rule may therefore be associated with $\gamma(ep_{i+1})$ for any episode attribute γ. The outcome attribute is the primary instance used for outcome-association analysis. Whether a constraint is designated as an inter-episode rule is part of its configuration.

The designation of the next episode outcome changes only the attribute used for association; it does not change conformance evaluation. The verdict X_i, its evaluation trace, and its per-episode diagnostics remain unchanged. The successor's attribute is used only in the downstream statistical analysis and does not affect the evaluation of either episode. Episode independence is therefore preserved.

4.4.2 Association Analysis

Once an outcome attribute is defined per episode (Definition 4.14), the per-episode verdicts become the exposure variable of a statistical association analysis and the designated outcome becomes the response.

Fix a constraint c and let its exposure–outcome pairs range over the episodes in which c is non-vacuously evaluated: (X_i, Y_i), where $X_i \in \{\mathit{satisfied}, \mathit{violated}\}$ is the episode verdict and $Y_i \in \{\mathit{pos}, \mathit{neg}\}$ is the associated outcome. The outcome is referenced against the constraint's own episode for ordinary constraints and against the successor for inter-episode rules (Definition 4.16). Vacuous episodes and episodes whose associated outcome is undefined are excluded, as fixed above, so the population under analysis is exactly those episodes in which the rule applied and its consequence is observable.

Cross-tabulating the two binary variables yields the 2×2 contingency table of Section 3.5, whose cells count episodes by verdict and outcome. From this table the framework computes an association test and an effect size per constraint: Fisher's exact test together with the odds ratio, the risk ratio, and their 95 % confidence intervals, the latter computed on the log scale and back-transformed (Section 3.5). Fisher's exact test is preferred over χ^2 because rarely violated constraints produce small cell counts. The risk ratio for the adverse outcome carries the interpretation of the result. A ratio above one means violating episodes fare worse, which certifies the rule as protective. A ratio indistinguishable from one means the log provides no evidence that the rule tracks the outcome. A ratio below one means adherence is associated with *worse* episodes, and the encoded rule may be miscalibrated. Because one test is run per constraint, the family-wise error rate is controlled by the Bonferroni correction [26] of Section 3.5. An association is reported as significant only if its adjusted p-value clears the target level.

Two instantiations illustrate the same machinery. In the ICU weaning study of Chapter 6, each spontaneous-breathing episode carries the binary outcome *successful* versus *failed* weaning. A readiness-assessment constraint is outcome-relevant if episodes that violate it fail significantly more often, with an odds ratio quantifying how much the omission raises the odds of failure. In a road-traffic fine-management log, each fine-handling episode carries the outcome *paid in full* versus *not paid*. A constraint requiring that a payment reminder be dispatched within the statutory deadline is assessed by whether violating episodes are associated with a lower full-payment rate. In both cases the rule, the outcome, and the direction of the favorable effect are domain inputs, while the contingency construction, the test, the effect size, and the multiplicity correction are fixed.

Example 4.18. Cross-tabulating the two pairs that c_1^* contributes in Example 4.17 gives the contingency table of Section 3.5 with $a = 1$ violating episode carrying the adverse outcome, $b = c = 0$, and $d = 1$ satisfying episode with the favorable outcome. The construction is the one applied per constraint across a full log; on two contributing episodes it supports no inference.

5 Evaluation

This chapter evaluates the framework of Chapter 4 along four axes. The first concerns *correctness*: synthetic logs with planted verdicts and episode outcomes allow the reported results to be compared with known ground truth. The second concerns *episode scoping*: on the same generated logs, episode-scoped evaluation is compared with case-scoped analysis of the same events, isolating the effect of the evaluation scope. The third concerns *applicability and configuration sensitivity*: the pipeline is instantiated on a public, non-clinical event log whose process semantics are documented independently of this thesis, while the trigger specification is varied to quantify its effect on the analyzable data. The fourth concerns *cost*: runtime and memory are measured as functions of log size, trace length, model size, and constraint complexity.

The first two axes provide the main empirical validation of the framework. They establish that the checker returns the prescribed verdicts, that the statistical stage recovers associations of known magnitude, and that episode-scoped evaluation can produce conformance verdicts and outcome associations that differ from those obtained at case level. The cost experiments assess computational feasibility and identify structural properties that influence evaluation cost. The applicability experiments provide an end-to-end demonstration on a public log and quantify the effect of trigger selection. The clinical instantiation on MIMIC-IV is presented separately in Chapter 6, where the domain-specific results are reported.

5.1 Experimental Setup

This section describes the setup shared by all experiments. It first states the evaluation questions and then introduces the event logs, the declarative models evaluated on them, the metrics used to answer each question, and the protocol of the performance benchmarks.

5.1.1 Evaluation Questions

Six questions structure the experiments.

EQ1 (Correctness). Does the checker return the verdict prescribed by Definition 3.20 across the implemented template families, data conditions, time windows, and target-selection modes?

EQ2 (Episode scoping). Does episode-scoped evaluation recover per-episode conformance and outcome associations that case-scoped analysis of the same log does not, and does the association stage recover an association of known magnitude?

EQ3 (Applicability). Does the pipeline produce interpretable rule-quality results on a second, structurally different log using the same checker and statistical analysis as the clinical study?

EQ4 (Trigger sensitivity). How does the choice of trigger activity affect the population of episodes available for analysis in terms of coverage, granularity, retained events, and outcome-label validity?

EQ5 (Scalability). How does evaluation cost grow with the number of cases, trace length, and number of constraints, and how does peak memory change over the same range?

EQ6 (Cost drivers). Which structural properties of a constraint drive its per-activation runtime, and does the resulting additive complexity score predict evaluation cost sufficiently well to estimate the cost of a specification before execution?

EQ1 and EQ2 are evaluated on synthetic logs, which provide the ground truth required to assess verdicts and estimated associations. EQ3 and EQ4 are evaluated on the public log described below, while EQ5 and EQ6 use dedicated benchmarks. The corresponding domain questions on MIMIC-IV are addressed in Chapter 6.

5.1.2 Log, Episodes, and Outcome

The primary evaluation log records fines issued by an Italian local police force [56], referred to below as the fine log. It contains 561 470 events in 150 370 cases over eleven activity labels, spanning January 2000 to June 2013. Table 5.1 lists the activity frequencies. The log was selected for three reasons. First, it is a public benchmark whose process semantics are documented independently of this thesis, providing an external basis for interpreting the process. Second, it contains numeric case and event attributes, including fine amounts, accrued expenses, and payment amounts, which exercise the data-aware evaluation path of the framework. Third, it admits a binary episode outcome, defined below, required by the outcome-association analysis.

Two properties of the log affect the experimental design. Timestamps have day-level resolution, so the order of events recorded on the same date cannot be determined reliably. In addition, repeated occurrences of an activity on the same date cannot be distinguished by timestamp-based segmentation. The case identifier represents an individual fine, and each fine is created once; consequently, multi-episode cases arise only for trigger activities that recur within a fine.

The default trigger activity is Create Fine, and each episode ends at the next trigger occurrence or at the end of the case. EQ4 treats the trigger as an experimental variable. The sweep considers six candidate triggers: Create Fine, Send Fine, Insert Fine Notification, Add penalty, Payment, and Appeal to Judge. For each trigger, the segmentation metrics of Section 5.1.5 are computed on a random sample of 20 000 cases. The conformance analysis uses the configured trigger on the full log.

An episode is labeled *Success* when the amount owed is settled in full before the process escalates, and *Failure* otherwise. Specifically, an episode is labeled *Success* if its recorded

Table 5.1: Activity frequencies in the fine log (561 470 events in 150 370 cases).

Activity	Events
Create Fine	150 370
Send Fine	103 987
Insert Fine Notification	79 860
Add penalty	79 860
Payment	77 601
Send for Credit Collection	59 013
Insert Date Appeal to Prefecture	4188
Send Appeal to Prefecture	4141
Receive Result Appeal from Prefecture	999
Notify Result Appeal to Offender	896
Appeal to Judge	555

payments cover the base fine amount registered at Create Fine plus recorded expenses, no Add penalty follows the last payment, and the episode contains no Send for Credit Collection. This operationalizes the process objective using the information recorded in the log. In the clinical instantiation, the corresponding outcome is the success or failure of the weaning episode, defined in Section 4.4.1 and instantiated in Chapter 6.

5.1.3 Synthetic Logs with Planted Ground Truth

Neither the fine log nor MIMIC-IV provides ground truth for constraint verdicts or episode outcomes. Two claims therefore cannot be evaluated directly on the observational logs: whether the checker returns the verdict prescribed by Definition 3.20, and whether episode scoping produces results that differ from case-scoped analysis in the expected way. On the fine log, the latter comparison is also unavailable because Create Fine occurs once per case, causing the two scopes to coincide. Two synthetic constructions address these limitations.

Verdict catalogue. The first construction is a catalogue of 95 test cases. Each case consists of a hand-written micro-trace, one constraint, and its expected verdict according to Definition 3.20. Every implemented template appears in satisfied and violated cases, and in a vacuous case where the semantics permit one. The catalogue also varies dimensions that occur only sparsely in the fine log: activation conditions, target correlation conditions, forward and backward time windows, combinations of these conditions, and the target-selection modes of Definition 4.10. Traces contain two to four events, allowing the expected verdict to be determined directly from the trace.

Multi-episode log. The second construction is a generated log with planted verdicts and outcomes. Each case is generated in three steps. First, the number of episodes is drawn from a geometric distribution with continuation probability 0.45, truncated at four. A case therefore starts with one episode and receives each additional episode with probability 0.45. Consecutive episodes are separated by ten days, preventing events from

Table 5.2: Constraints planted in the multi-episode log. φ_τ denotes the time condition relative to the activation, φ_a the activation condition. P_{viol} and P_{vac} are the probabilities with which an episode is assigned a violated or vacuous verdict; the remaining episodes are satisfied. RR is the planted risk ratio of *Success* under violation. The last column gives the error expected under case-scoped evaluation.

ID	Template	Activation	Target	Conditions	P_{viol}	P_{vac}	RR	Case scope
`SYN-R1`	Response	A1	B1		0.30	0.15	2.00	masked
`SYN-R2`	Response	A2	B2	$\varphi_\tau = [0, 1\,\mathrm{h}]$	0.30	0.15	0.50	none
`SYN-R3`	Precedence	C3	A3		0.30	0.15	3.00	masked
`SYN-R4`	AtMostOne	D4	—		0.25	0.00	0.25	invented
`SYN-R5`	Response	X5	Z5	φ_a: amount > 50	0.30	0.20	1.50	masked
`SYN-R6`	Response	P6	Q6		0.30	0.15	1.00	masked

one episode from entering the context window of another. Second, each episode begins with the trigger activity. Six constraints are then assigned verdicts independently from per-constraint distributions over satisfied, violated, and vacuous, and events realizing the assigned verdicts are added at fixed offsets within the episode. The expected verdict is therefore determined by construction. Third, the episode outcome is sampled from the model described below. The six constraints, listed in Table 5.2, cover the behaviors that distinguish episode- and case-scoped evaluation: an ordering constraint without a time window (`SYN-R1`), the same constraint with a time window (`SYN-R2`), a precedence constraint (`SYN-R3`), a unary occurrence constraint (`SYN-R4`), a data-aware constraint whose activation condition uses an event attribute (`SYN-R5`), and a constraint with no planted outcome effect (`SYN-R6`). The last column of Table 5.2 gives the error that case-scoped evaluation is expected to produce for each constraint. When episodes are pooled, an unanswered activation in one episode can be matched with a target from another, masking the violation. A counting constraint can see occurrences from several compliant episodes and report an invented violation. A time window confines the constraint to its own episode, so the verdict is unaffected.

Each episode outcome is sampled using a baseline success probability multiplied by the planted risk ratio of every constraint violated in that episode. The planted ratios range from 0.25 to 3, with one constraint fixed at 1. Because constraint verdicts are sampled independently, each constraint's marginal risk ratio equals its planted value in expectation. The baseline probability is chosen so that the resulting success probability remains at most one for every possible combination of violations. The generator rejects configurations that violate this bound rather than clipping the probability, since clipping would attenuate the planted effects. Activity labels are abstract to avoid introducing process assumptions into the generator. Generation is seeded, and the experiments use 20 replications of 1000 cases each.

5.1.4 Declarative Models

Two declarative models are used in the experiments. The first encodes the fine-management process and is evaluated on the fine log to assess conformance and rule quality. The second is a set of controlled rule variants, constructed over the activities and attributes of the fine log, that is used only in the rule-complexity experiment to vary constraint structure in a controlled way.

Fine log model. Eleven MP-Declare constraints encode the fine-management process. Table 5.3 lists them. `RTF-03` encodes the statutory notification deadline. `RTF-02`, `RTF-04`, `RTF-05`, `RTF-07`, `RTF-08`, and `RTF-10` encode orderings within the administrative procedure, such as the precedence of penalty over credit collection. `RTF-01` and `RTF-06` serve as sanity checks for the Init and AtMostOne templates: they hold throughout the log by construction because every case contains exactly one Create Fine and at most one Add penalty. `RTF-09` exercises the data-aware evaluation path through an activation condition on the fine amount. `RTF-03b` is calibrated against the log as described below. The constraint set is therefore intended as an evaluation model rather than as a validated normative model of Italian fine collection.

Three modeling choices in Table 5.3 require brief justification. `RTF-03` anchors the statutory notification deadline at Create Fine rather than dispatch, since the law counts from the offense, which the log does not record, and Create Fine is the earliest available proxy. `RTF-03b` separately constrains the dispatch-to-notification delay with a 30 d threshold, chosen as the observed 90th percentile so that it distinguishes late notification from complete absence instead of merely testing whether notification ever occurs. Finally, no constraint links notification of the prefecture result to an appeal to the judge. In this log, appeals frequently occur both with and without prior notification, so neither Response nor Precedence would express a clear obligation.

Controlled rule variants for cost measurement. The rule-complexity experiment requires constraints whose structure varies systematically, which the fine log model does not provide. A set of controlled rule variants is therefore constructed. They are not a process model and are not derived from domain knowledge. They are defined over the activities and attributes of the fine log and evaluated on the same fine log sample as the fine log model, not on the generated logs of Section 5.1.3. Their runtimes therefore reflect realistic trace lengths and attribute values, and they can be compared directly with those of the fine log constraints.

All variants start from a Response constraint from Create Fine to Send Fine. Each of the three sweeps (five constraints each) changes one component of the complexity score (Section 5.1.5). The template sweep varies only the template family over the same two activities, from a unary existence constraint on Send Fine through an unordered and an ordered binary template to the alternate and chain variants. The clause sweep adds an activation condition of one to five atomic predicates over a single attribute, the fine amount. The attribute sweep uses one to five predicates, each over a different attribute, so that the numbers of clauses and attributes grow together. No variant uses a time

Table 5.3: The fine log Declare model. φ_τ denotes the time condition relative to the activation, φ_a the activation condition where present.

ID	Template	Activation	Target	Conditions
RTF-01	Init	Create Fine	—	
RTF-02	Precedence	Send Fine	Create Fine	
RTF-03	Response	Create Fine	Insert Fine Notification	$\varphi_\tau = [0, 90\,\mathrm{d}]$
RTF-03b	Response	Send Fine	Insert Fine Notification	$\varphi_\tau = [0, 30\,\mathrm{d}]$
RTF-04	Precedence	Add penalty	Insert Fine Notification	
RTF-05	Precedence	Send for Credit Collection	Add penalty	
RTF-06	AtMostOne	Add penalty	—	
RTF-07	Response	Insert Date Appeal to Prefecture	Send Appeal to Prefecture	
RTF-08	Response	Receive Result Appeal from Prefecture	Notify Result Appeal to Offender	
RTF-09	Response	Create Fine	Send Fine	φ_a: amount $> 50\,\mathrm{EUR}$
RTF-10	NotCoExistence	Send for Credit Collection	Payment	

window. All predicates reference attributes recorded on Create Fine, so the activation conditions are actually evaluated rather than short-circuited because of missing values.

5.1.5 Metrics

Five metric families are used to answer the evaluation questions.

Ground-truth metrics (EQ1, EQ2). For the synthetic logs of Section 5.1.3, the planted verdict is compared directly with the observed verdict. For the verdict catalogue, the reported metric is the proportion of test cases whose verdict matches the expected result, broken down by template. A mismatch is treated as a defect unless the expected result is genuinely ambiguous under the implemented semantics, in which case the difference is reported and justified. For the multi-episode log, accuracy is reported per constraint under episode- and case-scoped evaluation. Case-scoped errors are classified as *masked*

violations, where an episode violation is not reported, and *invented* violations, where a case-level violation is reported although no episode violates the constraint. For the association stage, the estimated risk ratio is compared with the planted value across replications. Two quantities are reported: the proportion of replications whose 95 % confidence interval contains the planted value, and the proportion in which the constraint reaches the significance threshold. For the constraint planted at $RR = 1$, the latter is a false-positive rate rather than statistical power.

Conformance metrics (EQ3). For each constraint and episode, the framework reports the ternary verdict of Definition 3.20: *satisfied*, *violated*, or *vacuous*, together with the numbers of activation instances, fulfillments, and violations. These results yield, for each constraint, an *activation rate*, a *violation rate*, and a separate *vacuity rate*. Because treatment of vacuous episodes is a modeling choice, conformance results are reported under two treatments: activated episodes only, and vacuous episodes pooled with satisfied episodes. This makes the effect of the choice explicit.

Association metrics (EQ3). For each constraint, episode verdicts and episode outcomes form the 2×2 contingency table of Section 3.5, with violation as the exposure and effect sizes oriented toward the favorable outcome. Reported quantities are the four cell counts, the two-sided p-value of Fisher's exact test [32], the odds ratio and risk ratio of Section 3.5 [9], and 95 % confidence intervals for both ratios, computed on the log scale following Katz et al. [48]. Tables containing a zero cell use a continuity correction of 0.5, which is flagged in the output. Significance is assessed using a Bonferroni-adjusted threshold with family-wise error rate $\alpha = 0.05$, where the family is the constraint set of the respective model. Vacuous episodes and, for inter-episode rules, episodes whose successor is unobserved are excluded. The latter are counted as censored. The analyzed population therefore consists of episodes in which the rule applies and the referenced outcome is observable.

Segmentation metrics (EQ4). For each candidate trigger, six descriptive quantities characterize the resulting analysis population. Conformance checking is not required because these metrics describe the segmentation itself. Rule-quality results are not compared across triggers, since different triggers produce different episode populations.

Case coverage	Share of cases containing the trigger at least once. Cases without the trigger produce no episode and are excluded from the analysis.
Granularity	Episodes per case and events per episode.
Event coverage	Share of all log events contained in an episode. The excluded share is decomposed into events from cases without the trigger and prefix events preceding the first trigger in covered cases.
Label validity	Share of episodes for which the outcome can be computed, operationalized as the share containing the Create Fine event required

	to obtain the amount owed. Episodes without this event receive a degenerate label.
Outcome balance	Proportion of episodes labeled *Success*.

Cost metrics (EQ5, EQ6). Runtime is measured as wall-clock time and reported as the median over repetitions. This follows the recommendation of Georges et al. [35] to use robust statistics such as the median for performance measurements on managed runtimes, where garbage collection and operating-system scheduling can introduce right-skewed noise. Per-unit quantities are reported where applicable, including time per case, trace, constraint, and activation. Memory is reported using the `tracemalloc` peak and process resident set size.

Scaling behavior is summarized by the exponent of a log-log fit against the varied quantity. Following Hoos and Stützle [46], the fitted exponent is interpreted as an empirical growth rate over the measured range rather than as a formal asymptotic bound. Structural cost is summarized by the framework-specific complexity score

$$C(r) = w_{template} + n_{clauses} + n_{attributes} + t_{time},$$

where $w_{template} \in \{1, 2, 3, 4\}$ grades the template family according to its trace-scanning behavior: grade one for unary existence templates, grade two for unordered binary templates, grade three for ordered binary templates (Section 3.3.2), and grade four for the alternate and chain variants, which additionally inspect the events between activation and target: alternate templates require that no further activation occurs before the target, and chain templates require that the target immediately follows the activation. $n_{clauses}$ counts atomic predicates across the activation and correlation conditions, $n_{attributes}$ counts distinct referenced event attributes, and t_{time} indicates whether a time window is present. The score is not intended as a formal cost model. It assumes additive contributions from the four components and is evaluated empirically against measured per-constraint runtime in Section 5.2.5.

5.1.6 Benchmark Design and Protocol

Three ground-truth experiments precede the cost benchmark. *SE1* runs the verdict catalogue of Section 5.1.3 and compares each observed verdict with its prescribed result. *SE2* generates the multi-episode log and evaluates it both per episode and per case, comparing both results with the planted verdicts. *SE3* applies the association analysis of Section 5.1.5 to the verdicts produced by SE2 under both scopes and both vacuity treatments, comparing the estimated associations with the planted risk ratios. Both therefore operate on the same 20 generated logs (Section 5.1.3), so the association results in SE3 can be traced directly to the verdict errors measured in SE2. Replications are keyed by seed and written as they complete, allowing interrupted runs to resume without repeating completed measurements.

The scalability benchmark comprises five experiments, summarized in Table 5.4. *E1* varies log size and measures the full pipeline, including episode segmentation, outcome

Table 5.4: Benchmark experiments and their parameters. Case counts denote random samples drawn with a fixed seed. The same sample is reused across repetitions, so repetition spread reflects timing noise alone.

Experiment	Varied quantity	Configuration
E1 log size	cases	size ladder up to the full 150 370 cases; full pipeline, all eleven constraints
E2 trace length	events per trace	natural traces from a 10 000-case sample; synthetic replication factors $k \in \{1, 2, 5, 10, 25, 50\}$ over 100 traces, no segmentation
E3 model size	constraints	first k of eleven constraints, $k = 1, \ldots, 11$, on a 10 000-case sample
E4 rule complexity	constraint structure	per-constraint timing for the eleven fine log constraints and the fifteen controlled rule variants, on a 10 000-case sample
E5 memory	cases	E1 ladder up to the large-size threshold of 25 000 cases, under `tracemalloc`

derivation, and evaluation of all eleven constraints. *E2* varies trace length using both natural traces and synthetically lengthened traces produced by replicating an event sequence with shifted timestamps. Segmentation is disabled so that each lengthened trace remains a single evaluation unit. *E3* varies model size. *E4* measures per-constraint runtime for the fine log constraints and the controlled rule variants and relates it to the complexity components. *E5* repeats the E1 size ladder under `tracemalloc` to measure peak memory. Because `tracemalloc` affects runtime, its measurements are kept separate from E1. The size ladders, case samples, and replication counts are given in Table 5.4.

The protocol fixes the following conditions. Execution is single-threaded. Case samples are drawn with seed 42 and held fixed across repetitions, and one untimed warm-up pass precedes the first measurement. Each measurement is repeated and summarized by its median, with fewer repetitions at or above 25 000 cases in E1 and E5. E2, E3, and E4 use subsamples because they measure per-unit quantities that do not depend on the number of other cases in the log. E1 includes the complete log so that the main scaling result is based on a direct measurement rather than extrapolation from the fitted exponent.

Measurements are appended per unit and the runner supports resumption, so an interrupted run loses at most the measurement in progress. All experiments are executed on a single machine, whose configuration is recorded automatically with each run and reported in Table 5.5.

Table 5.5: Reference machine and software versions, recorded with every benchmark run.

Component	Configuration
CPU	Intel Core i5-10300H @ 2.50 GHz, 8 logical cores
Memory	15.5 GiB
Operating system	Linux `7.1.5`, x86-64, glibc `2.43`
Python	`3.14.6`
pandas	`3.0.1`

5.2 Results

The results are organized according to the six evaluation questions of Section 5.1.1. EQ1 and EQ2 are addressed in Section 5.2.1, EQ3 in Section 5.2.2, EQ4 in Section 5.2.3, EQ5 in Section 5.2.4, and EQ6 in Section 5.2.5.

5.2.1 Verdict Correctness and the Effect of Episode Scoping

The checker reproduces the prescribed verdict for all but two cases in the verdict catalogue. The two deviations correspond to the treatment of non-activated instances of unary negative templates described below. On the multi-episode log, episode-scoped evaluation recovers all planted verdicts, whereas case-scoped evaluation both hides existing violations and introduces violations that occur in no individual episode. These verdict differences also affect the estimated associations with the outcome.

Verdict correctness (EQ1). Table 5.6 reports the verdict results by template. Of the 95 cases covering all 24 implemented templates, 93 return the prescribed verdict according to the activation, correlation, time-window, and target-selection semantics of Definitions 3.19, 3.20 and 4.10. The two deviations occur in two different templates.

The two deviations concern the non-activation cases of Absence and AtMostOne. According to Definition 3.20, these cases are vacuously satisfied, whereas the checker reports them as satisfied rather than vacuous. For these unary negative templates, the absence of the relevant activity represents the compliant behavior, so treating the result as satisfied is consistent with their intended interpretation. The distinction nevertheless affects the reported vacuity rate and the population used for association analysis. In particular, `RTF-06` therefore has a reported vacuity rate of zero in Table 5.8, and its conformance and association results cover all episodes rather than only activated episodes. For the binary NotCoExistence, by contrast, the catalogue confirms that the checker follows Table 3.2: only the activation activity activates the constraint, so a trace containing neither activity, or only the target, is vacuous.

Episode scoping (EQ2). Figure 5.1 compares episode-scoped and case-scoped evaluation on the generated log. Episode-scoped evaluation reproduces every planted verdict. Case-scoped evaluation of the same events produces errors for cases containing multiple episodes, with two distinct error types.

Table 5.6: SE1. Verdict catalogue by template. *Cases* counts the hand-written test cases exercising the template, *Matching* those whose observed verdict equals the prescribed one, and *Deviations* the remainder. The two deviations are the non-activation cases discussed in the text.

Template	Cases	Matching	Deviations
Absence	2	1	1
Alternate Precedence	2	2	0
Alternate Response	2	2	0
Alternate Succession	3	3	0
At Most One	3	2	1
Chain Precedence	2	2	0
Chain Response	2	2	0
Chain Succession	3	3	0
Co-Existence	3	3	0
End	2	2	0
AtLeastOne	5	5	0
Init	2	2	0
Not Chain Precedence	2	2	0
Not Chain Response	2	2	0
Not Chain Succession	3	3	0
Not Co-Existence	4	4	0
Not Precedence	2	2	0
Not Responded Existence	2	2	0
Not Response	3	3	0
Not Succession	4	4	0
Precedence	9	9	0
Responded Existence	3	3	0
Response	24	24	0
Succession	6	6	0
Total	95	93	2

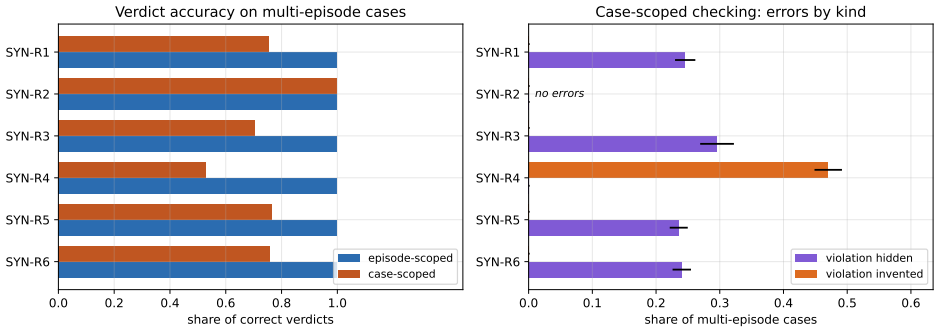


Figure 5.1: SE2, averaged over replications and restricted to cases carrying more than one episode. Left: share of correct verdicts under each scope. Right: the two kinds of error produced by case-scoped evaluation: violations hidden and violations invented, with the spread across replications. `SYN-R2` is the constraint carrying a time window, and `SYN-R4` is the unary constraint.

Ordering constraints without a time window can be *masked*. An activation in one episode may be satisfied by a target event from a later episode, causing an existing violation to be reported as satisfied. Masking affects between 23.6 % and 29.6 % of multi-episode cases for the four unwindowed constraints, reducing verdict accuracy to between 70.4 % and 76.4 %. The corresponding constraint with a time window is not masked because the temporal window excludes the later target, and its case-scoped accuracy remains 100 %. A temporal bound can therefore prevent this type of cross-episode fulfillment, but only for constraints that define such a bound. In the fine-log model of Table 5.3, only two of the eleven constraints use a time window.

The unary constraint produces the opposite error. When occurrences are pooled across episodes, their combined count may exceed the permitted limit even though every individual episode complies. Case-scoped evaluation therefore introduces a violation that occurs in no episode in 47.0 % of multi-episode cases, resulting in an overall accuracy of 53.0 %. Episode-scoped evaluation remains correct for all six constraints in every replication. Thus, case-scoped evaluation can both conceal existing violations and introduce violations that arise only from combining otherwise independent episodes.

Association recovery (EQ2). Figure 5.2 and Table 5.7 show how the choice of scope affects the estimated associations. Under episode-scoped evaluation, the planted risk ratios are recovered across the tested range. The median estimates are 2.04, 0.49, 3.02, 0.25, and 1.49, compared with planted values of 2, 0.5, 3, 0.25, and 1.5. Confidence-interval coverage ranges from 90 % to 100 %, compared with the nominal 95 %. The constraint planted at $RR = 1$ is significant in none of the 20 replications, while the four stronger effects are significant in every replication. The effect planted at 1.5 is detected in 75 % of replications.

Case-scoped evaluation distorts three of the five planted effects. The effect with a planted risk ratio of 3 is attenuated to 1.88, and its confidence interval contains the planted value in only 15 % of replications. The largest distortion occurs for the unary constraint, whose planted risk ratio of 0.25 is estimated at 0.69 and whose confidence interval contains the planted value in no replication. Because case-scoped evaluation introduces violations unrelated to the planted outcome association, power falls from 100 % to 40 % for both this constraint and the constraint planted at 0.5. The remaining two effects are recovered, showing that the distortion depends on the interaction between the constraint and the episode structure.

The two scopes use the same events, constraints, and statistical procedures; they differ only in the unit over which constraint verdicts are formed. The differences in the recovered associations therefore result from the choice of evaluation scope. The synthetic experiment thus shows that case-scoped evaluation can distort downstream statistical analysis even when the underlying event data and statistical method are unchanged.

Vacuity treatment. SE3 evaluates both vacuity treatments, and the two produce similar results on the generated log. Pooling vacuous episodes with satisfied ones gives risk-ratio estimates close to those obtained by excluding vacuous episodes, including 3.09 versus

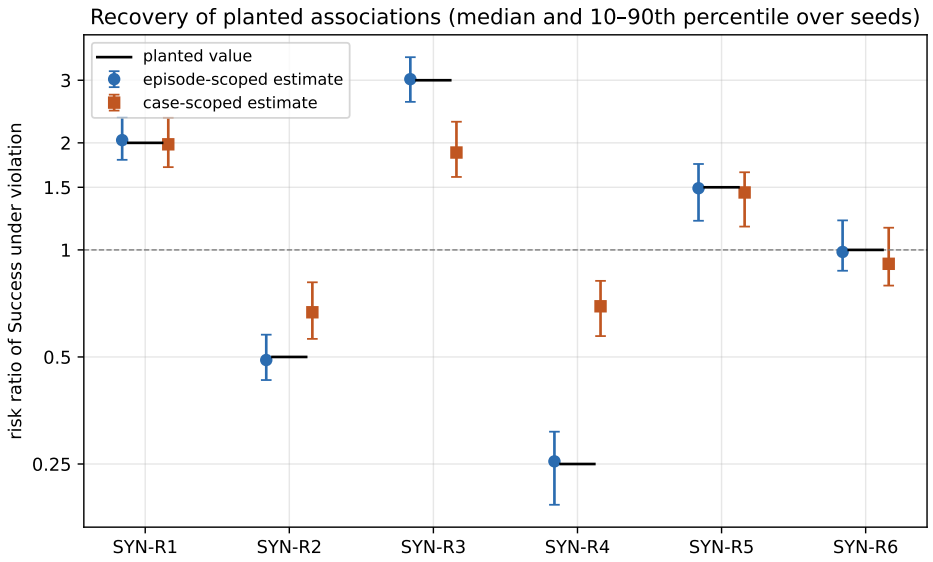


Figure 5.2: SE3. Estimated risk ratio of *Success* under violation against the planted value, per constraint and scope, on a logarithmic axis. Markers give the median over replications, bars the tenth to ninetieth percentile. `SYN-R6` is the constraint planted with no effect.

Table 5.7: SE3. Recovery of the planted risk ratio under each scope, over replications of the generated log. Coverage is the share of replications whose 95 % confidence interval contains the planted value. Power is the share clearing the Bonferroni threshold, which for `SYN-R6`, planted at $RR = 1$, is a false-positive rate.

		Episode-scoped			Case-scoped		
Rule	Planted RR	$\widehat{RR}$	Coverage	Power	$\widehat{RR}$	Coverage	Power
SYN-R1	2.00	2.04	95 %	100 %	1.98	100 %	100 %
SYN-R2	0.50	0.49	95 %	100 %	0.67	55 %	40 %
SYN-R3	3.00	3.02	100 %	100 %	1.88	15 %	95 %
SYN-R4	0.25	0.25	100 %	100 %	0.69	0 %	40 %
SYN-R5	1.50	1.49	90 %	75 %	1.45	95 %	40 %
SYN-R6	1.00	0.99	95 %	0 %	0.91	95 %	0 %

Table 5.8: Per-constraint verdicts over the 5000-episode sample. Activation is the share of episodes in which the constraint was activated. The violation rate is taken over activated episodes only.

ID	Satisfied	Violated	Vacuous	Activation	Violation rate
RTF-01	5000	0	0	100.0 %	0.0 %
RTF-02	3406	74	1520	69.6 %	2.1 %
RTF-03	1031	3969	0	100.0 %	79.4 %
RTF-03b	2332	1148	1520	69.6 %	33.0 %
RTF-04	2669	0	2331	53.4 %	0.0 %
RTF-05	1955	0	3045	39.1 %	0.0 %
RTF-06	5000	0	0	100.0 %	0.0 %
RTF-07	159	1	4840	3.2 %	0.6 %
RTF-08	33	4	4963	0.7 %	10.8 %
RTF-09	529	133	4338	13.2 %	20.1 %
RTF-10	1879	76	3045	39.1 %	3.9 %

3.02 for the constraint planted at 3. Confidence-interval coverage differs by at most one replication across the tested constraints. This agreement follows from the generator, which draws vacuity independently of the outcome. Under this construction, vacuous episodes have the same outcome distribution as satisfied episodes, so including them does not introduce an association bias.

The result does not imply that the two treatments are equivalent in general. It establishes that pooling is appropriate when vacuity is independent of the outcome. The fine-log results provide a case where this condition does not hold.

5.2.2 Conformance and Rule Quality on the Fine Log

On the fine log, the framework produces constraint-level conformance and outcome-association results using the same checker and statistical procedures as in the clinical study. Ordering and sanity constraints show no violations, while the two deadline constraints account for substantial deviation rates. Three constraints have associations with the outcome that remain significant after multiple-testing correction. The analysis demonstrates end-to-end applicability on a structurally different, non-clinical log; interpretation of guideline adherence is reserved for the ICU study of Chapter 6. Using Create Fine as the trigger on a random sample of 5000 cases produces 5000 episodes, of which 1629 (32.6 %) are labeled *Success*.

Table 5.8 shows the resulting constraint-level verdicts. The three ordering constraints and both sanity checks have zero violations. The two deadline constraints have the highest violation rates, with RTF-03 violated in 79.4 % of activated episodes and RTF-03b in 33.0 %. Activation rates vary substantially, from 100 % for Create Fine to 0.7 % for RTF-08. Reporting vacuity separately is therefore important. For RTF-08, pooling vacuous episodes with satisfied episodes would yield a satisfaction rate of 99.9 %, largely because prefecture appeals rarely reach a result.

The association results are reported in Table 5.9. With eleven tests, the Bonferroni

Table 5.9: Outcome association per constraint, vacuous episodes excluded. Exposure is violation of the constraint. *RR* is the risk ratio of the *Success* outcome under violation, so $RR < 1$ marks a protective constraint. Tables with a zero cell use a continuity correction of 0.5 and are marked †. Constraints with no violated episode, or with no *Success* episode among the activated ones, admit no estimate and are marked n/e. Significance is assessed against the Bonferroni threshold 4.5×10^{-3}.

ID	Violated	Satisfied	*RR*	95 % CI	*p* (Fisher)
RTF-01	0	5000	n/e	n/e	1.0
RTF-02†	74	3406	0.21	[0.01, 3.28]	0.17
RTF-03†	3969	1031	847	[53.0, 13 539]	3.6×10^{-204}
RTF-03b	1148	2332	221	[30.9, 1584]	6.7×10^{-53}
RTF-04	0	2669	n/e	n/e	1.0
RTF-05	0	1955	n/e	n/e	1.0
RTF-06	0	5000	n/e	n/e	1.0
RTF-07†	1	159	26.7	[1.50, 475]	1.0
RTF-08†	4	33	6.80	[0.15, 306]	1.0
RTF-09	133	529	21.4	[13.5, 34.1]	2.3×10^{-66}
RTF-10	76	1879	n/e	n/e	1.0

threshold is $\alpha/m = 4.5 \times 10^{-3}$. Three constraints are significant at this threshold: RTF-03, RTF-03b, and RTF-09.

Four constraints have no violated episodes, so no comparison between violating and satisfying episodes is possible and no effect size can be estimated. Although the continuity correction would produce a finite ratio for such tables, that value would result from the correction rather than the observed data; these rows are therefore reported as n/e. RTF-10 is not estimable for the complementary reason: none of its activated episodes is labeled *Success*. RTF-07 provides a related example: its single violated episode produces a confidence interval excluding 1, while Fisher's exact test gives $p = 1.0$. The interval uses corrected cell counts, whereas the exact test does not, so the test result is the appropriate basis for significance.

RTF-10 illustrates an entanglement between constraint and outcome. It is activated by Send for Credit Collection, which is itself part of the *Failure* definition, so every activated episode fails regardless of whether the constraint is satisfied. Its outcome is therefore determined by activation rather than by compliance, and no association can be estimated. All three significant associations are positive. RTF-03, RTF-03b, and RTF-09 occur along the escalation path, which is reached primarily by fines that are not settled immediately. Their association with *Success* therefore reflects differences between process populations rather than necessarily beneficial behavior. RTF-09 is additionally a synthetic data-aware constraint used to exercise the data-condition mechanism (Section 5.1.4) and has no normative interpretation.

These results illustrate that a constraint can be a valid representation of a procedural rule without being a useful standalone indicator of outcome quality. The synthetic

Table 5.10: Effect of the vacuity treatment on the association results, for the constraints where the two modes diverge. *RR* is again oriented toward *Success* under violation.

	Activated only		Vacuous pooled as satisfied	
ID	*RR*	*p*	*RR*	*p*
RTF-02	0.21	0.17	0.02	3.1×10^{-13}
RTF-03b	221	6.7×10^{-53}	0.24	1.0×10^{-93}
RTF-07	26.7	1.0	0.77	1.0
RTF-08	6.80	1.0	0.31	0.31
RTF-09	21.4	2.3×10^{-66}	2.32	5.2×10^{-22}
RTF-10	n/e	1.0	0.02	1.0×10^{-13}

experiment shows that the association procedure recovers planted effects when the data-generating process is controlled; the fine log shows that observed associations can instead reflect the structure of the process itself.

Table 5.10 shows that the treatment of vacuous episodes can substantially change the association result. For RTF-03b, excluding vacuous episodes gives $RR = 221$, whereas pooling them with satisfied episodes gives $RR = 0.24$. The difference is driven by the 1520 episodes that are settled before a Send Fine event. These episodes form a distinct population with a different outcome distribution. Pooling them with satisfied episodes therefore combines non-activation with compliance and changes the estimated association.

The synthetic results show that the two vacuity treatments agree when vacuity is independent of the outcome. The divergence observed for RTF-03b demonstrates that this condition does not hold here. Vacuous episodes should therefore be excluded from the primary association analysis, with pooled treatment reported as a sensitivity analysis.

This answers EQ3. On a structurally different, non-clinical log, the framework produces constraint-level conformance results and outcome associations using the same evaluation procedure as the clinical study. Their interpretation remains dependent on the process context, as illustrated by the escalation-related associations.

5.2.3 Sensitivity to the Trigger Specification

The trigger specification affects which cases and events enter the episode-based analysis. The candidate triggers differ in coverage, retained events, episode granularity, and availability of information required for outcome derivation. A suitable trigger should therefore provide broad case coverage, retain the information required for conformance and outcome evaluation, and produce episodes of sufficient length to contain the relevant behavior. The trigger sweep over the six candidate activities of Section 5.1.2, using a random sample of 20 000 cases, evaluates these properties.

Figure 5.3 shows case coverage. It decreases along the process from 100 % for Create Fine to 46.3 % for Payment and 0.3 % for Appeal to Judge. Cases without the selected trigger produce no episode and are therefore excluded from the analysis. For Payment,

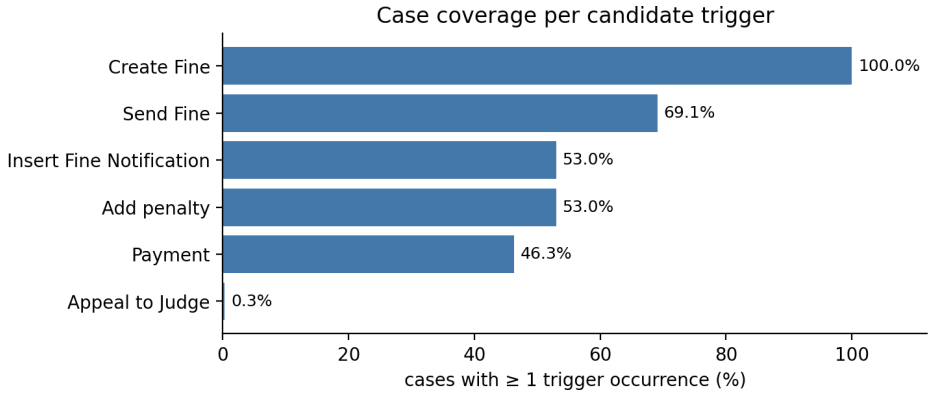


Figure 5.3: Share of cases containing the candidate trigger at least once. Cases without it contribute no episode.

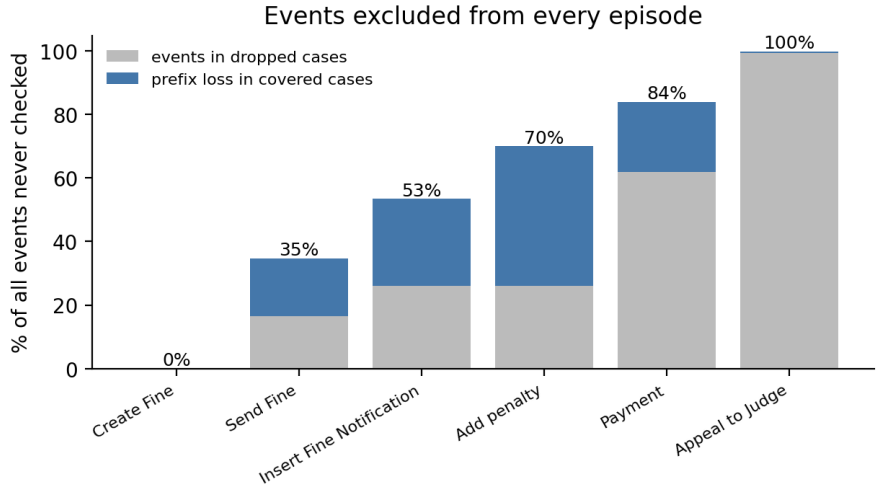


Figure 5.4: Events excluded from every episode, as a share of all log events, split by source. One part covers events in cases dropped for lack of the trigger, the other prefix events preceding the first trigger in covered cases.

this restricts the analysis to fines that received at least one payment, making the analyzed population dependent on process progression and potentially on the outcome.

Figure 5.4 shows the share of log events excluded from all episode windows, separated into events belonging to dropped cases and prefix events preceding the first trigger in covered cases. With Create Fine, the trigger is the first event of every case, so no events are excluded. Later triggers increase both forms of data loss, reaching 84 % for Payment. Pre-trigger events in covered cases can partly be recovered through the pre-context window of Section 4.2.1, whereas events in dropped cases cannot.

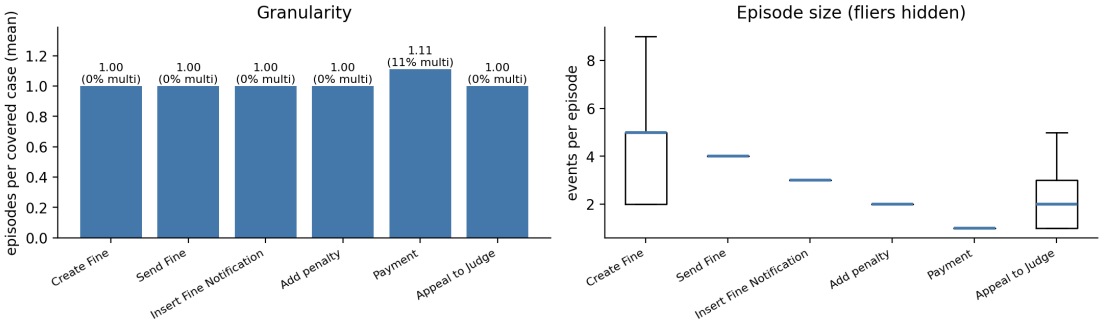


Figure 5.5: Left: mean episodes per covered case, annotated with the share of cases yielding more than one episode. Right: distribution of events per episode, outliers suppressed.

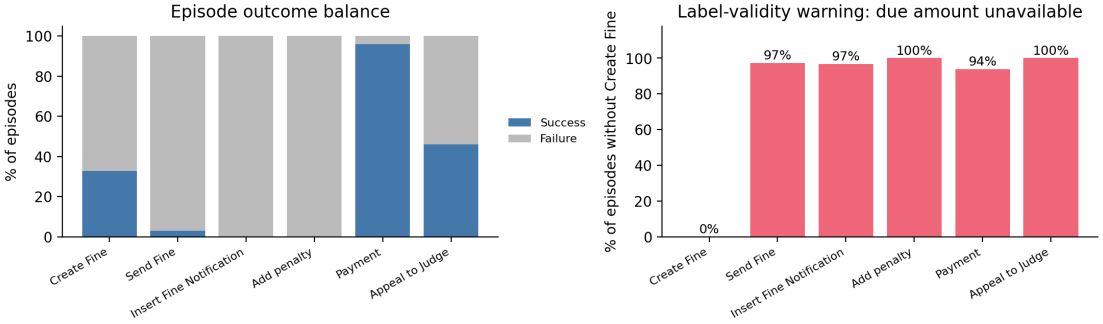


Figure 5.6: Left: outcome balance per trigger. Right: share of episodes for which the amount owed is unavailable because the episode lacks the Create Fine event, causing the outcome label to be dominated by *Success*.

Figure 5.5 shows the effect on episode granularity. Median episode size decreases along the process, from five events under Create Fine to one under Payment. Episodes containing a single event cannot provide the event ordering required by binary constraints, so late triggers reduce not only the amount of retained data but also the ability of an episode to express the modeled behavior.

The effect on retained information is shown in Figure 5.6. For every trigger except Create Fine, the episode lacks the Create Fine event required to determine the amount owed in 94 % to 100 % of cases. These missing values are assigned a near-zero due amount, causing the corresponding episodes to be labeled *Success*. The resulting success share is 96 % for Payment, compared with 32.9 % for Create Fine. This reflects the loss of information required for outcome derivation rather than a difference in process performance.

Create Fine provides the highest case coverage, retains the complete case prefix, and preserves the information required for outcome derivation. Its limitation is episode recurrence. Every case contains exactly one Create Fine event (`RTF-01`), so the resulting

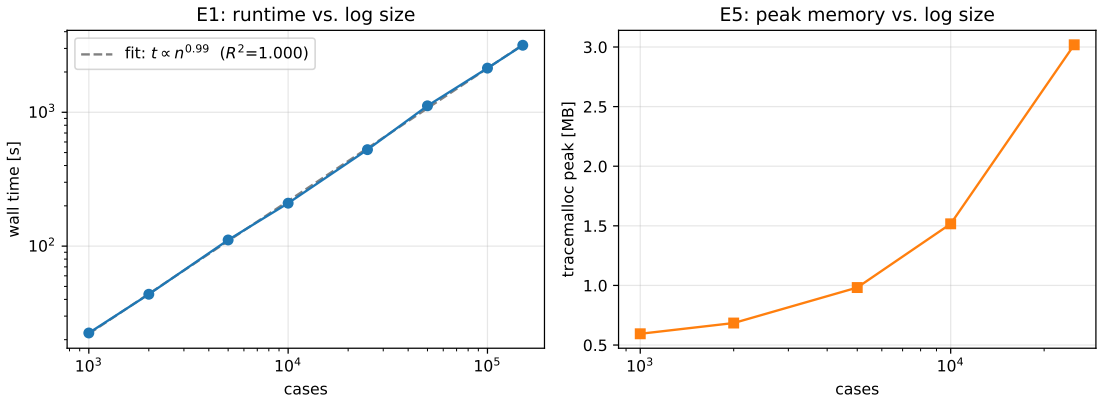


Figure 5.7: E1 and E5. Left: wall time against the number of cases, log-log, with the fitted power law. Right: `tracemalloc` peak against the number of cases.

segmentation contains one episode per case and does not exercise the multi-episode setting. The other candidate triggers do not provide a suitable alternative on this log. Payment is the only trigger that recurs, but it covers only 46.3 % of cases, yields more than one episode in 11 % of covered cases, and produces a median episode of one event together with substantial loss of outcome information.

For the fine log, Create Fine is therefore the most suitable trigger under the evaluated criteria. The clinical instantiation in Chapter 6 uses InvasiveVentilationStarted, which is universal and initial within a ventilated ICU stay and can recur on reventilation. The two logs therefore provide complementary settings: the fine log permits a controlled comparison of trigger properties, while the clinical log exercises recurring episode segmentation.

5.2.4 Scalability

Runtime and memory scale predictably with the number of cases, trace length, and model size. Runtime is approximately linear in the number of cases, sublinear over the tested range of synthetically lengthened traces, and approximately linear in the number of constraints. Memory remains largely independent of the number of cases at the tested scale.

Figure 5.7 shows the scaling with the number of cases. The fitted exponent is 0.99 at $R^2 = 1.000$ over the full ladder from 1000 cases to the complete log. Throughput remains between 45 and 48 cases per second, and the complete log of 150 370 cases is processed in 3168 s, or approximately 53 min, in a single-threaded run. Peak allocation increases from 0.59 MB at 1000 cases to 3.02 MB at 25 000 cases, while resident set size remains at 385 MB. The latter is dominated by the interpreter and loaded log rather than by the checker. The small spread between repetitions at smaller sizes supports using fewer repetitions for the most expensive configurations.

Trace length is evaluated separately because the natural fine log contains only a limited

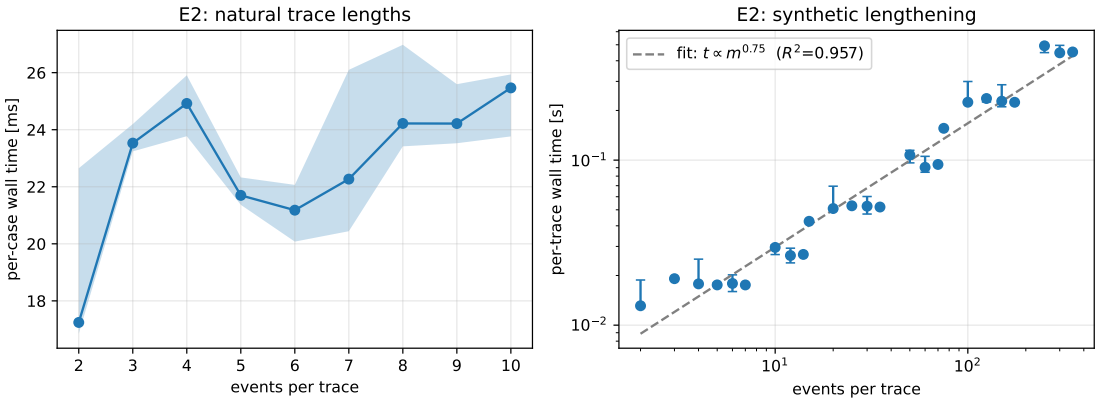


Figure 5.8: E2. Left: per-case wall time over the natural trace-length range, shaded by the interquartile spread. Right: per-trace wall time over synthetically lengthened traces, log-log, with the fitted power law.

range of trace lengths. Figure 5.8 shows that the natural range of two to twelve events per trace produces no clear runtime trend: per-case wall time remains between 17 and 25 ms. At these lengths, the fixed cost of setting up an evaluation dominates the cost of scanning the events. The synthetic lengthening extends the range to a mean of 187 events per trace and produces a clearer relationship, with a fitted exponent of 0.75 at $R^2 = 0.957$. Per-trace cost increases from 17 ms at a mean of 3.7 events to 438 ms at a mean of 187 events. The resulting sublinear scaling reflects the amortization of fixed per-trace overhead over longer traces; it does not imply sublinear evaluation work with respect to activation instances.

Figure 5.9 shows the effect of model size. On the 10 000-case sample, each additional constraint adds approximately 14.47 s, corresponding to 1.45 ms per constraint and case, with $R^2 = 0.949$. The intercept is 48 s, close to the 49.9 s measured for a single constraint. This indicates a substantial fixed pipeline cost from segmentation, outcome derivation, and result assembly. Deviations from the fitted line occur at $k = 3$ and $k = 10$, where the two most expensive constraints enter the model. The constraint at $k = 3$ uses a time window, while the constraint at $k = 10$ is the only one that accesses event attributes. Their higher costs are examined in Section 5.2.5.

These results answer EQ5. Runtime is predictable across the tested dimensions, while memory growth remains small relative to the loaded log. The measured costs therefore support the use of the framework at the data and model sizes considered in this thesis.

5.2.5 Determinants of Per-Constraint Evaluation Cost

The controlled experiments examine which structural properties of an MP-Declare constraint determine its evaluation cost. The results identify two main factors: template complexity and the presence of data access. Additional predicates and attributes have little effect once data access is already required.

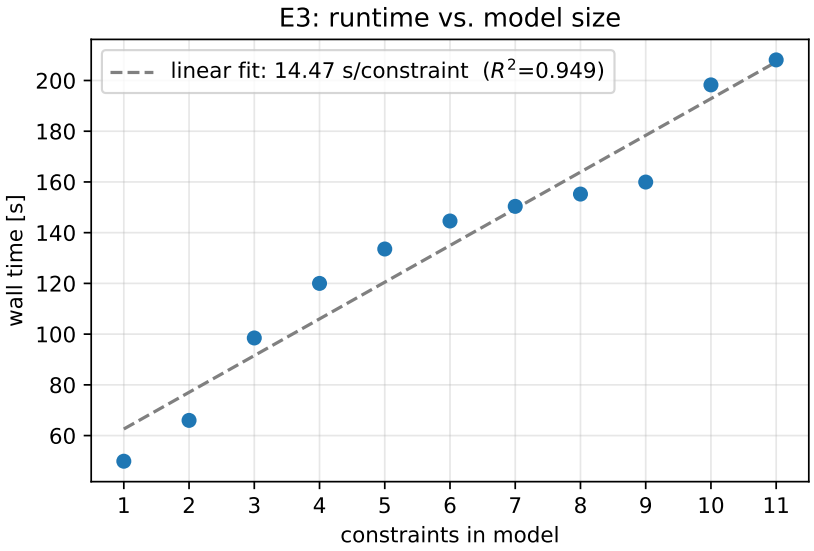


Figure 5.9: E3. Wall time on a fixed 10 000-case sample against the number of constraints in the model, with a linear fit.

The template sweep shows a clear increase in runtime with template grade. Runtime increases from 572 µs for the unary existence constraint (grade one) to 1.83 ms for the unordered and 2.90 ms for the ordered binary template over the same activities (grades two and three). At grade four, the chain variant requires 3.62 ms, whereas the alternate variant requires 2.89 ms and is therefore no more expensive than the ordered binary template. Its additional check bounds the candidate targets by the timestamp of the next activation instead of introducing a further scan, so grade four overestimates its cost. The data-aware sweeps behave differently. Across the clause sweep, increasing the number of atomic predicates from one to five at a fixed single attribute changes runtime from 6.11 ms to 6.15 ms, a difference below 1 %. Across the attribute sweep, increasing the number of distinct attributes from one to five produces no systematic change (Figure 5.10). All data-aware variants remain near the 6.1 ms level. The results therefore indicate that the main additional cost is associated with entering the data-aware evaluation path, while additional predicates and attributes contribute little within the tested range.

The complexity score $C(r)$ of Section 5.1.5 provides a useful ranking of the constraints. It correlates strongly with measured runtime across the eleven fine-log constraints and fifteen controlled rule variants, with $\rho = 0.91$ for the pooled set and $\rho = 0.80$ for the fine-log constraints alone (Figure 5.10). The score therefore distinguishes relatively cheap constraints from more expensive data-aware and higher-grade constraints.

The additive score is less suitable as a decomposition of absolute runtime. Because the cost of data access appears to arise primarily when the first data condition is evaluated, assigning a separate linear cost to every predicate or attribute does not reflect the observed sweep results. A modified score that represents data access as a single indicator

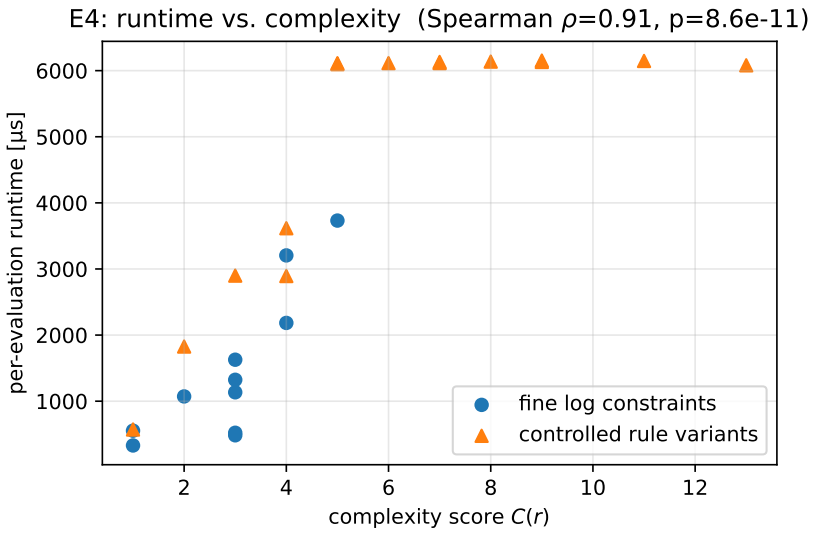


Figure 5.10: E4. Per-constraint evaluation cost as a function of the complexity score, for the fine log constraints and the controlled rule variants. Median per-evaluation runtime against the score; high-grade, data-aware constraints dominate, while the remaining constraints are substantially cheaper.

is therefore more consistent with the measurements:

$$C'(r) = w_{template} + \mathbb{1}[n_{clauses} > 0] + t_{time}, \tag{5.1}$$

where the indicator captures the presence of data-dependent evaluation. The controlled experiments support this formulation as a relative-cost measure, but not as a direct predictor of absolute runtime.

This answers EQ6. Template grade and the presence of data access are the main structural determinants of per-constraint evaluation cost in the tested configurations. The complexity score can therefore be used to anticipate relative cost before execution, while absolute runtime still depends on the implementation and its fixed evaluation overhead.

5.3 Threats to Validity

This section discusses threats to the validity of the evaluation, organized by internal, external, construct, and conclusion validity. Reproducibility is addressed separately. Threats specific to the clinical instantiation are discussed in Section 6.5.

Internal Validity The outcome label for the fine log is derived rather than recorded. *Success* is reconstructed from payment, expense, and penalty attributes, and an episode without Create Fine yields a degenerate amount owed that biases the label toward *Success*.

Label validity is therefore assessed under EQ4 rather than assumed. The configured trigger avoids this degenerate case because every case begins with Create Fine.

Episode segmentation and day-resolution timestamps also discard temporal detail. A trigger occurring after the initial activity leaves the case prefix outside the evaluation window; the pre-context window of Section 4.2.1 can recover such information for state-dependent conditions but not for activations. Day-resolution timestamps collapse repeated triggers occurring on the same date and make intra-day order unrecoverable. The evaluation therefore uses week-scale temporal windows in the fine-log constraints, and truncated episodes are marked explicitly.

External Validity The empirical evaluation uses one real-world event log, so results on process behavior and rule associations cannot be generalized to other processes or institutions without qualification. The synthetic experiments complement this limitation by varying structural properties in a controlled setting, but they test the method rather than its behavior across additional domains. The declarative model is likewise specific to the fine-processing procedure: a rule-quality verdict describes the encoded constraint, not the underlying norm in general.

The cost results are specific to the Python and pandas implementation and the hardware listed in Table 5.5. Absolute runtimes therefore do not generalize directly. The observed scaling behavior and relative ordering of constraint costs provide more transferable evidence, although these too remain bounded by the tested configurations.

Construct Validity The synthetic logs of Section 5.1.3 provide ground truth because the generator constructs it. They therefore establish that the checker implements the specified semantics of Definition 3.20 and that the association stage can recover planted effects under the assumptions built into the generator. They do not establish that these assumptions hold in real processes, and the correctness result covers only the constraint configurations represented in the catalogue.

The episode-scoping experiment likewise evaluates the behavior of the two scopes on logs whose episodes are constructed to carry independent verdicts. The absence of the same effect in the fine log is a property of that log rather than evidence against the mechanism. Similarly, synthetic trace lengthening replicates an event sequence with shifted timestamps rather than modelling genuinely longer processes. Its sublinear scaling therefore characterizes the implementation under the tested construction, not necessarily the behavior of real processes with hundreds of events per case.

The complexity score is a heuristic for ranking evaluation cost rather than a validated additive cost model. Section 5.2.5 shows that it ranks constraints reliably, although two of its four components have no measurable marginal effect in the tested configurations. The calibrated score C' is based on the synthetic sweeps because the eleven real constraints contain only one data-aware rule and therefore cannot distinguish the relevant variants.

Finally, associations between constraint violations and outcomes are observational. An association may reflect confounding rather than an effect of the violated rule. The results therefore show that a constraint verdict is associated with the outcome in the

evaluated population, not that enforcing the constraint would improve outcomes. The EQ4 segmentation metrics similarly characterize the properties of the resulting data rather than establish that a trigger defines the semantically correct unit of analysis. That judgment remains domain dependent.

Conclusion Validity Fisher's exact test combined with Bonferroni correction is conservative, so a non-significant result may reflect limited statistical power rather than the absence of an association. Episodes may also be correlated because multiple episodes can originate from the same case. Case and episode counts are therefore reported together to make the degree of clustering explicit.

Conversely, the large sample size of the fine log allows very small effects to become statistically significant. Associations are consequently reported with effect sizes and confidence intervals, and interpretation is based primarily on the estimated risk ratio rather than on statistical significance alone.

Reproducibility All sampling uses a fixed seed, and the same samples are reused across repetitions so that repetition differences primarily reflect measurement variability. Models, activity mappings, and analysis parameters are stored as versioned configuration files. Each benchmark records the machine configuration, library versions, and git commit under measurement, while results are written per measurement so interrupted runs can be resumed without mixing experimental conditions.

The synthetic experiments require no external data because their generators and seeds are versioned and the logs are reconstructed for each run. The fine log is publicly available, allowing the real-log evaluation to be reproduced when the corresponding implementation and configuration are available.

Taken together, the evaluation establishes four properties of the framework. First, the checker reproduces the prescribed constraint semantics across the tested catalogue and configurations. Second, episode-scoped evaluation preserves planted episode-level verdicts, whereas case-scoped evaluation can both mask violations and introduce violations across episode boundaries. Third, the association stage recovers planted effects when the underlying verdicts are correct. Fourth, the computational cost is predictable at the evaluated scale and is primarily determined by template grade and the presence of data access.

These results provide the method-level validation and computational evidence for the framework. The subsequent case study therefore focuses on the domain-specific question of how the same episode-based conformance analysis behaves when applied to a clinical process.

6 Case Study on the ICU Weaning Clinical Guideline in MIMIC-IV

This chapter applies the framework of Chapter 4 to the weaning of ICU patients from mechanical ventilation, using real-world data from the MIMIC-IV database. Section 6.1 motivates weaning as a case for episode-scoped conformance checking, and Section 6.2 introduces the necessary medical background. Section 6.3 formalizes the weaning guideline as a set of MP-Declare constraints. Section 6.4 reports the results, covering violation frequencies and their causes, the association of violations with the weaning outcome, and the effect of evaluating episodes rather than whole stays. Section 6.5 closes with the limitations of the case study.

6.1 Motivation

ICUs are complex healthcare environments where approximately 33% of patients require mechanical ventilation (MV) [28]. *Weaning*, the process of transitioning patients off ventilatory support, accounts for nearly 42% of total MV duration [29]. Ineffective weaning management is associated with substantial risks. Delayed weaning increases exposure to complications such as ventilator-associated pneumonia, while premature extubation is linked to ICU mortality rates of 45.5% [20, 29]. Clinical guidelines standardize weaning through readiness assessments and SBTs, and adherence to these protocols improves patient outcomes [10, 89].

Despite this evidence, monitoring compliance at scale remains an unresolved challenge, which makes weaning an especially demanding instance of the general problem addressed in this thesis. A persistent gap separates recommended guidelines from clinical practice. Yet not all deviations are erroneous, as some represent justified adaptations to patient-specific requirements [42, 49]. Traditional adherence metrics, such as length of stay or mortality, are insufficient for process-level analysis, since they do not capture whether the prescribed sequence, timing, and logic of care were followed [64]. Current ICU adherence assessment therefore relies heavily on manual chart reviews and retrospective audits, which are resource-intensive and subject to observer bias [62, 76]. While rule-based alignment methods offer some automation, they often lack the contextual sensitivity required to incorporate patient-specific factors such as comorbidities or clinician rationale, and consequently fail to distinguish *warranted* deviations (patient-centered care) from *unwarranted* deviations (potential harm) [41, 44, 72].

Weaning further exhibits the multi-perspective and episodic structure that motivates the framework. Its guidelines interleave event ordering, data conditions (physiological

thresholds), and temporal constraints (trial durations and re-attempt deadlines) [10], which is the combination of perspectives that MP-Declare is designed to express [16]. None of it is captured by procedural models that yield aggregate fitness scores over entire ICU stays. Moreover, a single admission may contain several distinct weaning attempts, so that global evaluation at the level of the full ICU stay conflates multiple SBTs into one trace and obscures deviations that are meaningful only at the level of an individual weaning episode. These characteristics make ICU weaning in MIMIC-IV a natural setting in which to instantiate and evaluate the episode-scoped, multi-perspective conformance-checking framework developed in Chapter 4.

6.2 Medical Background: Ventilator Weaning

The clinical process studied in this thesis is the *weaning* of intensive care unit (ICU) patients from invasive mechanical ventilation. As normative reference we use the practical weaning guideline by Akella et al. [10]. We read it as three consecutive phases: the assessment of a patient's *readiness* to wean, the execution of a *spontaneous breathing trial*, and, upon trial failure, a structured workup and treatment of the *reversible causes* of that failure. This section introduces the medical terminology underlying these phases and explains the role each concept plays in the weaning process. The encoding of the resulting rules as MP-Declare constraints is part of our approach and is described in later chapters.

6.2.1 Invasive Mechanical Ventilation and Weaning

Invasive mechanical ventilation is a form of life support in which a machine (the ventilator) fully or partially takes over a patient's breathing [88]. It requires *intubation*, the placement of an endotracheal tube into the trachea, through which the ventilator delivers gas to the lungs. The removal of this tube is called *extubation*. Two ventilator settings referenced throughout the guideline are the *fraction of inspired oxygen* (FiO_2), the oxygen concentration of the delivered gas (21 % in room air, up to 100 %), and the *positive end-expiratory pressure* (PEEP, in cmH_2O), a baseline airway pressure maintained at the end of expiration to keep the alveoli open [43, 90]. High FiO_2 or PEEP requirements indicate that a patient still depends substantially on ventilatory support.

While often life-saving, prolonged invasive ventilation carries its own risks (Section 6.1), whereas removing support too early risks extubation failure and reintubation [27, 60, 61]. *Weaning* denotes the process of progressively reducing and ultimately discontinuing ventilatory support, culminating in extubation. This trade-off between weaning too late and weaning too early motivates the guideline's central tenet. Readiness to wean should be assessed early and systematically, as soon as 48 h to 72 h after ICU admission, and re-evaluated regularly thereafter [10].

6.2.2 Readiness Assessment

The *readiness assessment* determines whether a patient is physiologically stable enough to attempt spontaneous breathing. It is not a single clinical action but a conjunction of criteria, each probing a different organ system, that must all be satisfied before a spontaneous breathing trial may be initiated. The criteria adopted from Akella et al. [10] are the following:

- **Resolution of the underlying illness.** The condition that necessitated ventilation must be improving. Weaning attempts during the acute phase of the illness are futile and potentially harmful.
- **Adequate oxygenation.** The patient must maintain sufficient blood oxygen levels at moderate levels of support, operationalized as a ratio of arterial oxygen partial pressure to inspired oxygen fraction (PaO_2/FiO_2) of at least 150, or a peripheral oxygen saturation (SpO_2, measured non-invasively by pulse oximetry) of at least 90 % on $FiO_2 \leq 40\,\%$ and PEEP $\leq 5\,\mathrm{cmH_2O}$. The PaO_2/FiO_2 ratio is the standard bedside index of the lungs' gas-exchange capacity [88]. Requiring it at low FiO_2 and PEEP ensures that oxygenation is intrinsic rather than an artifact of high ventilatory support.
- **Arterial pH above 7.25.** The acidity of arterial blood reflects the balance between carbon dioxide elimination and metabolic state. Marked acidemia (low pH) increases the respiratory drive and workload, making spontaneous breathing unsustainable.
- **Hemodynamic stability.** The patient must not be in shock. Circulatory support with *vasopressors* (drugs such as norepinephrine that raise blood pressure) must be absent or at a low dose. The guideline states this criterion qualitatively and gives no numeric cutoff. Because several vasopressor agents exist, we compare doses on a common scale as the *norepinephrine-equivalent dose* [39] and operationalize "low dose" as at most $0.1\,\mu\mathrm{g/kg/min}$, the conventional lower stratum in intensive care dosing studies. An unstable circulation cannot sustain the increased cardiac workload that spontaneous breathing imposes.
- **Ability to initiate inspiratory effort.** The patient must trigger breaths on their own, evidenced by a positive *spontaneous tidal volume*, i.e., the gas volume moved by patient-initiated (rather than ventilator-initiated) breaths. Without spontaneous respiratory drive, a breathing trial is meaningless.
- **Hemoglobin of at least $7\,\mathrm{g\,dL^{-1}}$.** Hemoglobin is the oxygen-carrying protein of the blood. Severe anemia limits oxygen delivery to the respiratory muscles and other tissues, undermining the tolerance of spontaneous breathing.
- **Core temperature of at most $38\,^{\circ}\mathrm{C}$.** Fever increases metabolic oxygen consumption and carbon dioxide production, and thereby the ventilatory demand placed on the patient.

- **Adequate mental status.** The patient must be awake and easily arousable. Sedation depth and agitation are quantified at the bedside by ordinal instruments such as the Richmond Agitation–Sedation Scale (RASS) [84] and the Riker Sedation–Agitation Scale (SAS) [79]. The guideline itself states the criterion qualitatively. A patient who is deeply sedated cannot protect their airway after extubation, whereas severe agitation precludes a controlled trial.

The importance of the readiness assessment for the weaning process is that it acts as a *gate*. A spontaneous breathing trial started while one of these criteria is unmet exposes the patient to a trial that is likely to fail, which fatigues the respiratory muscles and delays subsequent attempts. The guideline correspondingly requires a stable, non-fatiguing form of support after every failed trial [10]. In process terms, every trial start should therefore be preceded by recent measurements documenting that each criterion was satisfied.

6.2.3 Spontaneous Breathing Trial

The *spontaneous breathing trial* (SBT) is the diagnostic centerpiece of the weaning process. It is a time-limited test during which ventilatory support is reduced to a minimum and the patient breathes essentially on their own while remaining intubated and monitored.

During the trial, the patient is continuously monitored for signs of intolerance, and the guideline specifies bedside criteria whose occurrence terminates the trial as failed [10]. Monitoring these intra-trial criteria requires continuous bedside observation and is out of scope for this thesis. Instead, we adopt the outcome-based definitions used in the weaning literature. *Weaning success* is defined by extubation and the absence of ventilatory support 48 h after extubation, whereas *weaning failure* is defined as the inability to pass an SBT, the need for reintubation within 48 h following extubation, or death within the same period [10, 14].

The trial outcome dictates the subsequent process. A *successful* SBT is followed by evaluation for extubation. A *failed* SBT is terminated immediately, the patient is returned to full ventilatory support to rest the respiratory muscles, and the trial is repeated no earlier than the *following day*. Same-day retries risk trialing fatigued muscles, whereas longer delays unnecessarily prolong ventilation [10].

6.2.4 Management of Weaning Failure

A failed SBT is not merely a delay but a diagnostic finding. The guideline treats weaning failure as a symptom of an underlying, frequently *reversible* cause that must be sought and corrected before the next attempt [10]. The guideline organizes these causes and their management into pathways, each pairing a diagnostic step with a stabilizing treatment. Of the categories it lists, we consider those that are observable in charted ICU data and leave aside diaphragm dysfunction, cognitive dysfunction, and nutritional status, whose assessment relies on bedside measurements that are not routinely recorded:

- **Increased airway resistance**, e.g., due to bronchospasm or secretions, is treated with inhaled *bronchodilators*, drugs that relax the airway smooth muscle.

- **Reduced respiratory compliance** due to fluid overload is treated with *diuretics* (e.g., furosemide), which promote fluid excretion and restore compliance.
- **Weaning-induced cardiac dysfunction**, where the shift to spontaneous breathing raises the load on the left ventricle, is treated with *afterload-reducing* agents (e.g., nitroglycerin, beta-blockers).
- **Electrolyte derangements**, i.e., deficiencies of potassium, magnesium, or phosphate and acid–base disturbances that impair respiratory muscle function, are identified by an *arterial blood gas* analysis together with an electrolyte panel and corrected by targeted *electrolyte replacement*.
- **Endocrine dysfunction**, i.e., adrenal insufficiency or hypothyroidism causing generalized weakness, is detected by cortisol and thyroid function tests and treated by *hormonal replacement* (corticosteroids or thyroid supplementation), though the guideline reports the evidence for the thyroid pathway as anecdotal.

Two aspects of this phase are relevant from a process perspective. First, each failed trial creates an obligation. Within a bounded time window after the failure, the relevant diagnostic tests should be drawn and at least one stabilizing intervention should be documented, so that the reversible cause is addressed before the next trial. Second, the pathways impose an order. Replacement therapies presuppose the corresponding diagnostic test, e.g., electrolyte correction should follow, not precede, laboratory assessment. Both the temporal obligations and the orderings map naturally onto the constraint templates introduced in Section 3.3, which is precisely how they are operationalized in this thesis.

6.2.5 Operationalization in MIMIC-IV

The concepts introduced above are clinical notions. To check conformance against real ICU data, each must be grounded in what is actually recorded. The dataset used in this thesis is MIMIC-IV [47], a large database of de-identified ICU stays from the Beth Israel Deaconess Medical Center, available under a credentialed-access licence. MIMIC-IV does record events such as "invasive ventilation started" or "breathing trial begun", but these are clinician-reported entries whose timestamps, ordering, and values are not consistently reliable. Charting in critical care databases is known to carry timing error and temporal uncertainty [38], and in our data such entries are frequently missing or contradict the concurrently charted physiological record. A more dependable signal is the periodically charted *ventilator settings*, including the active *ventilator mode*, i.e., the algorithm by which the machine delivers or assists breaths. Phases of invasive ventilation and spontaneous breathing trials are therefore *reconstructed* from the sequence of charted modes instead of taken from the reported events. This subsection describes that reconstruction, which constitutes the domain-specific event-log preprocessing that the method assumes as input (Chapter 4). It maps raw records to activity labels, carries the physiological attributes that the constraint conditions reference, and materializes the ventilator-setting transitions as first-class value-change events.

From ventilator modes to weaning states. Each charted ventilator mode is assigned to one of two operational *weaning states*, reflecting the binary distinction that drives the weaning process:

- **Full Support:** modes with a mandatory or controlled breath component, in which the ventilator performs the work of breathing. This covers volume- and pressure-controlled ventilation (e.g., CMV, PCV), synchronized intermittent mandatory ventilation (SIMV), and airway pressure release ventilation (APRV). A patient in such a mode is under invasive mechanical ventilation in the sense of Section 6.2.1.
- **SBT Candidate:** purely spontaneous or pressure-supported modes, in which the patient initiates every breath and receives at most a low inspiratory assist. This covers pressure support ventilation (PSV), continuous positive airway pressure (CPAP) with or without pressure support, and spontaneous modes (SPONT). These are exactly the modalities in which an SBT is conducted (Section 6.2.3). Combination modes in which the patient breathes on pressure support are classified accordingly. The complete mode-to-state mapping is listed in Appendix A.

Modes indicating non-invasive or ambient conditions (e.g., non-invasive ventilation (NIV), Standby) are also mapped to *SBT Candidate* rather than to a separate "liberated" state, for two reasons. First, liberation from the ventilator manifests in MIMIC-IV as the *absence* of further charting and not as a positively recorded mode, so a liberation state defined over ventilator settings would systematically miss most true liberations. Second, these modes are ambiguous. A ventilator on standby may accompany a transport with the endotracheal tube still in place, and non-invasive ventilation occurs both after extubation and before intubation. Liberation is therefore treated as an *outcome* inferred from downstream evidence (extubation without subsequent reventilation), not as a ventilator mode. Charted timepoints whose mode cannot be classified are ignored for state transitions.

Value-change events from ventilator settings. The weaning state introduced above is not itself an event but a setting whose value evolves over the stay, and the normatively relevant occurrences are its *transitions*. A reduction of ventilatory support opens a breathing trial, and an increase back to full support marks its failure. Because Declare templates are predicated on the occurrence of labeled events and not on differing attribute values across records (Chapter 4), these transitions must be promoted to first-class *value-change events* before conformance can be checked. Scanning the chronological state sequence of each ICU stay therefore emits a synthetic event at each transition of the weaning state. Its label names the kind of change, either a support reduction or a support increase. Its timestamp is that of the new mode. Its payload carries the previous and the new state. The generation rule is directional. The reduction from *Full Support* to *SBT Candidate* and the increase in the reverse direction are distinct labels. A support reduction can therefore serve as the trigger opening an SBT episode while a support increase signals its termination, both under unmodified Declare templates.

Reconstructed episodes. These value-change events induce the episode structure. An *invasive ventilation episode* begins when the patient first enters *Full Support*, and every later re-entry into *Full Support* after an interruption opens a new episode, which captures reintubation after failed liberation. The episode ends on the transition out of *Full Support*. Within an episode, an *SBT episode* begins on the transition into *SBT Candidate*. A return to *Full Support* closes the trial as failed, subject to a 120 min debounce. If the patient reverts to *SBT Candidate* within this window, the interruption is treated as charting noise or a brief adjustment and the trial continues.

Trial outcomes. Each SBT episode receives an outcome label based on a 48 h window beginning at its SBTStarted event, following the definitions of weaning success and failure in Section 6.2.3. The base rule is that a trial *failed* if invasive ventilation resumes within the window.

Some traces end after extubation without a subsequent re-ventilation event. This does not necessarily indicate successful weaning, as extubation may occur as part of a transition to comfort care. According to the domain expert, patients in these cases may die shortly after extubation without being re-ventilated. To account for this, we also classify a trial as a failure if the patient dies in hospital within the same 48 h window. The time of death t_d is obtained from the `deathtime` field in the MIMIC-IV `admissions` table, and the death criterion applies only when re-ventilation has not occurred. Since this field records in-hospital deaths only, patients discharged before death, including to hospice, remain classified as successful if no re-ventilation occurs within the window.

All remaining trials are classified as *successful*. Episodes without an SBT receive no outcome label. Only re-ventilation failures generate an SBTFailed event; failures identified by the death criterion affect the outcome label but do not add an event to the log. Both types of failure are treated identically in the statistical analyses.

6.3 Guideline Constraints

This section presents the MP-Declare constraints used for the weaning analysis. Each normative statement of the guideline (Section 6.2) is represented as a Declare template from Table 3.2, instantiated with concrete activities and refined by an activation condition φ_a, a correlation condition φ_c, and a time condition φ_τ, as defined in Definition 3.19. The referenced activities are produced by the event-log construction: reconstructed trial events SBTStarted, SBTFailed, and SBTSucceeded from Section 6.2.5, measurement events for charted physiological observations, and intervention events derived from medication administrations, laboratory orders, and prescriptions. Constraint identifiers indicate their provenance: *RC* denotes readiness constraints, *SBT-TM* trial-timing constraints, and *INT* failure-management constraints; the suffixes *DX*, *TR*, and *SEQ* distinguish diagnostic, treatment, and sequencing obligations.

Table 6.1: Readiness constraints. Each is a PRECEDENCE constraint activated by SBTStarted, requiring the listed measurement event within the preceding 24 h with a payload satisfying φ_c. Thresholds follow Akella et al. [10].

ID	Criterion	Target activity	Correlation condition φ_c
RC-DP-01	Oxygenation	OxygenationMeasured	$PaO_2/FiO_2 \geq 150$, or $SpO_2 \geq 90\,\%$ on $FiO_2 \leq 40\,\%$ and PEEP $\leq 5\,cmH_2O$
RC-DP-02	Arterial pH	pHMeasured	pH > 7.25
RC-DP-03	Hemoglobin	HemoglobinMeasured	Hgb $\geq 7\,g\,dL^{-1}$
RC-DP-04	Temperature	TemperatureMeasured	Temp $\leq 38\,°C$
RC-DP-05	Hemodynamics	VasopressorMeasured	NE-equivalent dose $\leq 0.1\,\mu g/kg/min$
RC-DP-06	Mental status	MentalStatusMeasured	RASS level of consciousness (itemid 228302) = "No", or SAS (itemid 223753) $\in \{3, 4\}$
RC-DP-07	Inspiratory effort	InspiratoryEffortMeasured	spontaneous tidal volume $> 0\,mL$

6.3.1 Readiness Constraints

The readiness assessment is not represented by a discrete activity in MIMIC-IV. Instead, it consists of physiological conditions that should hold when an SBT begins. Each criterion of Section 6.2.2 is therefore encoded as a separate PRECEDENCE constraint of the form

$$\text{PRECEDENCE}(\langle \text{Criterion} \rangle \text{Measured}, \text{SBTStarted}),$$

with $\varphi_a = \mathit{true}$, φ_c as specified in Table 6.1, and $\varphi_\tau = [0, 24\,h]$. As the target of PRECEDENCE refers to the past, this interval bounds the time from the measurement to the activation (Section 3.3.4). The trial start thus activates the constraint and requires a qualifying measurement within the preceding 24 h. The pre-context window is set to $W_{pre} = 24\,h$, so that this lookback is never truncated at the start of the ventilation episode (Section 4.2.1). The correlation condition represents the corresponding clinical threshold. Table 6.1 lists the seven resulting constraints.

Four design decisions are relevant. First, the trial start, rather than the measurement, acts as the activation. This reflects the role of readiness as a precondition for trialing rather than an obligation in itself. Episodes without a trial therefore leave the readiness constraints inactive, while the obligation to attempt a trial is represented separately in Section 6.3.2.

Second, the 24 h lookback reflects the guideline's requirement that readiness be re-assessed daily. A measurement older than one day cannot substantiate current readiness, while requiring a measurement at the exact trial start would be incompatible with the temporal granularity of routine ICU documentation.

The lookback window also provides the basis for resolving state-dependent criteria at trial start. A criterion such as whether the patient is receiving pressure-support ventilation may be represented by a measurement recorded before the trial rather than by an attribute of the trial-start event itself. For an attribute x and trial start at time t,

let

$$O(x,t) = \{\, e \in \sigma_{ev} \mid \pi_x(e) \text{ is defined},\ t - 24\,\mathrm{h} \leq \pi_{time}(e) \leq t \,\}$$

denote the qualifying observations of x in the evaluation trace of Section 4.2.1. The value of x at t is taken from the most recent element of $O(x,t)$ and is undefined when the set is empty. Resolution is performed separately for each attribute, so a criterion involving multiple attributes may be only partially resolvable. An unresolved attribute is treated as undefined, and its atomic comparison evaluates to false. An activation for which no required attribute can be resolved is therefore neither classified as a fulfillment nor as a violation. This distinction provides the basis for the cause decomposition in Section 6.4.3.

Third, the correlation condition is evaluated on the most recent measurement before the activation rather than existentially over the lookback window. Under standard MP-Declare semantics, any qualifying target event can satisfy the condition [16]. For physiological measurements, this can be too permissive because an earlier qualifying value may mask a later violation. The *latest* target-selection mode of Definition 4.10 therefore evaluates the most recent measurement, reflecting that readiness concerns the patient's current state. This is a refinement of the standard existential semantics: every trace satisfying the refined condition also satisfies the existential condition.

Fourth, the threshold resides in the correlation condition rather than in the definition of the measurement activity. Measurement events are emitted for *every* charted observation, unconditionally. This separates two failure modes that would otherwise be conflated: a criterion that was *assessed but not met* (a qualifying activity exists, but its payload fails φ_c) and a criterion that was *never assessed* (no target event in the window). While both constitute violations, they do so for distinct clinical reasons.

Operationalizing the mental status criterion. The guideline states the mental status criterion only qualitatively, and MIMIC-IV records it through two instruments. The first is the item `CAM-ICU RASS LOC` (itemid 228302), a text-valued entry whose answer "No" denotes a patient without impaired consciousness. It is charted in only a small fraction of stays. The second is the SAS [79] (itemid 223753), which is charted far more often in this cohort. Using the first item alone would leave `RC-DP-06` without a qualifying measurement in most episodes, so the second serves as a fallback.

SAS scores are stored in MIMIC-IV as text labels. During event-log construction each label is trimmed, lower-cased, and mapped to the ordinal score in Table 6.2, which is stored as the numeric value of the measurement. The label "calm" is treated as a variant of "calm/cooperative". A label with no entry in the mapping and no numeric fallback is left undefined and yields no event. Every RASS entry with a non-empty answer and every SAS entry with a defined score then produces one MentalStatusMeasured event, unconditionally, as for the other criteria.

The correlation condition of `RC-DP-06` is the disjunction $RASS\ LOC = \text{"No"} \vee SAS \in \{3,4\}$. Each attribute is resolved separately from the lookback window as defined above, so an episode is assessable through whichever instrument was charted, and an instrument

Table 6.2: Mapping of the charted SAS label (itemid 223753) to the ordinal score used by `RC-DP-06`.

Charted label	Score	Criterion met
unarousable	1	no
very sedated	2	no
sedated	3	yes
calm/cooperative, calm	4	yes
agitated	5	no
very agitated	6	no
danger agitation	7	no

that was not charted evaluates to false without invalidating the other. The scores $\{3, 4\}$ operationalize "awake and easily arousable": scores of 1 and 2 correspond to excessive sedation, and scores of 5 to 7 to agitation severe enough to preclude a controlled trial (Section 6.2.2). This cut-off is an operational choice, since the guideline does not name a scale or a score.

Of the eight guideline criteria, the resolution of the underlying illness is not encoded. It is a global clinical judgment with no structured correlate in MIMIC-IV, and any proxy would inject more noise than signal. The remaining criteria are encoded as seven separate constraints instead of one conjunctive rule so that the activation-level diagnostics of Definition 3.20 identify *which* criterion was unmet at each non-conformant trial start.

6.3.2 Trial Timing Constraints

The guideline specifies two temporal requirements for trial execution (Sections 6.2.1 and 6.2.3): a failed trial should be repeated on the following day, and weaning attempts should begin early in the ICU stay. The re-attempt requirement is encoded as

$$\texttt{SBT-TM-01:}\quad \textsc{Response}(\mathsf{SBTFailed}, \mathsf{SBTStarted}) \quad \text{with} \quad \varphi_c : \operatorname{date}(t_\tau) = \operatorname{date}(t_\alpha) + 1,$$

where t_α and t_τ denote the timestamps of the activation and target events. Two additional constraints distinguish the direction of deviation. `SBT-TM-01-early` uses NotResponse with $\varphi_c : \operatorname{date}(t_\tau) = \operatorname{date}(t_\alpha)$ to identify same-day retries, while `SBT-TM-01-late` uses Response with $\varphi_c : \operatorname{date}(t_\tau) \leq \operatorname{date}(t_\alpha)+1$ to identify retries occurring after the following day. This decomposition allows early and late deviations to be reported separately.

The re-attempt requirement is expressed in terms of calendar days rather than a metric 24 h interval. This follows the wording of the guideline and the organization of ICU care around daily clinical review [45, 53]. A retry the morning after an afternoon failure, for example, is on the following calendar day even though fewer than 24 h have elapsed. The requirement is therefore represented in the correlation condition rather than φ_τ, which expresses metric temporal intervals.

The early-attempt obligation uses NotRespondedExistence (Table 3.2), activated

by the start of the trial:

$$\texttt{SBT-TM-02:}\quad \textsc{NotRespondedExistence}(\mathsf{SBTStarted}, \mathsf{SBTSucceeded}),$$

with activation condition $\varphi_a : t_\alpha - t_i > 72\,\text{h}$, where t_i is the timestamp of the episode's trigger, i.e., the start of the current invasive ventilation. The guideline formulates this requirement relative to the ICU admission. Measuring from the start of the current ventilation episode instead is a design choice. It keeps the condition within the episode, whose trigger is part of every evaluation trace, and it gives the condition a meaning for repeated ventilation phases, for which the time since admission would include earlier, already concluded episodes. The constraint therefore activates only for trials initiated more than 72 h after the episode's ventilation start and is violated when such a late trial subsequently succeeds. The threshold operationalizes the guideline's recommendation to begin weaning early, rather than a literal deadline stated by Akella et al. [10]. The resulting verdict identifies late successful trials as potential delays but does not establish that the delay caused harm.

6.3.3 Failure-Management Constraints

The failure-management pathways of Section 6.2.4 produce reactive and sequencing constraints. The reactive constraints require diagnostic or stabilizing interventions to be documented after a failed trial:

$$\textsc{Response}(\mathsf{SBTFailed}, \langle \mathsf{Intervention} \rangle) \quad \text{with} \quad \varphi_a = \mathit{true},\ \varphi_c = \mathit{true},\ \varphi_\tau = [0, W],$$

with the window W listed in Table 6.3. A failed trial typically leads to re-ventilation, which opens the next episode, so most interventions are documented after the failed trial's episode has ended. The post-context window is therefore set to $W_{post} = W$, which keeps the full time condition observable (Section 4.2.1). Intervention activities are derived from MIMIC-IV administration, laboratory, and prescription records [47]; the concrete item mappings are part of the event log construction.

The response windows were selected to reflect the expected tempo of each intervention. Most obligations use 24 h, consistent with guidance to reassess the patient and repeat an SBT daily after reversible causes of failure have been addressed [10]. The electrolyte and blood-gas workup uses a shorter operational window of 6 h. Point-of-care systems can return blood-gas and electrolyte results within approximately 5–15 minutes [24]; thus, six hours provides a conservative allowance for ordering, sampling, and documentation rather than representing a guideline-prescribed deadline. Hormonal replacement uses a pragmatic 48 h window because thyroid-function abnormalities in critical illness require cautious interpretation and the benefit of thyroid-hormone replacement remains debated [65]. The 48-hour threshold is an operational modeling choice rather than a treatment deadline specified by the cited review.

An important design decision is that *no cause classification is attempted.* Every constraint in Table 6.3 activates on every trial failure, although the guideline requires

Table 6.3: Failure-management response constraints. Each is activated by SBTFailed and requires the target event within the window W.

ID	Target activity	W	Kind	Reversible-cause pathway
INT-TR-01	BronchodilatorAdmin	24 h	treatment	airway resistance
INT-TR-02	DiureticAdmin	24 h	treatment	fluid overload / compliance
INT-TR-03	AfterloadReductionAdmin	24 h	treatment	weaning-induced cardiac dysfunction
INT-DX-04	ElectrolyteBGTest	6 h	diagnostic	electrolyte / acid–base derangement
INT-TR-04	ElectrolyteCorrectionAdmin	24 h	treatment	electrolyte derangement
INT-DX-05	HormonalLabTest	24 h	diagnostic	endocrine dysfunction
INT-TR-05	HormonalReplacementAdmin	48 h	treatment	endocrine dysfunction
INT-COMPOSITE-01	AnyFailureIntervention	24 h	composite	any of the above

only the pathway matching the actual cause of failure to be pursued. This is a concession to the data: MIMIC-IV contains no structured documentation of *why* a trial failed, so activation cannot be conditioned on the underlying cause. Each pathway constraint consequently activates after every failure and should be interpreted descriptively: its violation rate indicates how often the corresponding pathway is documented after a failure, not how often the guideline was breached. The composite constraint INT-COMPOSITE-01 instead represents the normative requirement that at least one reversible-cause pathway be pursued within 24 h. Its violations identify failures followed by no documented intervention from the modeled pathways.

The second family represents ordering within the pathways. Replacement therapy is required to be preceded by the corresponding diagnostic test:

$$\texttt{INT-SEQ-01:}\quad \textsc{Precedence}(\mathsf{ElectrolyteBGTest}, \mathsf{ElectrolyteCorrectionAdmin}),$$

$$\texttt{INT-SEQ-02:}\quad \textsc{Precedence}(\mathsf{HormonalLabTest}, \mathsf{HormonalReplacementAdmin}),$$

Both constraints are activated by the administration event and require a prior diagnostic event. No time condition is imposed, since an earlier laboratory result may legitimately inform a later treatment decision; as no pre-context window is configured, the lookback extends to the start of the episode.

6.4 Results

The results first characterize the analyzed population and the resulting constraint verdicts. The subsequent analyses decompose readiness violations, relate constraint verdicts to weaning outcomes, examine the structure of deviations, assess the effect of the vacuity treatment, and compare episode-local with trace-global evaluation.

6.4.1 Analyzed Population

Table 6.4 summarizes the data used for the analysis, drawn from 16 MIMIC-IV source tables. Only a small fraction of the reconstructed events carry one of the activity labels referenced by the constraint model. The remaining events are not evaluated directly but provide the clinical context for data conditions and derived attributes.

Segmenting on InvasiveVentilationStarted produces 66 848 episodes. Two thirds of the stays contain exactly one episode, while a smaller group contains repeated ventilation phases, in extreme cases several dozen.

For stays with multiple episodes, episode-local evaluation keeps activations and their targets within the corresponding weaning attempt. A trace-global evaluation instead combines multiple attempts into one verdict and can allow events from one attempt to satisfy obligations raised during another. The synthetic evaluation in Section 5.2.1 showed that this can both mask and introduce violations; the present section examines the same issue in the clinical data.

Most episodes contain a breathing trial and therefore receive an outcome (Section 6.2.5), with successful trials somewhat more frequent than failed ones. Almost all failures are re-ventilations; only 977 trials (1.8 %) failed because the patient died in hospital within 48 h without being re-ventilated. A considerable share of the successful trials is right-censored: the ICU stay ended before the 48 h window closed, so the success is observed over a shorter period.

The inter-episode analysis is restricted to episodes with an observable successor outcome, which holds for only about a third of the episodes. Because a successor exists only when ventilation resumes, this population is selected by a subsequent event related to the weaning outcome. Inter-episode effect sizes should therefore not be interpreted as directly comparable to same-episode associations.

6.4.2 Violation Frequencies

The three constraint families produce distinct patterns. Readiness constraints activate when an SBT begins and therefore provide direct information about adherence to readiness criteria. The individual failure-management pathway constraints activate after every failed trial and are expected to have high violation rates because only the pathway corresponding to the actual cause should normally be pursued. Their rates therefore describe pathway use rather than overall guideline adherence. The composite constraint `INT-COMPOSITE-01` provides the corresponding normative measure.

The trial-timing constraints occupy a different position. A failed trial activates the re-attempt requirement, while the satisfying retry normally occurs in the subsequent episode. Whether it is visible to the constraint therefore depends on the configured post-context window (Section 4.2.1).

Table 6.5 shows the resulting verdicts. All readiness constraints activate in every episode containing a trial, but their violation rates differ widely, from 20.7 % for arterial pH to 92.6 % for mental status. These differences largely follow documentation coverage. Oxygenation and pH are recorded in nearly all stays, whereas hemoglobin (56.0 %),

Table 6.4: The analyzed population. The upper part lists counts; shares are relative to the total of the same unit (events, stays, or episodes), and the indented rows are relative to the episodes containing a breathing trial. The lower part lists distributions per ICU stay, with Q_1 and Q_3 the lower and upper quartiles. Model activities are the 24 activity labels referenced by the constraint set. All other events serve as attribute evidence only.

Quantity	Count	Share (%)			
ICU stays	35 692				
Distinct patients	30 713				
Events	348 514 716				
Distinct activity labels	3027				
Events carrying a model activity	16 402 437	4.7			
Episodes	66 848				
Stays with exactly one episode	23 902	67.0			
Episodes containing a breathing trial	54 988	82.3			
thereof failed	24 622	44.8			
re-ventilated within 48 h	23 645	43.0			
died within 48 h	977	1.8			
thereof successful	30 366	55.2			
thereof successful and right-censored	10 464	19.0			
Episodes with an observed successor outcome	24 883	37.2			
Per stay	**Median**	**Q_1**	**Q_3**	**Mean**	**Max**
Events	4760.5	2446	11 102	—	—
Model-activity events	247	141	522	—	—
Episodes	1	—	—	1.87	49

temperature (23.6 %), and mental status (9.7 %) are charted far less often. The cause decomposition in Section 6.4.3 distinguishes missing documentation from measurements that were recorded and failed the threshold, a distinction that matters most for criteria with low coverage. `RC-DP-06` is the extreme case. It is the most frequently violated readiness constraint, yet a RASS or SAS score [79] is rarely charted. In 98.0 % of its violated episodes no mental-status score was charted in the 24 h before the trial, and only 2.0 % carry a score that did not meet the criterion. Its violations therefore reflect missing documentation almost entirely, and its verdict says little about the patient's actual mental status.

The post-context window of $W_{post} = 48$ h allows the re-attempt constraints to observe retries in the subsequent episode. About half of the failed trials are not followed by a retry on the next calendar day, and delayed retries are more frequent than premature same-day ones.

The individual failure-management pathways show the expected wide spread of violation rates, since each pathway applies only to some causes of failure. The normative composite constraint `INT-COMPOSITE-01` is violated after 15.7 % of failures, in which no modeled reversible-cause intervention was documented within 24 h. The two sequencing constraints behave very differently: electrolyte corrections are almost always preceded by a diagnostic

Table 6.5: Per-constraint verdicts over the 66 848 episodes. Activation is the share of episodes in which the constraint was activated; the remaining episodes are vacuous, that is the constraint never activated in them (Definition 3.20). The violation rate is taken over activated episodes only. The constraint marked ‡ is violated almost exclusively in successful episodes, so its verdict and the outcome label are entangled by construction.

ID	Satisfied	Violated	Vacuous	Activation	Violation rate
RC-DP-01	39 654	15 334	11 860	82.3 %	27.9 %
RC-DP-02	43 607	11 381	11 860	82.3 %	20.7 %
RC-DP-03	15 038	39 950	11 860	82.3 %	72.7 %
RC-DP-04	7189	47 799	11 860	82.3 %	86.9 %
RC-DP-05	16 914	38 074	11 860	82.3 %	69.2 %
RC-DP-06	4080	50 908	11 860	82.3 %	92.6 %
RC-DP-07	22 692	32 296	11 860	82.3 %	58.7 %
SBT-TM-01	11 769	11 876	43 203	35.4 %	50.2 %
SBT-TM-01-early	18 171	5474	43 203	35.4 %	23.2 %
SBT-TM-01-late	15 800	7845	43 203	35.4 %	33.2 %
SBT-TM-02‡	2274	1467	63 107	5.6 %	39.2 %
INT-TR-01	6814	16 831	43 203	35.4 %	71.2 %
INT-TR-02	6180	17 465	43 203	35.4 %	73.9 %
INT-TR-03	2832	20 813	43 203	35.4 %	88.0 %
INT-DX-04	15 657	7988	43 203	35.4 %	33.8 %
INT-TR-04	6144	17 501	43 203	35.4 %	74.0 %
INT-DX-05	727	22 918	43 203	35.4 %	96.9 %
INT-TR-05	1095	22 550	43 203	35.4 %	95.4 %
INT-COMPOSITE-01	20 891	3898	0	100.0 %	15.7 %
INT-SEQ-01	25 901	2438	38 509	42.4 %	8.6 %
INT-SEQ-02	614	4858	61 376	8.2 %	88.8 %

test, whereas the comparatively rare hormonal replacements mostly are not.

Across the model, an episode violates about four applicable constraints on average. Constraint applicability differs substantially between families: readiness constraints are vacuous in fewer than one in five episodes, whereas failure-management and trial-timing constraints are vacuous in the majority. Reporting activation and violation rates separately is therefore necessary to distinguish lack of applicability from non-conformance.

6.4.3 Decomposing Violations by Cause

A violated verdict indicates that an obligation was not discharged but does not identify the reason. For the data-aware readiness constraints, each violation is therefore classified as *not documented*, *partly documented*, or *unmet*. *Not documented* denotes that no measurement of the criterion was charted in the 24 h lookback window, regardless of its value, and *partly documented* an incompletely observable disjunctive criterion. *Unmet* denotes a criterion assessed on complete readings but not satisfied. Only the last category directly indicates that the recorded clinical state failed the criterion. Table 6.6 and

Table 6.6: Why each readiness constraint was violated, over its violated episodes. *Not doc.* is no measurement in the 24 h lookback, *partly doc.* a disjunctive criterion decided on incomplete readings, and *unmet* a criterion assessed on complete readings and not met. Only the last column measures care. `RC-DP-06` is not part of this table, which covers the six criteria with a graded reading; its criterion is a categorical bedside score. Its own split is given in Section 6.4.2.

ID	Criterion	Violated	Not doc.	Partly doc.	Unmet
`RC-DP-01`	Oxygenation	15 334	0.2 %	55.6 %	44.2 %
`RC-DP-05`	Hemodynamics	38 074	82.2 %	—	17.8 %
`RC-DP-02`	Arterial pH	11 381	89.2 %	—	10.8 %
`RC-DP-07`	Inspiratory effort	32 296	97.0 %	—	3.0 %
`RC-DP-04`	Temperature	47 799	98.7 %	—	1.3 %
`RC-DP-03`	Hemoglobin	39 950	98.7 %	—	1.3 %
All six criteria		184 834	90.9 % documentation		9.1 %

Figure 6.1 report this decomposition over the 184 834 readiness violations.

Across the readiness family, 90.9 % of violations are attributable to missing, stale, or incomplete documentation, and only 9.1 % correspond to a criterion that was assessed and not met. The composition differs by criterion. Temperature, hemoglobin, and inspiratory effort violations are almost entirely documentation gaps. Arterial pH has a somewhat larger unmet share, which nevertheless remains minor.

Only oxygenation and hemodynamics have a substantial share of documented but unmet criteria, reaching 44.2 % of the violations for oxygenation; for hemodynamics, these are trials started while the norepinephrine-equivalent dose exceeded the threshold. For these two criteria, the verdicts therefore provide more direct evidence of deviation from the encoded readiness conditions than their aggregate violation rates alone suggest.

The remaining oxygenation violations are almost all *partly documented*. Its correlation condition is disjunctive and requires multiple physiological readings for the second alternative. When only part of the required information is available, the criterion cannot be fully assessed. Treating such cases as unmet would therefore conflate incomplete documentation with a documented physiological deviation.

6.4.4 Association with the Weaning Outcome

The constraint verdicts are next related to the outcome of the corresponding episode. Figure 6.2 compares violation rates between successful and failed episodes, while Table 6.7 reports the corresponding risk ratios, confidence intervals, and Fisher exact-test results. Vacuous episodes are excluded. For inter-episode constraints, the reference outcome is the outcome of the successor episode as defined in Section 4.4.1. With 21 constraints, the Bonferroni threshold is $\alpha/m = 2.4 \times 10^{-3}$.

The readiness constraints generally show the expected direction. Six of the seven constraints have risk ratios below one, five of them statistically significant after correction: episodes in which a readiness criterion was violated succeed less often. The association

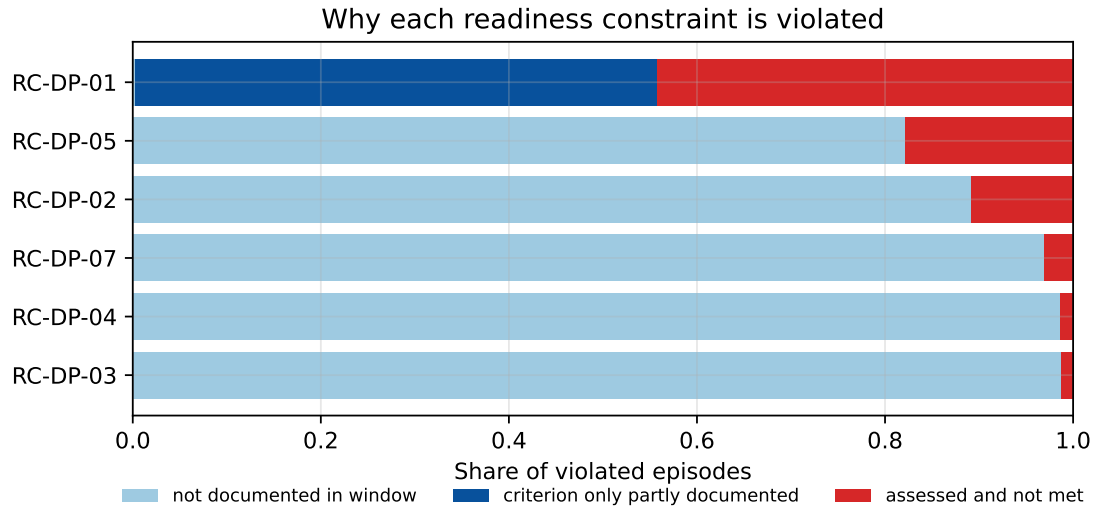


Figure 6.1: Composition of the readiness violations by cause. The two shades of blue are documentation states; only the red segment records a criterion assessed and not met.

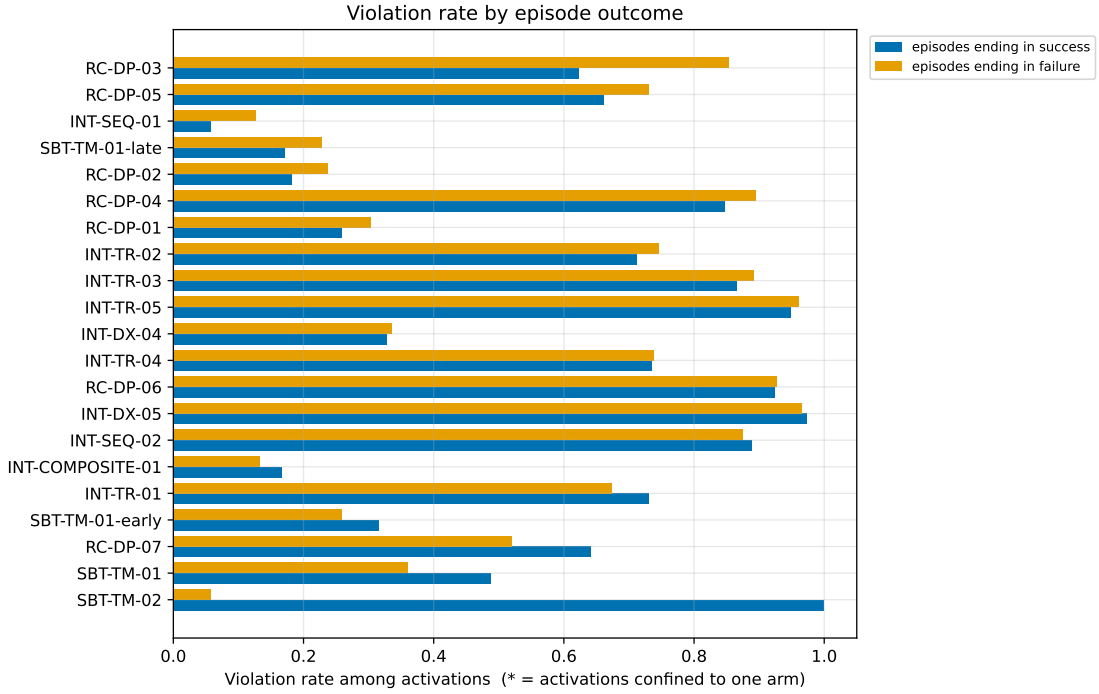


Figure 6.2: Violation rate among activations, split by the outcome of the reference episode, which is the successor episode for the inter-episode rules of Section 4.4.1. Constraints marked with an asterisk activate in only one outcome class and have no comparison to show.

is strongest for hemoglobin (RC-DP-03), where violation goes along with a success rate lower by more than a third ($RR = 0.62$), and moderate for the remaining criteria. Mental status (RC-DP-06) shows no association ($RR = 0.98$, 95 % confidence interval (CI) $[0.95, 1.01]$). Since its satisfying group is sizeable and the confidence interval is narrow, any true association is likely to be small. This is consistent with the cause split above: a violation almost always indicates that no score was charted. The constraint therefore primarily distinguishes episodes by documentation rather than clinical state, while the documentation of mental status appears unrelated to the outcome.

RC-DP-07 is the exception: its violation is significantly associated with a *higher* success rate ($RR = 1.26$). This result has to be read in light of what a violation of this constraint actually records. In almost all cases (97.0 %, Table 6.6), it does not mean that the patient was found to lack inspiratory effort, but only that no qualifying measurement was documented in the 24 h before the trial. A violation of RC-DP-07 therefore mostly reflects missing information, not a deviation of the care process from the guideline: the readiness assessment may well have taken place without leaving a trace in the data. The positive association consequently says nothing about whether trialing without inspiratory effort is beneficial. A more plausible explanation is that the measurement is charted more often for patients who are harder to wean, so that whether it is recorded reflects patient severity [8, 37, 82]. The available data do not allow this mechanism to be tested directly. The result illustrates that a constraint can faithfully encode a guideline requirement while its violations, taken alone, are not a reliable indicator of non-conformant care.

The failure-management constraints show mixed associations. INT-TR-02, INT-TR-03, INT-TR-05, and INT-SEQ-01 are associated with lower success rates in the successor episode. INT-SEQ-01 has the strongest association in this group: corrections without a preceding electrolyte test go along with markedly lower success. INT-DX-04, INT-TR-04, INT-DX-05, and INT-SEQ-02 do not reach the multiple-testing threshold.

INT-TR-01 and INT-COMPOSITE-01 show the opposite direction, with $RR = 1.18$ and 1.17: constraint violations are associated with a higher probability of success in the successor episode. This pattern should not be interpreted as evidence that withholding investigation or treatment improves weaning outcomes. Management after a failed SBT is cause-directed: clinicians assess the apparent mechanism and clinical presentation of the failure and select diagnostic or therapeutic actions accordingly, including bronchodilators when increased airway resistance or bronchoconstriction is suspected [10, 25]. Consequently, the clinical findings that prompt an intervention may themselves be associated with a lower probability of success at the next attempt, resulting in confounding by indication [52, 85]. The observed relative risks are therefore associational rather than causal estimates. The constraints nevertheless retain their normative meaning because clinical guidance recommends identifying and correcting reversible causes after SBT failure before reassessment [10, 14].

SBT-TM-01 and its early variant are associated with higher successor success rates when violated, whereas the late variant is associated with lower successor success. These associations are again subject to confounding because retry timing is a clinical decision rather than a randomized exposure. SBT-TM-02 remains structurally degenerate because

Table 6.7: Outcome association per constraint, vacuous episodes excluded. Exposure is violation of the constraint; *RR* is the risk ratio of the *Success* outcome under violation, so $RR < 1$ marks a protective constraint. *Ref.* names the reference episode: *own* for same-episode constraints and *next* for the inter-episode rules of Section 4.4.1, for which 4610 episodes (10 850 for `INT-COMPOSITE-01`) are censored for lack of an observed successor. Significance is assessed against the Bonferroni threshold 2.4×10^{-3}; significant rows are marked *. `SBT-TM-02` is omitted: it is violated exactly when its own trial ends in SBTSucceeded, which fixes the outcome except for the 139 such trials relabelled as failures by the death criterion, so exposure and outcome nearly coincide by definition and no risk ratio is meaningful.

ID	Ref.	Violated	Satisfied	*RR*	95 % CI	*p* (Fisher)
`RC-DP-01`*	own	15 334	39 654	0.90	[0.89, 0.92]	4.2×10^{-32}
`RC-DP-02`*	own	11 381	43 607	0.86	[0.84, 0.87]	2.8×10^{-55}
`RC-DP-03`*	own	39 950	15 038	0.62	[0.62, 0.63]	$< 1 \times 10^{-300}$
`RC-DP-04`*	own	47 799	7189	0.84	[0.82, 0.85]	8.5×10^{-62}
`RC-DP-05`*	own	38 074	16 914	0.87	[0.85, 0.88]	4.5×10^{-70}
`RC-DP-06`	own	50 908	4080	0.98	[0.95, 1.01]	0.20
`RC-DP-07`*	own	32 296	22 692	1.26	[1.24, 1.28]	4.0×10^{-180}
`SBT-TM-01`*	next	7860	11 175	1.35	[1.31, 1.40]	2.0×10^{-68}
`SBT-TM-01-early`*	next	5371	13 664	1.18	[1.14, 1.22]	2.6×10^{-18}
`SBT-TM-01-late`*	next	3898	15 137	0.80	[0.77, 0.84]	5.1×10^{-22}
`INT-TR-01`*	next	13 277	5758	1.18	[1.13, 1.22]	6.5×10^{-17}
`INT-TR-02`*	next	13 939	5096	0.91	[0.87, 0.94]	3.9×10^{-7}
`INT-TR-03`*	next	16 774	2261	0.87	[0.83, 0.91]	2.7×10^{-8}
`INT-DX-04`	next	6327	12 708	0.98	[0.94, 1.01]	0.21
`INT-TR-04`	next	14 036	4999	0.99	[0.95, 1.03]	0.58
`INT-DX-05`	next	18 441	594	1.14	[1.03, 1.27]	0.012
`INT-TR-05`*	next	18 186	849	0.86	[0.80, 0.92]	9.9×10^{-5}
`INT-COMPOSITE-01`*	next	2028	11 911	1.17	[1.11, 1.24]	4.3×10^{-8}
`INT-SEQ-01`*	own	1983	22 209	0.69	[0.66, 0.73]	2.2×10^{-72}
`INT-SEQ-02`	own	4058	530	1.04	[0.98, 1.11]	0.18

its violation is defined by a successful late trial; no choice of reference episode removes this relationship.

6.4.5 Structure of Detected Deviations

The association analysis treats each constraint violation as a binary exposure, but clinical guidelines are consensus documents calibrated to the average patient, and a clinician who knows more about a particular case may deliberately and defensibly depart from them [42, 49]. Such considered departures should behave differently from cases where the guideline was simply not followed. Because verdicts are localized to a single weaning attempt, this is testable directly, along four dimensions: breadth of deviation, its cause, its magnitude, and its position in the stay.

Table 6.8: Outcome by the shape of the readiness deviation. The reference arm throughout is the 315 episodes that violated none of the six criteria with graded readings, in which 74.0 % of trials succeeded; *RR* is the risk ratio of *Success* against that arm. `RC-DP-06` is excluded because it is violated in 92.6 % of activations, almost always due to missing documentation, and would therefore appear in nearly every episode's profile without distinguishing between episodes. The lower block restricts to episodes whose *only* readiness violation was the constraint named.

Deviation	Episodes	Success rate	*RR*	95 % CI	*p* (Fisher)
By breadth: criteria violated in the episode					
1 criterion	3300	74.5 %	1.01	[0.94, 1.08]	0.84
2 criteria	10 533	64.9 %	0.88	[0.82, 0.94]	7.6×10^{-4}
3 criteria	17 218	53.7 %	0.73	[0.68, 0.78]	3.2×10^{-13}
4 criteria	12 722	48.6 %	0.66	[0.61, 0.70]	1.8×10^{-19}
5 criteria	7474	45.9 %	0.62	[0.58, 0.67]	6.0×10^{-23}
6 criteria	3426	57.8 %	0.78	[0.73, 0.84]	1.4×10^{-8}
By criterion: the episode's sole readiness violation					
`RC-DP-07` only	1728	85.0 %	1.15	[1.07, 1.23]	5.2×10^{-6}
`RC-DP-01` only	42	78.6 %	1.06	[0.90, 1.26]	0.58
`RC-DP-04` only	861	69.5 %	0.94	[0.87, 1.02]	0.15
`RC-DP-05` only	289	67.8 %	0.92	[0.83, 1.02]	0.11
`RC-DP-03` only	379	42.7 %	0.58	[0.51, 0.66]	9.4×10^{-17}

Breadth. Table 6.8 and Figure 6.3 group episodes by the number of the six interpretable readiness criteria they violate. Episodes violating a single criterion succeed as often as episodes violating none ($RR = 1.01$). From two violated criteria onward, the success rate declines with each additional criterion, down to $RR = 0.62$ for five criteria, before partially recovering for episodes that violate all six.

Readiness non-conformance is not all the same. Missing one criterion does not lower the outcome rate, but missing several does. This fits the idea that missing one criterion reflects a deliberate exception, while missing many reflects a readiness assessment that was not properly done or documented. The reference group of fully conformant episodes is small, however, so its outcome rate is less precise than those of the larger groups.

Looking at each criterion alone shows the same pattern. Episodes whose only violation is temperature or hemodynamics do not differ significantly from fully conformant episodes. Hemoglobin is the exception: as a sole violation, it already goes along with a much lower success rate. Episodes whose only violation is inspiratory effort instead succeed more often. The latter is consistent with the monitoring explanation for `RC-DP-07` given earlier.

Isolated deviations are uncommon: 6.0 % of episodes violate exactly one criterion and 0.6 % violate none, while the average episode violates 3.4 criteria. Violations also cluster across criteria. For example, among episodes violating the pH criterion, hemoglobin is also violated in 97 % and temperature in 96 %. Together with the cause decomposition,

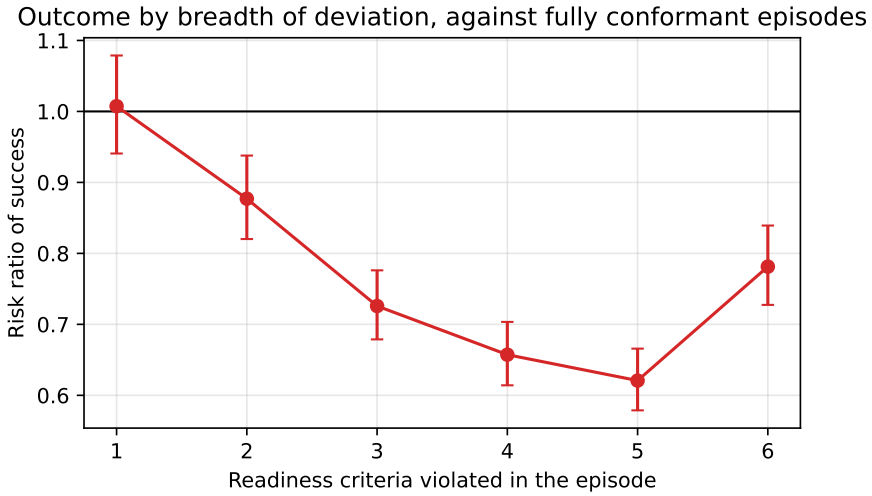


Figure 6.3: Risk ratio of a successful trial by the number of readiness criteria the episode violated, against the fully conformant episodes. A single violated criterion carries no detectable penalty. The penalty appears from two onward and deepens.

this indicates that violations do not occur independently. Instead, failing one criterion strongly suggests failing several others.

Cause. The cause decomposition of Section 6.4.3 raises the question of whether readiness violations are associated with trial outcomes differently depending on whether a criterion was assessed and not met or could not be adequately assessed. Figure 6.4 compares each type of violation with episodes satisfying the corresponding criterion. Relative to the same 315 fully conformant episodes, episodes with only documentation-related violations have lower success rates ($RR = 0.79$, 95 % CI $[0.74, 0.84]$). The association is stronger for episodes with at least one unmet criterion ($RR = 0.63$, $[0.59, 0.67]$).

This pattern is also evident for criteria with directly measured physiological values. For arterial pH, the risk ratio is lower for unmet criteria than for undocumented ones (0.62 versus 0.88); for hemodynamics, the corresponding values are 0.71 and 0.90. For oxygenation, partly documented violations have a similar success rate to conformant episodes ($RR = 0.99$, $[0.97, 1.01]$), whereas episodes with unmet criteria have a lower success rate ($RR = 0.79$). Classifying partial documentation as unmet would therefore have obscured this distinction.

For hemoglobin, the lower success rate ($RR = 0.62$) is driven almost entirely by episodes without a blood-gas hemoglobin measurement in the lookback window, which account for 98.7 % of violations. Episodes with a measured value below the threshold show a much weaker association ($RR = 0.93$, $[0.87, 0.98]$), which is not significant after correction for the 15 cause-specific tests.

These findings show that, for oxygenation, arterial pH, hemodynamics, and temperature,

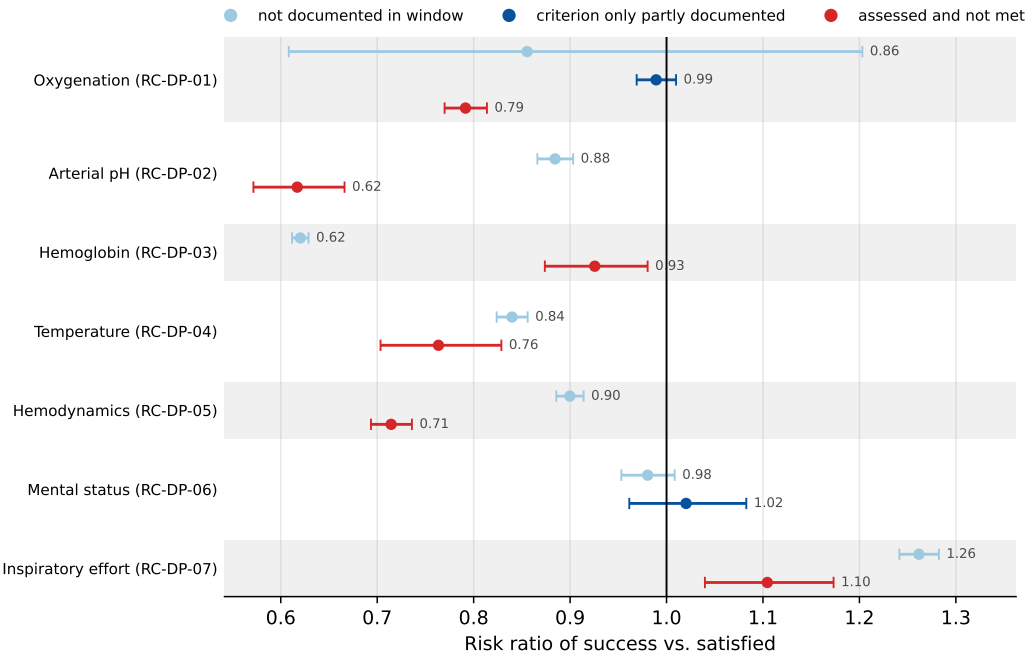


Figure 6.4: Risk ratio of a successful trial for readiness violations split by cause, against the episodes satisfying the same criterion. Causes with fewer than 30 violated episodes are omitted. Unmet violations carry the stronger association for oxygenation, arterial pH, and hemodynamics; for hemoglobin and inspiratory effort the association comes from undocumented violations.

failing a measured criterion is associated with lower success. However, a substantial part of the overall violation–outcome association comes from documentation status, which is informative in itself and accounts for most of the association for hemoglobin and inspiratory effort. For hemoglobin, the undocumented category reflects the absence of a blood-gas reading specifically, since the criterion is evaluated on blood-gas hemoglobin only; it is therefore likely to capture whether arterial blood gases were drawn rather than whether hemoglobin was assessed at all. Differences in documentation intensity may in turn reflect patient severity or monitoring practices [8, 37], although this cannot be established from the available data. Missing documentation should therefore be treated neither as evidence that a threshold was unmet nor as outcome-neutral, and violation verdicts are more interpretable when reported together with their cause. All cause-specific associations are not adjusted for patient characteristics and remain observational.

Magnitude. For criteria with a numeric reading, an *unmet* violation measures how far the observed value lies from the guideline threshold. This makes it possible to compare violations by severity rather than simply counting them. Figure 6.5 groups violations into tertiles based on this distance and compares each group with episodes in which the

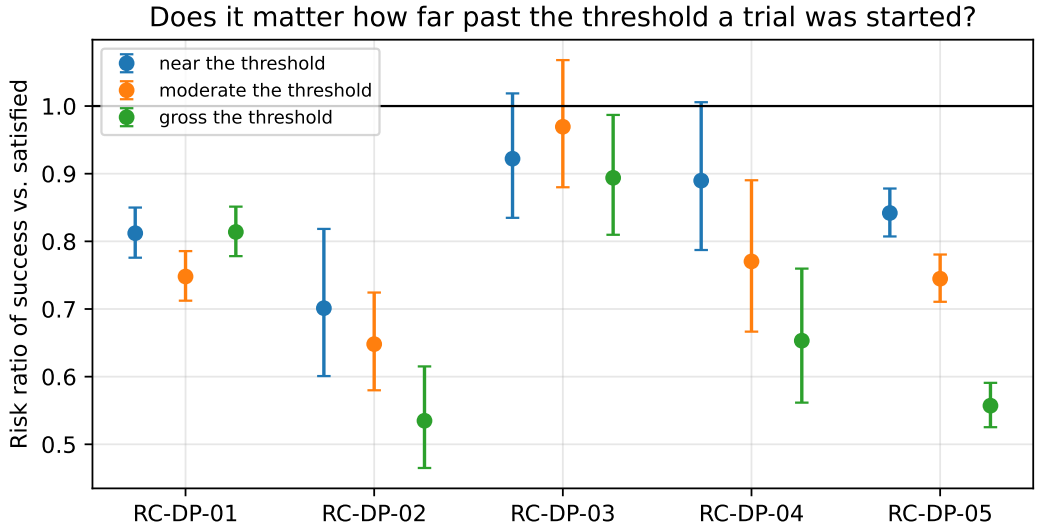


Figure 6.5: Risk ratio of a successful trial for readiness violations stratified into tertiles by the distance of the reading from the guideline threshold, against the episodes satisfying the same criterion. Hemodynamics (`RC-DP-05`) and pH (`RC-DP-02`) show a monotone dose-response throughout; temperature (`RC-DP-04`) is monotone but its mildest tertile is not significant; oxygenation and hemoglobin do not show a consistent gradient.

criterion was met.

Hemodynamics and arterial pH show the clearest gradients. For hemodynamics, the risk ratio declines monotonically from 0.84 in the mildest to 0.56 in the most severe tertile; for pH, from 0.70 to 0.53. All six estimates are significant. Temperature follows a similar pattern, but the confidence interval of its mildest tertile includes one, so the dose-response holds only from the second tertile onward.

Oxygenation shows no gradient: all three tertiles are associated with lower success to a similar degree (RR between 0.75 and 0.81). Hemoglobin shows no dose-response pattern either. Only its most severe tertile is associated with lower success, and only weakly ($RR = 0.89$, 95 % CI $[0.81, 0.99]$), while the confidence intervals of the two milder tertiles include one.

The magnitude analysis therefore provides evidence of a graded association for hemodynamics and pH and, more weakly, temperature. In particular, these results do not suggest a sharp distinction between harmless near-threshold deviations and more severe ones; even the smallest deviations are associated with lower success. The evidence for isolated, potentially distinct deviations comes instead from the breadth analysis.

Position in the stay. Because each verdict is associated with a specific weaning attempt, violations can also be examined by attempt index. Figure 6.6 shows the violation rate of each readiness criterion across successive attempts. Five of the six criteria become

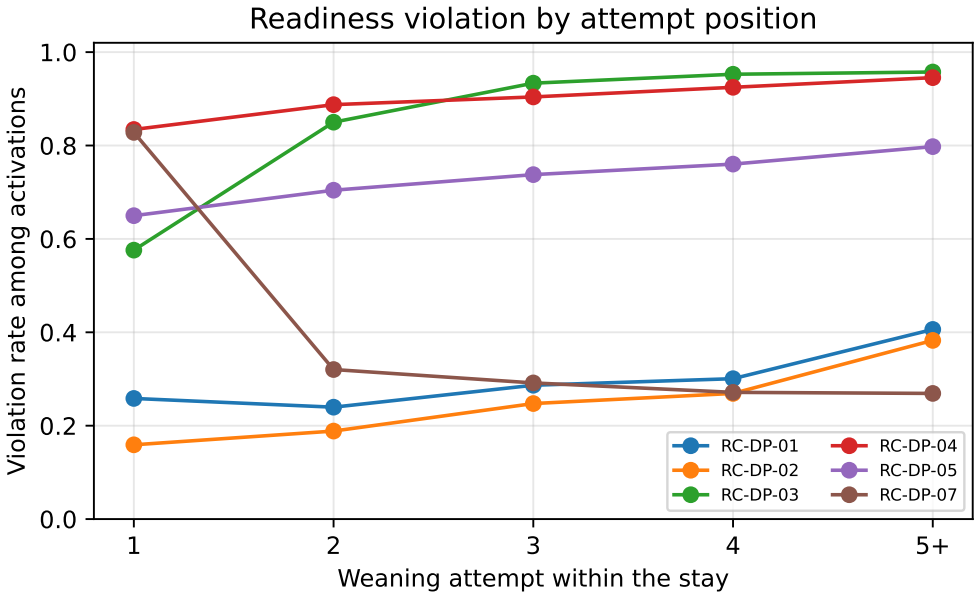


Figure 6.6: Readiness violation rate among activations by the index of the weaning attempt within the stay, pooling the fifth and later attempts. `RC-DP-06` is omitted because its violation rate stays between 91.6 % and 93.0 % across all attempts and shows no gradient.

more frequently violated in later attempts; hemoglobin, for instance, is violated in nearly every trial from the fifth attempt onward. Inspiratory effort shows the opposite pattern and is violated far less often in later attempts.

These trends are consistent with the documentation patterns identified above. In repeatedly ventilated patients, the last measurement more often lies outside the lookback of later trials, while inspiratory-effort measurements are more frequently available in difficult weaning courses. The analysis also illustrates the localization provided by episode-scoped evaluation: a trace-global verdict cannot assign a violation to a specific weaning attempt.

6.4.6 Sensitivity to the Vacuity Treatment

The main association analysis excludes vacuous episodes so that the exposure is defined only where the corresponding constraint activates. Table 6.9 repeats the analysis by pooling vacuous episodes into the satisfied category for constraints where this changes the result. Readiness constraints are excluded because they activate in essentially every trial, while `INT-COMPOSITE-01` never becomes vacuous.

For the individual failure-management constraints, pooling substantially changes the associations. The 43 203 episodes without a re-ventilation failure are added to the satisfied category, so the resulting exposure no longer represents only conformance to a post-failure

Table 6.9: Effect of the vacuity treatment on the association results, for the constraints where the two modes diverge. *RR* is again the risk ratio of *Success* under violation. Rows marked ‡ have degenerate tables in at least one mode.

	Activated only		Vacuous pooled as satisfied	
ID	*RR*	*p*	*RR*	*p*
INT-TR-01	1.18	6.5×10^{-17}	1.03	0.049
INT-TR-02	0.91	3.9×10^{-7}	0.86	2.1×10^{-21}
INT-TR-03	0.87	2.7×10^{-8}	0.84	8.4×10^{-27}
INT-DX-04	0.98	0.21	0.94	1.2×10^{-4}
INT-TR-04	0.99	0.58	0.91	5.3×10^{-9}
INT-DX-05	1.14	0.012	0.87	2.1×10^{-14}
INT-TR-05	0.86	9.9×10^{-5}	0.83	3.9×10^{-24}
INT-SEQ-01	0.69	2.2×10^{-72}	0.83	1.0×10^{-16}
INT-SEQ-02	1.04	0.18	1.28	3.4×10^{-81}
SBT-TM-01	1.35	2.0×10^{-68}	1.23	2.0×10^{-40}
SBT-TM-01-early	1.18	2.6×10^{-18}	1.11	1.6×10^{-9}
SBT-TM-01-late	0.80	5.1×10^{-22}	0.78	5.5×10^{-30}
SBT-TM-02‡	4.1×10^{3}	$< 1 \times 10^{-300}$	1.67	9.8×10^{-198}

obligation; it also captures whether a failure occurred. Several associations consequently change direction or become significant; INT-DX-05, for instance, reverses from a positive to a negative association, and previously non-significant constraints such as INT-TR-04 and INT-SEQ-02 become significant. These changes arise from the altered comparison population rather than from different constraint verdicts.

The trial-timing constraints are less sensitive to this treatment. Pooling vacuous episodes attenuates the associations of SBT-TM-01 and its early variant and slightly strengthens that of the late variant, but changes no direction. SBT-TM-02 remains structurally entangled because its violation is defined by the same event that determines the outcome.

6.4.7 Effect of Episode-Local Evaluation

The preceding analyses evaluate constraints per episode. This section compares that evaluation with trace-global evaluation on the same multi-episode stays. The analysis first examines disagreement between episode outcomes within a stay and then repeats the conformance analysis with segmentation disabled.

Mixed outcomes within a stay. Among the 9698 stays with at least two episodes carrying an evaluable outcome, 7767 (80.1 %) contain both a failed and a successful trial (Figure 6.7), and this share grows with the number of episodes. Even assigning each mixed stay its majority outcome would misclassify a third of the episodes in these stays.

Ablation design. The outcome analysis shows that a single stay-level outcome can lose information when several weaning attempts have different outcomes. The same question

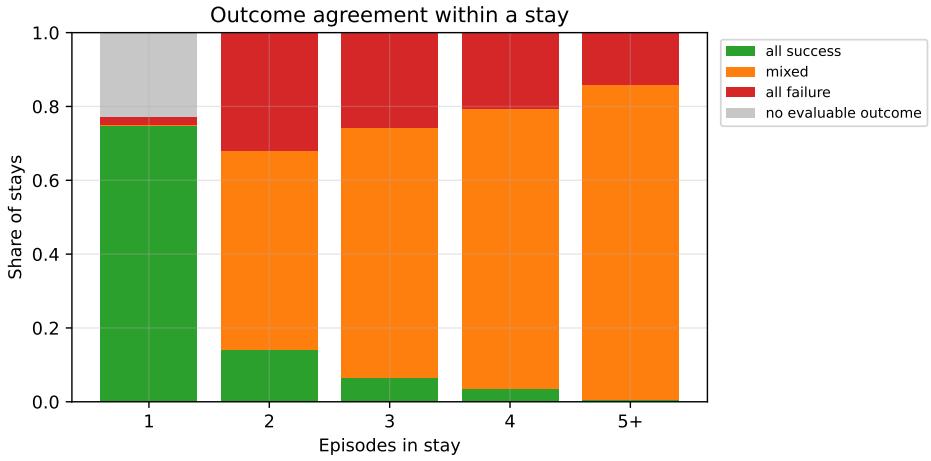


Figure 6.7: Outcome agreement within a stay, by number of episodes. A trace-global verdict cannot represent the *mixed* share without discarding information.

can be asked of conformance verdicts. To isolate the effect of segmentation, only the episode-extraction step was disabled. Evaluation-trace construction, context windows, constraint evaluation, and outcome reading remained unchanged.

A fixed-seed sample of 2000 multi-episode stays was evaluated both with episode segmentation and with each stay treated as one trace. The two verdict sets were then compared constraint by constraint. `INT-COMPOSITE-01` was excluded because its activation condition requires a prior trial within the stay and therefore cannot activate when the whole stay is represented as a single trace. The remaining twenty constraints were included.

Verdict relocation. Removing segmentation changes verdicts in both directions. In 1182 of the 2000 stays (59.1 %), at least one constraint was violated under one evaluation level but not the other.

A violation is *masked* when it occurs under episode-level evaluation but not under trace-level evaluation. This occurs when evidence from another episode can discharge an obligation after the episode boundary has been removed. Three constraints account for the 434 masked flags(Table 6.10). `INT-SEQ-01` provides the clearest example: because its precedence relation has no time bound, a diagnostic event from an earlier trial can satisfy an obligation associated with a later correction, so 88.2 % of its episode-level violations disappear under trace-global evaluation. The other constraints with masking are similarly affected by cross-episode evidence.

Trace-global evaluation also introduces violations that do not occur in any individual episode. There are 1448 such trace-only flags, compared with 434 masked flags. `SBT-TM-02` accounts for most of them: it is anchored to the start of the ICU stay rather than to the individual trial, so a trace-global evaluation can classify a stay as violating the deadline even when no individual episode contains such a violation. Segmentation therefore

Table 6.10: Segmentation ablation on the 2000-stay sample, sorted by total disagreement. *Episode-level* and *Trace-level* are counts of stays flagged violated under each evaluation level. *Masked* is flagged violated at episode level and not at trace level; *Trace-only* is the reverse. Rates are taken against the episode-level and trace-level counts respectively. *Ep./stay* is the mean number of episodes in a stay flagged violated at trace level. `INT-COMPOSITE-01` never activates on a whole-stay trace and is omitted.

ID	Episode-level	Trace-level	Masked	Masking rate	Trace-only	Trace-only rate	Ep./stay
SBT-TM-02	145	1118	0	0.0 %	973	87.0 %	4.5
RC-DP-01	933	817	187	20.0 %	71	8.7 %	4.1
INT-SEQ-01	237	49	209	88.2 %	21	42.9 %	4.4
INT-SEQ-02	300	290	38	12.7 %	28	9.7 %	3.8
RC-DP-07	1744	1810	0	0.0 %	66	3.6 %	3.7
RC-DP-02	777	827	0	0.0 %	50	6.0 %	4.6
INT-TR-04	1470	1495	0	0.0 %	25	1.7 %	3.9
INT-TR-01	1468	1492	0	0.0 %	24	1.6 %	3.8
SBT-TM-01-late	994	1018	0	0.0 %	24	2.4 %	4.3
SBT-TM-01	1345	1368	0	0.0 %	23	1.7 %	4.0
INT-TR-02	1493	1514	0	0.0 %	21	1.4 %	3.9
SBT-TM-01-early	689	709	0	0.0 %	20	2.8 %	4.9
INT-TR-03	1644	1663	0	0.0 %	19	1.1 %	3.9
INT-TR-05	1764	1783	0	0.0 %	19	1.1 %	3.8
INT-DX-05	1801	1819	0	0.0 %	18	1.0 %	3.8
RC-DP-05	1728	1744	0	0.0 %	16	0.9 %	3.8
INT-DX-04	809	823	0	0.0 %	14	1.7 %	4.8
RC-DP-06	1881	1888	0	0.0 %	7	0.4 %	3.7
RC-DP-04	1876	1882	0	0.0 %	6	0.3 %	3.7
RC-DP-03	1768	1771	0	0.0 %	3	0.2 %	3.8

changes the location and meaning of verdicts rather than simply increasing or decreasing their frequency.

Activation loss and localization. Trace-global evaluation also collapses repeated activations within a stay. The median constraint loses 53.4 % of its episode-level activations when multiple episodes are represented as one trace. This reduction varies across constraints because some obligations can activate repeatedly in successive episodes, whereas others depend on events that occur only once within the stay. `SBT-TM-02` is the main exception, gaining 134.5 % of activations because its activation condition is evaluated against the start of the ICU stay rather than the individual trial. Under trace-global evaluation, it can therefore activate on stays where no individual episode would activate it.

The loss of activation instances also reduces the granularity of the conformance result. At episode level, each activation and its verdict can be associated with a specific weaning attempt. At trace level, these instances are aggregated into a single stay-level verdict. Stays flagged as violated at trace level contain about four episodes on average (Table 6.10). A clinician reviewing such a verdict must therefore inspect several weaning attempts to identify the source of the deviation, whereas the episode-level verdict identifies

the corresponding attempt directly. Trace-global evaluation thus loses both repeated activation information and the localization of a deviation within the weaning course.

6.4.8 Summary

Five findings summarize the case study.

First, the readiness verdicts can be decomposed into different sources of non-conformance. About nine in ten readiness violations involve missing or incomplete documentation, and only about one in ten corresponds to a criterion that was assessed and not met. This distinction varies by criterion and cannot be recovered from aggregate violation rates alone.

Second, readiness non-conformance is heterogeneous. Episodes violating a single criterion do not show a lower success rate than fully conformant episodes, whereas broader non-conformance is associated with lower success. The magnitude analysis provides a more limited result: a graded association is observed for hemodynamics and pH and, less consistently, temperature, but not for oxygenation or hemoglobin.

Third, conformance and outcome associations need to be interpreted jointly. Several faithfully encoded guideline constraints show associations in the opposite direction from their normative interpretation, including `RC-DP-07`, `INT-TR-01`, and `INT-COMPOSITE-01`. These patterns are consistent with selection and confounding mechanisms in the observed clinical process and therefore do not provide causal evidence against the corresponding recommendations. The analysis also shows that the choice of reference episode can determine whether an association is even estimable, as demonstrated by the trial-timing constraints.

Fourth, episode-local verdicts allow deviations to be localized within the weaning course. Readiness violations change across successive attempts, and multiple criteria frequently fail together. These patterns would be obscured by a single verdict for the complete ICU stay.

Fifth, episode-local evaluation changes both the verdicts and the information available for interpretation. Among multi-episode stays, most stays with evaluable outcomes contain both successful and failed attempts. Trace-global evaluation masks some episode-level violations, introduces others, collapses repeated activations, and removes the localization of a deviation to the corresponding trial.

Together, these findings show that episode-scoped conformance checking provides information beyond an aggregate adherence score. It distinguishes documentation gaps from documented deviations, characterizes the breadth and magnitude of deviations, and associates each verdict with the weaning attempt in which it occurred. Outcome information is useful for assessing whether an encoded constraint functions as a meaningful quality indicator, but the conformance analysis itself remains based on the structure and content of the episode.

6.5 Limitations

The limitations discussed here are specific to the MIMIC-IV weaning instantiation and complement the general threats to validity discussed in Section 5.3.

The analysis uses a single-center, retrospective, credentialed-access dataset (Section 6.2.5). The reported violation frequencies, outcome associations, and deviation patterns therefore reflect the patient population and documentation practices of one hospital system and may not generalize to other institutions.

Several inputs to the constraint evaluation are reconstructed rather than directly observed. Ventilator-mode states are inferred as proxies for clinically reported events, with explicit ambiguity for Standby- and NIV-mapped states (Section 6.2.5). These variables should therefore not be treated as validated ground truth. The dataset also contains no structured representation of the cause of a failed trial. Consequently, the failure-management constraints cannot condition their activation on the underlying cause (Section 6.3.3), and their violation rates indicate whether the corresponding pathway was documented rather than whether the guideline was breached.

Charting and timestamp imprecision are inherent to ICU data [38]. From the recorded data alone, their effects cannot always be distinguished from genuine deviations, so both false violations and false satisfactions are possible. Some analyses also rely on small reference groups, such as the 315 fully conformant episodes in Table 6.8 and the 42 `RC-DP-01-only` episodes. Estimates based on these groups are consequently less precise than those based on the full population.

Two constraints have additional limitations. `SBT-TM-02` is definitionally entangled with its reference outcome: its violation is itself defined by a successful late trial. A risk ratio is therefore not meaningful for this constraint, regardless of the choice of reference episode. `RC-DP-06` rests on an operational choice, since the guideline names neither a scale nor a score and the SAS cut-off of $\{3, 4\}$ is our own operationalization. The chosen data points are also charted in only 9.7 % of stays, so its verdicts mostly reflect missing documentation and say little about the patients' actual mental status.

More generally, the time-condition and context-window sizes used across the constraints (a 24 h readiness lookback and 6 h, 24 h, and 48 h intervention windows) are derived from guideline text and literature proxies. They were not subjected to a systematic sensitivity analysis and should therefore be regarded as configuration choices rather than validated parameters.

7 Discussion

This chapter interprets the results of Chapters 5 and 6. Section 7.1 discusses the main findings of both studies and their implications. Section 7.2 then examines the methodological trade-offs of the framework and the limits they place on the conclusions that can be drawn.

7.1 Key Findings and Implications

Five findings stand out. The first and the last two are supported by both the controlled experiments and the clinical data.

Episode scope changes verdicts, and the effect depends on the constraint. Removing the episode boundary changes verdicts in both directions, and the constraints affected can be identified from their semantics. On the generated logs (Section 5.2.1), case-scoped evaluation masked violations of the four ordering constraints without a time window in 23.6 % to 29.6 % of multi-episode cases, whereas the windowed variant was unaffected. It also invented violations of the unary constraint in 47.0 % of these cases by pooling occurrences across episodes. The clinical ablation (Section 6.4.7) shows the same two patterns. Masking is concentrated in the unbounded precedence constraints `INT-SEQ-01` (88.2 % of its episode-level flags) and `INT-SEQ-02` (12.7 %), and most trace-only violations (973 of 1448) stem from `SBT-TM-02`, whose activation is anchored to the start of the stay rather than to the episode. Among the constraints with a bounded target, only `RC-DP-01` is masked (20.0 %). Scope sensitivity thus depends on whether an obligation can be discharged, or an activation counted, outside the episode. The distortion extends to the outcome analysis: on the generated logs, case-scoped verdicts attenuated a planted risk ratio of 3 to 1.88 and reduced power from 100 % to 40 % for two constraints. In the clinical data, 80.1 % of the stays with at least two evaluable episodes contain both a failed and a successful trial, so a single stay-level outcome cannot represent them.

Documentation gaps dominate apparent non-conformance. Aggregate violation rates alone are insufficient to characterize the observed deviations. Of the 184 834 readiness violations, 90.9 % involved missing or incomplete documentation, while only 9.1 % corresponded to criteria that were assessed and not met (Section 6.4.3). The same violation rate can therefore arise from very different evidence states. Separating these cases changes the interpretation of the diagnostic: much of the apparent non-conformance concerns what was documented rather than a recorded physiological state that failed the guideline threshold. The gaps are not outcome-neutral. Splitting the association by cause shows

that unmet criteria carry the stronger association where the criterion is directly measured, for example $RR = 0.62$ against 0.88 for undocumented violations of the pH criterion, but the strongest single association, that of hemoglobin ($RR = 0.62$), comes entirely from episodes without a charted reading (Section 6.4.5). Whether a criterion is charted therefore carries outcome information of its own. The fine log illustrates the converse risk (Section 5.2.3). When the event required to derive the amount owed is missing, the outcome label defaults to *Success*, yielding a success share of 96 % under Payment against 32.9 % under Create Fine. Missing evidence that is not represented as a distinct state is thus converted into a favorable value. The implication is that a violated verdict should be reported together with its cause category, and that the categories, although informative, remain dependent on the structure and documentation practices of the MIMIC-IV data.

Non-conformance is not a homogeneous exposure. Isolated readiness violations were not associated with lower success rates, whereas broader violations were (Table 6.8). The magnitude analysis provided a more limited pattern, with graded associations for hemodynamics, pH, and temperature but not for oxygenation or hemoglobin (Figure 6.5). Breadth and, for some criteria, magnitude therefore carry information that is lost when every violation is treated as equivalent. This supports severity-aware reporting, in which episodes are ranked for review rather than only enumerated.

Conformance and outcome are not interchangeable. Several constraints were associated with outcomes in directions that do not correspond to their normative interpretation. For `RC-DP-07`, the violation primarily represents missing documentation of inspiratory effort, making the association sensitive to monitoring intensity. For `INT-TR-01` and `INT-COMPOSITE-01`, the decision to intervene after a failed trial is itself related to the clinical circumstances surrounding that failure, creating the potential for confounding by indication. `SBT-TM-02` illustrates a different problem: its violation is defined by the same successful late trial that determines the outcome, making an outcome association structurally meaningless. The fine log shows both patterns (Section 5.2.2). `RTF-10` is activated by handover to credit collection, which is itself part of the *Failure* definition, so its outcome is fixed by activation and no association can be estimated, which parallels the entanglement of `SBT-TM-02`. `RTF-03`, `RTF-03b`, and `RTF-09` are positively associated with success because their activation and violation reflect how far a fine progressed along the escalation path, that is, differences between process populations. On the generated logs, the association stage recovered planted risk ratios between 0.25 and 3 when the data-generating process was controlled. The observed patterns therefore originate in the process rather than in the estimator. A constraint can be correctly formalized and still be unsuitable as a standalone outcome indicator, and outcome associations should be read as diagnostics for the constraint set rather than as evidence of guideline efficacy.

Vacuity is informative. Whether a constraint applies to an episode often depends on how the episode unfolded, and this dependence can be related to the outcome. On the generated logs, pooling vacuous with satisfied episodes gave estimates close to those

obtained by excluding them (3.09 versus 3.02 for a planted risk ratio of 3), because vacuity was drawn independently of the outcome. Both real logs violate this condition. On the fine log, RTF-03b changes from $RR = 221$ to 0.24 when the 1520 episodes settled before Send Fine (30.4 %) are pooled as satisfied. In the clinical data, the 43 203 episodes without a re-ventilation failure (64.6 %) enter the satisfied group of the failure-management constraints. INT-DX-05 then changes from $RR = 1.14$ to 0.87, INT-SEQ-02 from 1.04 to 1.28, and INT-TR-01 loses significance after correction (1.18, $p = 6.5 \times 10^{-17}$ to 1.03, $p = 0.049$) (Table 6.9). In each case, the pooled exposure partly indicates whether the episode reached the point at which the obligation arises. Excluding vacuous episodes is therefore the appropriate primary analysis, with the pooled treatment reported as a sensitivity analysis. A disagreement between the two indicates that the association concerns the population rather than conformance.

Implications. Taken together, the results show that the value of episode-scoped conformance checking lies less in a single adherence score than in preserving the structure of the evidence. A verdict can be tied to a particular attempt, decomposed by evidence state, examined across repeated attempts, and related to outcomes without treating the outcome as a substitute for conformance. The controlled experiments establish that evaluation scope and vacuity treatment alter results when ground truth is known, and the clinical data show that the same effects arise in real records. The limitations trade-offs of these conclusions are discussed next.

7.2 Trade-Offs

The configuration-driven architecture (Chapter 4) shifts part of the methodological effort from implementation to configuration. The framework provides the evaluation machinery, but applying it to a new process requires domain-specific choices about activity mappings, trigger definitions, constraints, thresholds, context windows, and outcomes.

This configuration dependence is evident in the trigger-sensitivity results (Section 5.2.3). Coverage, episode granularity, retained information, and outcome-label validity vary substantially with the trigger, and no single trigger satisfies all criteria equally. Trigger selection is therefore part of the modeling process rather than a purely technical preprocessing step. The same applies to the thresholds and temporal windows used by individual constraints. The clinical study shows this dependence as well. The satisfying retry for the trial-timing constraints is visible only through the post-context window (Section 6.4.2), and inter-episode rules can be analyzed for only 37.2 % of episodes because a successor exists only when ventilation resumes (Section 6.4.1).

The framework also makes explicit a trade-off between expressive constraint evaluation and computational cost. The scalability results (Section 5.2.4) show near-linear growth in runtime with the number of cases and constraints at the tested scale, while the complexity analysis (Section 5.2.5) identifies template grade and data access as the main determinants of per-constraint cost. These results characterize the observed implementation rather

than providing a general performance guarantee, but they provide a basis for anticipating the cost of a new constraint set.

A further trade-off concerns the interpretation of outcome associations. The association analysis does not adjust for covariates or explicitly model confounding. The explanations discussed for individual constraints are therefore interpretations of observed patterns rather than causal estimates. In addition, the definition of the comparison population affects the resulting effect size, as the vacuity analyses of Sections 5.2.2 and 6.4.6 show.

These trade-offs limit what can be concluded from the present study. The framework provides a configurable mechanism for episode-scoped conformance checking and for relating its results to outcomes, but the configuration, domain interpretation, and validation remain process-specific. The results therefore do not establish clinical validity, general applicability across guidelines, or readiness for deployment as a clinical monitoring tool.

8 Conclusion

Conformance checking relates observed event data to a normative specification in order to detect and diagnose deviations. This thesis addressed two obstacles that arise when the specification is a clinical guideline and the data are routinely collected hospital records. Procedural techniques evaluate whole traces and return aggregate fitness scores, which obscures which obligation was breached and when. Independently of that, a single case identifier frequently bundles several enactments of the same process, so a trace-level verdict conflates episodes that should be judged separately.

The response developed here is a declarative, multi-perspective conformance-checking framework organized as four steps. Normative rules are expressed as MP-Declare constraints spanning control-flow, data, and temporal dimensions while preserving the open-world assumption (RQ1). A trigger-based extraction procedure extracts episodes from each case trace, each opened by a recurrent domain-meaningful event and bounded by a configurable observation window, an end-event condition, or the next trigger, so that the episode and not the trace becomes the unit of evaluation (Section 4.1; RQ2). Bounded pre- and post-context windows then admit the measurements and the late-recorded evidence that an episode boundary would otherwise cut off (Section 4.2). Constraint evaluation is thereby localized to the episode, and standard MP-Declare semantics are conservatively extended with a target-selection mode that decides an obligation on the last measurement in the window instead of on any satisfying occurrence, for the common case of a guideline regulating where a repeatedly charted quantity stands at the end of a phase (Section 4.3). A final step relates the per-episode verdicts to a designated episode outcome, estimating per constraint whether adherence is associated with the favorable result and thereby checking the rules against the data as well as the data against the rules (Section 4.4). All domain knowledge enters through externalized configuration, so transferring the framework requires authoring configuration and not modifying the method.

The evaluation in Chapter 5 established the semantic correctness and practical behavior of the framework under controlled conditions. The checker reproduced the prescribed verdicts in 93 of 95 hand-written cases covering all 24 implemented templates, with the two divergences arising from the documented treatment of non-activated unary negative constraints. With respect to RQ2, the experiments showed that episode-level evaluation is more precise than case-level evaluation, whose results deviate from the episode-level verdicts in both directions. On generated multi-episode logs, episode-scoped evaluation recovered all planted verdicts, whereas case-scoped evaluation both masked existing violations and introduced violations that occurred in no individual episode. Temporal bounds can prevent some of these effects, but only for constraints whose semantics impose such bounds. The same distinction carried through to outcome associations: episode-

scoped evaluation recovered the planted effects closely, while case-scoped evaluation substantially attenuated the strongest effect and reduced coverage of the planted value. The benchmark on a public event log further showed that the complete pipeline is computationally feasible at realistic scale.

Chapter 6 applied the framework to ventilator weaning in MIMIC-IV, covering 35 692 ICU stays, 66 848 episodes, and 21 constraints derived from a practical weaning guideline. This addresses RQ3. Every activated obligation received a verdict tied to a specific episode and constraint, with coverage limited to stays containing the configured trigger. The identified deviations contain substantially more structure than an aggregate violation rate suggests. Of the 184 834 readiness violations, 90.9 % involved missing or incomplete documentation, whereas only 9.1 % represented criteria that were assessed and not met. The breadth of deviation also mattered: isolated readiness violations showed no corresponding outcome difference, whereas associations emerged as multiple criteria were violated. At the same time, the outcome analysis demonstrated that a correctly formalized constraint is not necessarily a valid quality indicator. The observed associations could reflect monitoring intensity, confounding by indication, or, in the case of `SBT-TM-02`, a definitional relationship between the constraint and the outcome itself.

These findings depend on retaining the episode as the unit of evaluation. Across multi-episode stays with an observable outcome, episodes frequently contained different outcomes, making a single stay-level label an inadequate representation of individual weaning attempts. Removing episode segmentation also changed the conformance results in both directions, masking 434 violations and introducing 1448 violations that occur in no episode in the clinical ablation. More fundamentally, trace-level evaluation loses the connection between a verdict and the attempt in which it arose: a stay flagged at trace level contains, on average, 3.7 to 4.9 episodes. Episode-level evaluation preserves this localization while still allowing bounded context to incorporate evidence immediately outside the episode, which makes its diagnostics more interpretable (RQ2).

Taken together, the results support a specific contribution: localizing multi-perspective declarative conformance checking to recurrent process episodes makes deviations more interpretable by preserving their constraint, temporal, and episode-level context. The framework does not determine whether a deviation was clinically warranted, nor do its outcome associations establish causality. Its contribution is instead to make the structure of observed conformance deviations explicit: what requirement was violated, in which episode, and on the basis of what available evidence. This provides a more informative basis for subsequent clinical interpretation than an aggregate trace-level adherence measure alone.

Future Work. Four directions follow from the open limitations of this thesis, ordered by how much each would strengthen its main claim.

Sensitivity and transfer. The context windows of the case study were taken from guideline text and literature and were not varied. Re-running the analysis over a grid of lookback and response windows would show which findings depend on them, in particular the cause decomposition and the trial-timing verdicts. The claim that transfer requires

only new configuration is likewise supported by a single guideline. Instantiating a second guideline, ideally in a different clinical setting, would test it and would show how much effort the authoring of constraints and event mappings actually requires. Tooling that proposes activity mappings from a terminology, for review by a domain expert, could reduce that effort.

Association analysis. The outcome associations treat episodes as independent, although several episodes come from the same stay and several stays from the same patient, and they are unadjusted for patient characteristics. Cluster-robust or mixed-effects estimates, together with stratification or covariate adjustment, would show which of the observed associations survive. Several inverted associations, such as those of `RC-DP-07` and `INT-TR-01`, are plausibly explained by selection effects that such an analysis could partly separate. The results would remain associational, since the framework makes no causal claim.

Prioritization of violations. All violations currently count equally, whereas the breadth analysis shows that they are not equally consequential: a single violated readiness criterion carried no penalty in success rate, while several did. Severity weights derived from guideline metadata or from the observed relationship between breadth and outcome would allow a report to rank episodes for review. It could be evaluated by whether a ranking places the episodes that reviewers judge unwarranted ahead of the rest.

Explainability. A verdict currently reports its cause category and, where available, its margin. Reconstructing the minimal evidence that would have discharged an obligation, such as the missing measurement or the value that would have satisfied the threshold, and presenting it with the flag would let a clinician act on a verdict without consulting the underlying log. Whether this shortens review time or improves agreement between reviewers is a testable question.

A Ventilator Mode Mapping

This appendix lists the concrete mapping from charted ventilator mode strings in MIMIC-IV to the two operational weaning states introduced in Section 6.2.5. Two mode fields are used. These are the general *ventilator mode* (chartevents itemid 223849) and the Hamilton-ventilator-specific mode (itemid 229314). When both are charted at a timepoint, the Hamilton mode takes precedence, as it is the more granular of the two. The general mode serves as a fallback when the Hamilton mode is missing or unmapped. Each mode string is first assigned a clinical category, which then collapses to one of the two operational states. *Full Invasive Support* maps to *Full Support*, while both *SBT* and *Liberated/Non-Invasive* map to *SBT Candidate* (see Section 6.2.5 for the rationale). Tables A.1 and A.2 list the mappings for the general and the Hamilton mode field, respectively. Charted mode strings not listed there are ignored for state transitions.

Table A.1: Mapping of general ventilator modes (itemid 223849) to clinical categories and operational weaning states.

Ventilator mode	**Clinical category**	**Weaning state**
Apnea Ventilation	Full Invasive Support	Full Support
APRV, APRV/Biphasic+ApnPress, APRV/Biphasic+ApnVol	Full Invasive Support	Full Support
APV (cmv)	Full Invasive Support	Full Support
CMV, CMV/ASSIST, CMV/AutoFlow, CMV/ASSIST/AutoFlow, (S) CMV, P-CMV	Full Invasive Support	Full Support
DuoPaP	Full Invasive Support	Full Support
MMV, MMV/AutoFlow	Full Invasive Support	Full Support
PCV+, PCV+Assist	Full Invasive Support	Full Support
PRES/AC, PRVC/AC, PRVC/SIMV, VOL/AC	Full Invasive Support	Full Support
SIMV, SIMV/AutoFlow, SIMV/PRES, SIMV/VOL, P-SIMV	Full Invasive Support	Full Support
SYNCHRON MASTER, SYNCHRON SLAVE	Full Invasive Support	Full Support
CPAP, CPAP/PPS, CPAP/PSV, CPAP/PSV+ApnPres, CPAP/PSV+Apn TCPL, CPAP/PSV+ApnVol	SBT	SBT Candidate
MMV/PSV, MMV/PSV/AutoFlow	SBT	SBT Candidate
PCV+/PSV	SBT	SBT Candidate
PSV/SBT	SBT	SBT Candidate
SIMV/PSV, SIMV/PSV/AutoFlow	SBT	SBT Candidate
SPONT	SBT	SBT Candidate
Ambient, Standby	Liberated/Non-Invasive	SBT Candidate

Table A.2: Mapping of Hamilton-specific ventilator modes (itemid 229314) to clinical categories and operational weaning states.

Ventilator mode	**Clinical category**	**Weaning state**
APRV, APV (cmv), APV (simv), ASV, DuoPaP, P-CMV, P-SIMV, (S) CMV, SIMV	Full Invasive Support	Full Support
SPONT, VS	SBT	SBT Candidate
Ambient, nCPAP-PS, NIV, NIV-ST	Liberated/Non-Invasive	SBT Candidate

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